

AN INTRODUCTION TO FLOW-MATCHING AND DIFFERENTIAL TOPOLOGY

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Abstract

These notes begin as an expanded, didactic introduction to the multilinear algebra of tensors and differential forms, and then develop the basics of smooth manifolds, integration, Stokes’ theorem, and de Rham cohomology. We include a self-contained “diagram-chasing” toolkit from homological algebra (exact sequences, Snake Lemma, Mayer–Vietoris, and related constructions), and then turn to core ideas in differential topology (regular values, transversality, degree, and applications).

This unified edition also adds an applications part: a carefully translated and notation-aligned case study on *temperature-conditioned iterated energy-based flow matching* (iEFM) for simulation-free enzyme path and sequence design with a learned quantum-accuracy potential. A bridge section explains how the language of differential forms naturally encodes densities, change of variables, continuity equations, and the dynamics of continuous normalizing flows.

The guiding principle throughout is to keep the original spirit and computations of the notes, while filling in missing steps, adding clarifying examples, and weaving the foundational and applied pieces into a single cohesive narrative.

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Part I

Foundations: Forms, Manifolds, Cohomology, and Differential Topology

1 The Tensor Algebra (Free Algebra)

The Tensor Algebra

Here we give a brief review of the tensor algebra and some of its subalgebras, subspaces, and quotient spaces.

Definition 1.1 (Tensor algebra). *Let V be a finite-dimensional complex vector space. The tensor algebra over V is*

$$T(V) := \bigoplus_{n \geq 0} V^{\otimes n},$$

the direct sum of all tensor powers of V , where by convention $V^{\otimes 0} := \mathbb{C}$. The multiplication in $T(V)$ is the bilinear map

$$V^{\otimes p} \times V^{\otimes q} \longrightarrow V^{\otimes(p+q)}, \quad (v, w) \longmapsto v \otimes w,$$

extended by linearity to all of $T(V)$.

Remark 1.2. The product in $T(V)$ is *concatenation* of tensors. It is not the “tensor product of tensors” (which is not defined between *elements* of different tensor powers). Concretely, if $v \in V^{\otimes p}$ and $w \in V^{\otimes q}$ then the multiplication in $T(V)$ produces an element of $V^{\otimes(p+q)}$ by simply writing the symbols next to each other: $vw := v \otimes w$.

The algebra $T(V)$ is naturally $\mathbb{N}$ -graded: the homogeneous degree- p subspace is $V^{\otimes p}$.

In the same way we define the tensor algebra $T(V^*)$ on the dual space. Elements of $T(V^*)$ are tensors of covectors; they can also be interpreted as multilinear functions on V via

$$(\varepsilon^{i_1} \otimes \cdots \otimes \varepsilon^{i_p})(v_1, \dots, v_p) = \varepsilon^{i_1}(v_1) \cdots \varepsilon^{i_p}(v_p),$$

where $\{\varepsilon^1, \dots, \varepsilon^k\}$ is a basis of V^* and $v_j \in V$.

If we choose a basis of V , say

$$B_V = \{x_1, \dots, x_k\},$$

then $T(V^*)$ can be identified with the free associative (noncommutative) algebra $\mathbb{C}\langle x_1, \dots, x_k \rangle$, i.e. noncommutative polynomials in the variables x_i .

Example 1.3 (Generating tensor monomials in $T(V^*)$). Take $(V^*)^{\otimes 0} = \mathbb{C}$ and the basis elements $\{\varepsilon^1, \dots, \varepsilon^k\}$ of $(V^*)^{\otimes 1} = V^*$ (dual to B_V). These degree-one tensors generate $T(V^*)$ under the multiplication in the tensor algebra. Indeed, any decomposable (simple) tensor $\varepsilon^{i_1} \otimes \cdots \otimes \varepsilon^{i_p} \in (V^*)^{\otimes p}$ can be produced as a product of degree-one generators.

For example, if $\dim V^* = 3$ and $B_{V^*} = \{\varepsilon^1, \varepsilon^2, \varepsilon^3\}$, then the basis of $(V^*)^{\otimes 2}$ can be generated in the following way:

$$\begin{array}{lll} \varepsilon^1 \cdot \varepsilon^1 \mapsto \varepsilon^1 \otimes \varepsilon^1 & \varepsilon^1 \cdot \varepsilon^2 \mapsto \varepsilon^1 \otimes \varepsilon^2 & \varepsilon^1 \cdot \varepsilon^3 \mapsto \varepsilon^1 \otimes \varepsilon^3 \\ \varepsilon^2 \cdot \varepsilon^1 \mapsto \varepsilon^2 \otimes \varepsilon^1 & \varepsilon^2 \cdot \varepsilon^2 \mapsto \varepsilon^2 \otimes \varepsilon^2 & \varepsilon^2 \cdot \varepsilon^3 \mapsto \varepsilon^2 \otimes \varepsilon^3 \\ \varepsilon^3 \cdot \varepsilon^1 \mapsto \varepsilon^3 \otimes \varepsilon^1 & \varepsilon^3 \cdot \varepsilon^2 \mapsto \varepsilon^3 \otimes \varepsilon^2 & \varepsilon^3 \cdot \varepsilon^3 \mapsto \varepsilon^3 \otimes \varepsilon^3. \end{array}$$

One can then easily see how to generate a basis $\{\varepsilon^{i_1} \otimes \cdots \otimes \varepsilon^{i_n} : i_j \in \{1, \dots, k\}\}$ of $(V^*)^{\otimes n}$, and thus any homogeneous degree- n component of $T(V^*)$.

Definition 1.4 (Ideal). *Let A be an algebra. An ideal $I \subseteq A$ is a vector subspace such that $aI \subseteq I$ and $Ia \subseteq I$ for all $a \in A$.*

Exterior and symmetric quotients

Two important quotients of $T(V)$ are obtained by imposing commutativity or anticommutativity.

Definition 1.5 (Exterior algebra as a quotient). *Assume $\text{char}(\mathbb{C}) = 0$. Let $I_\wedge \subseteq T(V)$ be the ideal generated by elements*

$$v \otimes v, \quad v \in V.$$

The exterior algebra of V is the quotient

$$\bigwedge V := T(V)/I_\wedge.$$

The image of $v_1 \otimes \cdots \otimes v_p$ in this quotient is written $v_1 \wedge \cdots \wedge v_p$.

Remark 1.6 (Equivalent generators). Over a field of characteristic $\neq 2$, the ideal generated by all $v \otimes v$ is the same as the ideal generated by all $v \otimes w + w \otimes v$ (for $v, w \in V$). Indeed,

$$(v + w) \otimes (v + w) = v \otimes v + v \otimes w + w \otimes v + w \otimes w,$$

so if $v \otimes v$ and $w \otimes w$ vanish in the quotient, then so does $v \otimes w + w \otimes v$. Conversely, setting $w = v$ shows $2(v \otimes v) = 0$, hence $v \otimes v = 0$.

In the exterior algebra, permuting factors introduces a sign: if $\sigma \in S_p$, then

$$v_1 \wedge \cdots \wedge v_p = \text{sgn}(\sigma) v_{\sigma(1)} \wedge \cdots \wedge v_{\sigma(p)}.$$

Definition 1.7 (Symmetric algebra as a quotient). *Let $I_{\text{sym}} \subseteq T(V)$ be the ideal generated by*

$$v \otimes w - w \otimes v, \quad v, w \in V.$$

The symmetric algebra of V is

$$\text{Sym}(V) := T(V)/I_{\text{sym}}.$$

Remark 1.8 (Polynomial functions). If V is finite-dimensional and we choose a basis, then $\text{Sym}(V^*)$ identifies with the algebra $\mathbb{C}[x_1, \dots, x_k]$ of commutative polynomials and can be viewed as the coordinate ring of V .

1.1 Exterior Algebra

We now focus on the exterior product, alternating forms, and compute a few examples.

Definition 1.9 (Tensor power and exterior power). *Let V be a finite-dimensional vector space. Define*

$$V^{\otimes k} := \underbrace{V \otimes \cdots \otimes V}_{k \text{ factors}}.$$

Define the k th exterior power as the quotient

$$\bigwedge^k V := V^{\otimes k} / R,$$

where R is the subspace generated by all tensors of the form

$$v_1 \otimes \cdots \otimes v_k - \text{sgn}(\sigma) v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(k)}, \quad \sigma \in S_k.$$

We denote the image of $v_1 \otimes \cdots \otimes v_k$ by $v_1 \wedge \cdots \wedge v_k$.

Example 1.10. For $v_1 \wedge v_2 \in \wedge^2 V$ we have

$$v_1 \wedge v_2 = -v_2 \wedge v_1.$$

Remark 1.11 (Linear dependence forces vanishing). If $v_1, \dots, v_k$ are linearly dependent, then $v_1 \wedge \cdots \wedge v_k = 0$. In particular, if $k > \dim V$, then $\wedge^k V = 0$.

Remark 1.12 (Basis of $\wedge^k V$). If $\dim V = n$ and $\{e_1, \dots, e_n\}$ is a basis, then a basis of $\wedge^k V$ is

$$\{e_{i_1} \wedge \cdots \wedge e_{i_k} \mid 1 \leq i_1 < \cdots < i_k \leq n\}.$$

Hence $\dim(\wedge^k V) = \binom{n}{k}$.

Definition 1.13 (Induced map on exterior powers). Let $T \in \text{End}(V)$, i.e. $T : V \rightarrow V$ is linear. The induced map

$$\wedge^k T : \wedge^k V \longrightarrow \wedge^k V$$

is defined on decomposable elements by

$$v_1 \wedge \cdots \wedge v_k \longmapsto T(v_1) \wedge \cdots \wedge T(v_k),$$

and extended linearly.

Remark 1.14 (Matrix entries are minors). If T is represented by a matrix A in the basis $\{e_i\}$, then the matrix of $\wedge^k T$ in the induced basis of $\wedge^k V$ is given by $k \times k$ minors of A .

Example 1.15 (A 3×3 computation). Let K be any algebraically closed field of characteristic 0. Let $T : K^3 \rightarrow K^3$ be represented by the matrix

$$A = \begin{pmatrix} 1 & 3 & 1 \\ 2 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

with respect to the standard basis $\{e_1, e_2, e_3\}$ of K^3 . Form the basis of the tensor power $(K^3)^{\otimes 2}$:

$$\{e_1 \otimes e_1, e_1 \otimes e_2, e_1 \otimes e_3; e_2 \otimes e_1, e_2 \otimes e_2, e_2 \otimes e_3; e_3 \otimes e_1, e_3 \otimes e_2, e_3 \otimes e_3\}.$$

Let $\pi : (K^3)^{\otimes 2} \rightarrow \wedge^2 K^3$ be the quotient map. Then $\ker(\pi)$ is spanned by the symmetric tensors:

$$\ker(\pi) = \text{Span}(e_1 \otimes e_1, e_2 \otimes e_2, e_3 \otimes e_3, e_1 \otimes e_2 + e_2 \otimes e_1, e_1 \otimes e_3 + e_3 \otimes e_1, e_2 \otimes e_3 + e_3 \otimes e_2).$$

There are several natural choices of basis for $\wedge^2 K^3$. The “standard” choice is $\{e_1 \wedge e_2, e_1 \wedge e_3, e_2 \wedge e_3\}$, but in these notes we choose

$$\mathcal{B} = \{e_2 \wedge e_3, e_3 \wedge e_1, e_1 \wedge e_2\}$$

(for reasons that will become clear later when we identify $\bigwedge^2(\mathbb{R}^3)^*$ with $\mathbb{R}^3$). We compute the matrix of

$$\bigwedge^2 T : \bigwedge^2 K^3 \longrightarrow \bigwedge^2 K^3$$

with respect to $\mathcal{B}$ by computing the images of the basis wedges.

Column computations. Using $T(e_1) = e_1 + 2e_2 + 3e_3$, $T(e_2) = 3e_1 + 2e_2 + e_3$, and $T(e_3) = e_1 + 3e_2 + 2e_3$, we obtain:

$$\begin{aligned} T(e_2) \wedge T(e_3) &= (3e_1 + 2e_2 + e_3) \wedge (e_1 + 3e_2 + 2e_3) \\ &= (e_2 \wedge e_3) - 5(e_3 \wedge e_1) + 7(e_1 \wedge e_2), \\ T(e_3) \wedge T(e_1) &= (e_1 + 3e_2 + 2e_3) \wedge (e_1 + 2e_2 + 3e_3) \\ &= 5(e_2 \wedge e_3) - 1(e_3 \wedge e_1) - 1(e_1 \wedge e_2), \\ T(e_1) \wedge T(e_2) &= (e_1 + 2e_2 + 3e_3) \wedge (3e_1 + 2e_2 + e_3) \\ &= -4(e_2 \wedge e_3) + 8(e_3 \wedge e_1) - 4(e_1 \wedge e_2). \end{aligned}$$

Therefore, the matrix of $\bigwedge^2 T$ in the basis $\mathcal{B}$ is

$$[\bigwedge^2 T]_{\mathcal{B}} = \begin{pmatrix} 1 & 5 & -4 \\ -5 & -1 & 8 \\ 7 & -1 & -4 \end{pmatrix}.$$

Minors and cofactors. One notices that the entries above are exactly the signed 2×2 minors of A (the cofactors). Explicitly,

$$\bigwedge^2 T = \begin{pmatrix} \det \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} & -\det \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix} & \det \begin{pmatrix} 2 & 2 \\ 3 & 1 \end{pmatrix} \\ -\det \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix} & \det \begin{pmatrix} 1 & 1 \\ 3 & 2 \end{pmatrix} & -\det \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix} \\ \det \begin{pmatrix} 3 & 1 \\ 2 & 3 \end{pmatrix} & -\det \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} & \det \begin{pmatrix} 1 & 3 \\ 2 & 2 \end{pmatrix} \end{pmatrix}.$$

In particular, $[\bigwedge^2 T]_{\mathcal{B}}$ is the cofactor matrix of A . Hence $(\bigwedge^2 T)^*$ is given by the transpose, and equals the classical adjoint (adjugate) of T .

Definition 1.16 (Determinant via top exterior power). *Let $\dim V = n$. For any $A \in \text{End}(V)$ we get a map*

$$\bigwedge^n A : \bigwedge^n V \longrightarrow \bigwedge^n V, \quad v_1 \wedge \cdots \wedge v_n \longmapsto A(v_1) \wedge \cdots \wedge A(v_n).$$

Since $\dim(\bigwedge^n V) = 1$, this map is multiplication by a scalar; that scalar is $\det(A)$. Equivalently,

$$A(v_1) \wedge \cdots \wedge A(v_n) = \det(A) (v_1 \wedge \cdots \wedge v_n).$$

1.2 Multilinear Forms

We now shift focus to alternating multilinear maps and their coordinate expressions. From now on we often take $V = \mathbb{R}^n$ (but the linear-algebra statements hold in general).

Definition 1.17 (Multilinear 1-forms). Let V be a finite-dimensional real vector space and V^* its dual. A multilinear 1-form is an element $\alpha \in V^* = \text{Hom}(V, \mathbb{R})$. If $B = \{e_1, \dots, e_n\}$ is a basis and $B^* = \{\varepsilon^1, \dots, \varepsilon^n\}$ is the dual basis, then any $\alpha \in V^*$ can be written

$$\alpha = \sum_{i=1}^n a_i \varepsilon^i.$$

By convention $\bigwedge^0 V = \mathbb{R}$.

Definition 1.18 (Multilinear k -forms). A (multilinear) k -form on $\mathbb{R}^n$ is an element $\beta \in \bigwedge^k(\mathbb{R}^n)^*$. Thus

$$\beta = \sum_{1 \leq i_1 < \dots < i_k \leq n} b_{i_1, \dots, i_k} \varepsilon^{i_1} \wedge \dots \wedge \varepsilon^{i_k}.$$

Write $\varepsilon^I = \varepsilon^{i_1} \wedge \dots \wedge \varepsilon^{i_k}$ for a multi-index $I = (i_1 < \dots < i_k)$.

Each ε^I defines an alternating k -linear map $(\mathbb{R}^n)^k \rightarrow \mathbb{R}$ by the determinant pairing:

$$(\varepsilon^{i_1} \wedge \dots \wedge \varepsilon^{i_k})(v_1, \dots, v_k) = \det(\varepsilon^{i_r}(v_s))_{r,s=1}^k.$$

Extend linearly to get $\beta(v_1, \dots, v_k)$.

Example 1.19. Let $V = \mathbb{R}^5$, so $\varepsilon^i(v) = v_i$ is the i th coordinate. Let

$$u = (1, 3, 5, 7, 9), \quad v = (2, 4, 6, 8, 10), \quad w = (1, 0, 1, 0, 1).$$

Then for $\varepsilon^{1,3,5} = \varepsilon^1 \wedge \varepsilon^3 \wedge \varepsilon^5 \in \bigwedge^3(\mathbb{R}^5)^*$,

$$\varepsilon^{1,3,5}(u, v, w) = \det \begin{pmatrix} \varepsilon^1(u) & \varepsilon^1(v) & \varepsilon^1(w) \\ \varepsilon^3(u) & \varepsilon^3(v) & \varepsilon^3(w) \\ \varepsilon^5(u) & \varepsilon^5(v) & \varepsilon^5(w) \end{pmatrix} = \det \begin{pmatrix} 1 & 2 & 1 \\ 5 & 6 & 1 \\ 9 & 10 & 1 \end{pmatrix} = 1.$$

Remark 1.20 (Geometric meaning). In general, $\varepsilon^{i_1, \dots, i_k}(v_1, \dots, v_k)$ gives the *oriented k -dimensional volume* of the parallelepiped spanned by the projections of $v_1, \dots, v_k$ onto the coordinate subspace $\text{span}\{e_{i_1}, \dots, e_{i_k}\} \cong \mathbb{R}^k$.

Definition 1.21 (Wedge product of multilinear forms). Let $\alpha \in \bigwedge^r(\mathbb{R}^n)^*$ and $\beta \in \bigwedge^s(\mathbb{R}^n)^*$. Then $\alpha \wedge \beta \in \bigwedge^{r+s}(\mathbb{R}^n)^*$ is defined by bilinearity and the rule

$$(\varepsilon^I) \wedge (\varepsilon^J) = \varepsilon^I \wedge \varepsilon^J,$$

together with antisymmetry. If $\alpha = \sum_I a_I \varepsilon^I$ and $\beta = \sum_J b_J \varepsilon^J$, then

$$\alpha \wedge \beta = \sum_{I,J} a_I b_J (\varepsilon^I \wedge \varepsilon^J).$$

Example 1.22. Let $V = \mathbb{R}^5$ and

$$\alpha = \varepsilon^{1,2} + \varepsilon^{3,5} \in \bigwedge^2(\mathbb{R}^5)^*, \quad \beta = 2\varepsilon^{1,2,4} + \varepsilon^{1,3,5} \in \bigwedge^3(\mathbb{R}^5)^*.$$

Then

$$\begin{aligned} \alpha \wedge \beta &= (\varepsilon^{1,2} + \varepsilon^{3,5}) \wedge (2\varepsilon^{1,2,4} + \varepsilon^{1,3,5}) \\ &= \varepsilon^{1,2} \wedge (2\varepsilon^{1,2,4}) + \varepsilon^{1,2} \wedge \varepsilon^{1,3,5} + \varepsilon^{3,5} \wedge (2\varepsilon^{1,2,4}) + \varepsilon^{3,5} \wedge \varepsilon^{1,3,5}. \end{aligned}$$

The first, second, and last terms vanish because of repeated indices, leaving

$$\alpha \wedge \beta = 2\varepsilon^{3,5} \wedge \varepsilon^{1,2,4} = -2\varepsilon^{1,2,3,4,5}$$

(after reordering and keeping track of the sign of the permutation).

Example 1.23 (A quick computation). Let $\gamma \in \wedge^2(\mathbb{R}^4)^*$ with $\gamma = 3\varepsilon^{1,2} - 2\varepsilon^{3,4}$. For $a = (a_1, a_2, a_3, a_4)$ and $b = (b_1, b_2, b_3, b_4)$,

$$\gamma(a, b) = 3(a_1b_2 - b_1a_2) - 2(a_3b_4 - b_3a_4).$$

One solution to $\gamma(a, b) = 0$ is $a = (2, 4, 2, 8)$ and $b = (1, 2, 1, 4)$.

1.3 The Pullback

Definition 1.24 (Pullback of 1-forms for linear maps). Let $\alpha \in W^*$ and $T : V \rightarrow W$ be linear. The pullback of α by T is the 1-form $T^*\alpha \in V^*$ defined by

$$T^*\alpha := \alpha \circ T, \quad \text{i.e.} \quad (T^*\alpha)(v) = \alpha(Tv).$$

Definition 1.25 (Pullback of k -forms for linear maps). Let $\beta \in \wedge^k W^*$ and $T : V \rightarrow W$ be linear. Define

$$\bigwedge^k T : \bigwedge^k V \rightarrow \bigwedge^k W, \quad v_1 \wedge \cdots \wedge v_k \mapsto T(v_1) \wedge \cdots \wedge T(v_k).$$

The pullback is the map

$$\bigwedge^k T^* : \bigwedge^k W^* \rightarrow \bigwedge^k V^*$$

given by

$$(\bigwedge^k T^*\beta)(v_1, \dots, v_k) := \beta(Tv_1, \dots, Tv_k).$$

Example 1.26 (Pullback of a 2-form: a direct determinant computation). Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the linear map

$$(a, b, c) \mapsto (2a - c, b - 3c).$$

Then $\bigwedge^2 T : \bigwedge^2 \mathbb{R}^3 \rightarrow \bigwedge^2 \mathbb{R}^2$ and $\bigwedge^2 T^* : \bigwedge^2(\mathbb{R}^2)^* \rightarrow \bigwedge^2(\mathbb{R}^3)^*$. Let $\{\varepsilon^1, \varepsilon^2\}$ be the standard basis of $(\mathbb{R}^2)^*$ and $\{\eta^1, \eta^2, \eta^3\}$ the standard basis of $(\mathbb{R}^3)^*$. Let $\alpha = \varepsilon^1 \wedge \varepsilon^2 \in \bigwedge^2(\mathbb{R}^2)^*$ and let $v, w \in \mathbb{R}^3$. Then

$$\begin{aligned} (\bigwedge^2 T^*\alpha)(v, w) &= \alpha(T(v), T(w)) \\ &= \det \begin{pmatrix} \varepsilon^1(T(v)) & \varepsilon^1(T(w)) \\ \varepsilon^2(T(v)) & \varepsilon^2(T(w)) \end{pmatrix} \\ &= \det \begin{pmatrix} 2v_1 - v_3 & 2w_1 - w_3 \\ v_2 - 3v_3 & w_2 - 3w_3 \end{pmatrix} \\ &= (2v_1 - v_3)(w_2 - 3w_3) - (2w_1 - w_3)(v_2 - 3v_3) \\ &= 2(v_1w_2 - w_1v_2) - 6(v_1w_3 - w_1v_3) + (v_2w_3 - w_2v_3) + 3(v_3w_3 - w_3v_3) \\ &= 2 \det \begin{pmatrix} \eta^1(v) & \eta^1(w) \\ \eta^2(v) & \eta^2(w) \end{pmatrix} - 6 \det \begin{pmatrix} \eta^1(v) & \eta^1(w) \\ \eta^3(v) & \eta^3(w) \end{pmatrix} + \det \begin{pmatrix} \eta^2(v) & \eta^2(w) \\ \eta^3(v) & \eta^3(w) \end{pmatrix}, \end{aligned}$$

since the last term is identically zero. Therefore

$$\bigwedge^2 T^*(\varepsilon^1 \wedge \varepsilon^2) = 2(\eta^1 \wedge \eta^2) - 6(\eta^1 \wedge \eta^3) + (\eta^2 \wedge \eta^3).$$

Remark 1.27 (A useful shortcut). The previous computation can be shortened by first computing T^* on 1-forms:

$$T^*\varepsilon^1 = 2\eta^1 - \eta^3, \quad T^*\varepsilon^2 = \eta^2 - 3\eta^3,$$

and then using $\bigwedge^2 T^*(\varepsilon^1 \wedge \varepsilon^2) = (T^*\varepsilon^1) \wedge (T^*\varepsilon^2)$.

Example 1.28. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be $T(a, b, c) = (2a - c, b - 3c)$. Let $\{\varepsilon^1, \varepsilon^2\}$ be the standard basis of $(\mathbb{R}^2)^*$ and $\{\eta^1, \eta^2, \eta^3\}$ the standard basis of $(\mathbb{R}^3)^*$. Let $\alpha = \varepsilon^1 \wedge \varepsilon^2 \in \wedge^2(\mathbb{R}^2)^*$. Then

$$T^*\varepsilon^1 = 2\eta^1 - \eta^3, \quad T^*\varepsilon^2 = \eta^2 - 3\eta^3,$$

so

$$\wedge^2 T^*(\alpha) = (T^*\varepsilon^1) \wedge (T^*\varepsilon^2) = 2\eta^1 \wedge \eta^2 - 6\eta^1 \wedge \eta^3 + \eta^2 \wedge \eta^3.$$

Example 1.29. Let $\alpha = 2\varepsilon^2 + \varepsilon^3 \in (\mathbb{R}^3)^*$ and let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be

$$T(x, y, z) = (3y + 2z, -x - y + z, x - 4z).$$

Then

$$T^*\alpha(v) = \alpha(Tv) = -(\varepsilon^1 + 2\varepsilon^2 + 2\varepsilon^3)(v),$$

so $T^*\alpha = -(\varepsilon^1 + 2\varepsilon^2 + 2\varepsilon^3)$.

Example 1.30. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be $T(x, y) = (12x - 13y, x + 2y, -x - y)$ and let $\alpha = 2\varepsilon^2 + \varepsilon^3$. Then

$$T^*\alpha(x, y) = 2(x + 2y) + (-x - y) = x + y,$$

so $T^*\alpha = \eta^1 + \eta^2 \in (\mathbb{R}^2)^*$.

Example 1.31 (Eigenvector condition for pullback). Let $\alpha = c\varepsilon^2 + \varepsilon^3 \in (\mathbb{R}^3)^*$ for some $c \in \mathbb{R}$, and let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ have matrix $A = (a_{ij})$ in the standard basis. We seek conditions for $T^*\alpha = k\alpha$. Since T^* corresponds to the transpose A^t on coordinate vectors, this is

$$A^t \begin{pmatrix} 0 \\ c \\ 1 \end{pmatrix} = k \begin{pmatrix} 0 \\ c \\ 1 \end{pmatrix}.$$

Equivalently,

$$c a_{21} + a_{31} = 0, \quad c a_{22} + a_{32} = kc, \quad c a_{23} + a_{33} = k.$$

2 Differential Forms

2.1 Differential Forms and Operations on Differential Forms

We now move from multilinear algebra at a fixed vector space to objects that vary smoothly from point to point in Euclidean space.

Definition 2.1 (Tangent space of $\mathbb{R}^n$ at a point). Let $p \in \mathbb{R}^n$. The tangent space $T_p\mathbb{R}^n$ may be identified with $\mathbb{R}^n$ itself, but it is conceptually useful to keep track of the basepoint. One way is to think of $T_p\mathbb{R}^n$ as the set of “arrows” $q - p$ based at p .

Equivalently, tangent vectors can be viewed as derivations at p : a tangent vector $v_p \in T_p\mathbb{R}^n$ is a linear map assigning to each smooth function f (defined on a neighborhood of p) a number $v_p[f] \in \mathbb{R}$ such that the Leibniz rule holds. In coordinates $(x^1, \dots, x^n)$ the standard basis is

$$\left\{ \left. \frac{\partial}{\partial x^1} \right|_p, \dots, \left. \frac{\partial}{\partial x^n} \right|_p \right\}.$$

Definition 2.2 (Vector fields). A (smooth) vector field on $\mathbb{R}^n$ is a smooth map X that assigns to each p a tangent vector $X(p) \in T_p\mathbb{R}^n$. In coordinates,

$$X = \sum_{i=1}^n X^i \frac{\partial}{\partial x^i},$$

where the coefficient functions $X^i : \mathbb{R}^n \rightarrow \mathbb{R}$ are smooth.

Definition 2.3 (Cotangent space and 1-forms). The cotangent space $T_p^*\mathbb{R}^n$ is the dual space $(T_p\mathbb{R}^n)^*$. Its dual basis is written

$$\{(\mathrm{d}x^1)_p, \dots, (\mathrm{d}x^n)_p\},$$

characterized by $(\mathrm{d}x^i)_p \left(\frac{\partial}{\partial x^j} \Big|_p \right) = \delta_j^i$. A (smooth) differential 1-form on $\mathbb{R}^n$ is a smooth assignment

$$\omega(p) \in T_p^*\mathbb{R}^n,$$

i.e. a smooth section of the cotangent bundle. In coordinates

$$\omega = \sum_{i=1}^n a_i \mathrm{d}x^i$$

with smooth coefficient functions $a_i : \mathbb{R}^n \rightarrow \mathbb{R}$.

Definition 2.4 (Differential k -forms). For each p , let $\bigwedge^k T_p^*\mathbb{R}^n$ be the space of alternating k -linear maps $(T_p\mathbb{R}^n)^k \rightarrow \mathbb{R}$. A smooth differential k -form on $\mathbb{R}^n$ is a smooth assignment

$$\omega(p) \in \bigwedge^k T_p^*\mathbb{R}^n.$$

Equivalently, in coordinates one can write

$$\omega = \sum_{1 \leq i_1 < \dots < i_k \leq n} a_{i_1 \dots i_k}(x) \mathrm{d}x^{i_1} \wedge \dots \wedge \mathrm{d}x^{i_k}.$$

We denote the space of smooth k -forms on $\mathbb{R}^n$ by $\Omega^k(\mathbb{R}^n)$. We also set $\Omega^0(\mathbb{R}^n) = C^\infty(\mathbb{R}^n)$.

Example 2.5 (2-forms on $\mathbb{R}^3$). A basis of $\bigwedge^2 T_p^*\mathbb{R}^3$ is

$$\{(\mathrm{d}x)_p \wedge (\mathrm{d}y)_p, (\mathrm{d}x)_p \wedge (\mathrm{d}z)_p, (\mathrm{d}y)_p \wedge (\mathrm{d}z)_p\}.$$

Thus a 2-form on $\mathbb{R}^3$ can be written

$$\omega = a_{12} \mathrm{d}x \wedge \mathrm{d}y + a_{13} \mathrm{d}x \wedge \mathrm{d}z + a_{23} \mathrm{d}y \wedge \mathrm{d}z.$$

Example 2.6. Let $\alpha = xz (\mathrm{d}y \wedge \mathrm{d}z) \in \Omega^2(\mathbb{R}^3)$ and let

$$u = a_y \frac{\partial}{\partial y} + a_z \frac{\partial}{\partial z}, \quad v = b_x \frac{\partial}{\partial x} + b_y \frac{\partial}{\partial y}.$$

Then

$$\alpha(u, v) = xz \det \begin{pmatrix} \mathrm{d}y(u) & \mathrm{d}y(v) \\ \mathrm{d}z(u) & \mathrm{d}z(v) \end{pmatrix} = xz \det \begin{pmatrix} a_y & b_y \\ a_z & 0 \end{pmatrix} = -xz b_y a_z.$$

Definition 2.7 (Addition and wedge product of differential forms). If $\alpha, \beta \in \Omega^k(\mathbb{R}^n)$ are k -forms,

$$\alpha = \sum_I a_I dx^I, \quad \beta = \sum_I b_I dx^I,$$

their sum is $\alpha + \beta = \sum_I (a_I + b_I) dx^I$.

If $\alpha \in \Omega^r(\mathbb{R}^n)$ and $\beta \in \Omega^s(\mathbb{R}^n)$ then $\alpha \wedge \beta \in \Omega^{r+s}(\mathbb{R}^n)$ is defined by bilinearity and the rule $dx^I \wedge dx^J = dx^I \wedge dx^J$, together with $dx^i \wedge dx^i = 0$ and graded-antisymmetry.

Example 2.8. Let $\alpha = x dx - y dy$, $\beta = z dx \wedge dy + x dy \wedge dz$, and $\omega = z dy$ be forms on $\mathbb{R}^3$. Then

$$\begin{aligned} \alpha \wedge \beta &= (x dx - y dy) \wedge (z dx \wedge dy + x dy \wedge dz) \\ &= x dx \wedge x dy \wedge dz \\ &= x^2 dx \wedge dy \wedge dz, \end{aligned}$$

and

$$\omega \wedge \alpha \wedge \beta = z dy \wedge x^2 dx \wedge dy \wedge dz = 0$$

because $dy \wedge dy = 0$.

2.2 The Pullback of Forms and Change of Variables

Definition 2.9 (Pullback of forms). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be smooth. The pullback of a k -form $\omega \in \Omega^k(\mathbb{R}^m)$ is the k -form $f^*\omega \in \Omega^k(\mathbb{R}^n)$ defined by

$$(f^*\omega)_p(v_1, \dots, v_k) := \omega_{f(p)}((df)_p(v_1), \dots, (df)_p(v_k)),$$

where $(df)_p : T_p\mathbb{R}^n \rightarrow T_{f(p)}\mathbb{R}^m$ is the derivative. For a 0-form (function) $g \in C^\infty(\mathbb{R}^m)$ we set $f^*g = g \circ f$.

Remark 2.10 (Algebraic properties). For $\omega, \eta \in \Omega^k(\mathbb{R}^m)$, $\alpha \in \Omega^r(\mathbb{R}^m)$, $\beta \in \Omega^s(\mathbb{R}^m)$, and $g \in C^\infty(\mathbb{R}^m)$:

1. $f^*(\omega + \eta) = f^*\omega + f^*\eta$.
2. $f^*(g\omega) = (f^*g)(f^*\omega)$.
3. $f^*(\alpha \wedge \beta) = f^*\alpha \wedge f^*\beta$.

Remark 2.11 (Pullback is substitution). Write coordinates on $\mathbb{R}^n$ as $(x^1, \dots, x^n)$ and on $\mathbb{R}^m$ as $(y^1, \dots, y^m)$. If $f = (f^1, \dots, f^m)$, then

$$f^*(dy^i) = d(y^i \circ f) = df^i = \sum_{j=1}^n \frac{\partial f^i}{\partial x^j} dx^j.$$

Thus if $\omega = \sum_I a_I(y) dy^I$, then

$$f^*\omega = \sum_I a_I(f(x)) df^{i_1} \wedge \dots \wedge df^{i_k}.$$

Example 2.12. Let $\omega \in \Omega^2(\mathbb{R}^3)$ be

$$\omega = z^2 dx \wedge dy + y^2 dx \wedge dz + x^2 dy \wedge dz.$$

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be

$$f(u, v) = (u^2v, u, v) = (x, y, z).$$

Compute

$$dx = d(u^2v) = 2uv du + u^2 dv, \quad dy = du, \quad dz = dv.$$

Hence

$$\begin{aligned} f^*\omega &= v^2 (2uv du + u^2 dv) \wedge du + u^2 (2uv du + u^2 dv) \wedge dv + u^4 v^2 du \wedge dv \\ &= (u^2v^2 + 2u^3v + u^4v^2) du \wedge dv. \end{aligned}$$

2.3 The Exterior Derivative of Multilinear k -Forms

Definition 2.13 (Exterior derivative on functions). *For a smooth function $g : \mathbb{R}^n \rightarrow \mathbb{R}$ (a 0-form), the exterior derivative is*

$$dg = \sum_{i=1}^n \frac{\partial g}{\partial x^i} dx^i \in \Omega^1(\mathbb{R}^n).$$

Definition 2.14 (Exterior derivative on k -forms). *Let $\omega \in \Omega^k(\mathbb{R}^n)$ be written*

$$\omega = \sum_I a_I dx^I.$$

Define

$$d\omega := \sum_I da_I \wedge dx^I \in \Omega^{k+1}(\mathbb{R}^n).$$

This yields a map $d : \Omega^k(\mathbb{R}^n) \rightarrow \Omega^{k+1}(\mathbb{R}^n)$ for all $k \geq 0$.

Example 2.15. Let $\varphi = xyz dx + yz dy + (x + z) dz \in \Omega^1(\mathbb{R}^3)$. Then

$$\begin{aligned} d\varphi &= d(xyz) \wedge dx + d(yz) \wedge dy + d(x + z) \wedge dz \\ &= (yz dx + xz dy + xy dz) \wedge dx + (z dy + y dz) \wedge dy + (dx + dz) \wedge dz \\ &= -xz dy \wedge dx - xy dx \wedge dz - y dy \wedge dz + dx \wedge dz \\ &= xz dx \wedge dy + (1 - xy) dx \wedge dz - y dy \wedge dz. \end{aligned}$$

Proposition 2.16 ($d^2 = 0$). *For all k and all $\omega \in \Omega^k(\mathbb{R}^n)$ we have $d(d\omega) = 0$.*

Proof. Write $\omega = \sum_I a_I dx^I$. Then $d\omega = \sum_I da_I \wedge dx^I$. Applying d again yields $d^2\omega = \sum_I d(da_I) \wedge dx^I$ since dx^I is constant. But $d(da_I) = 0$ for each coefficient function a_I because mixed partial derivatives commute. $\square$

Remark 2.17 (The de Rham complex). The maps $d : \Omega^k(\mathbb{R}^n) \rightarrow \Omega^{k+1}(\mathbb{R}^n)$ form a cochain complex

$$0 \longrightarrow \Omega^0(\mathbb{R}^n) \xrightarrow{d} \Omega^1(\mathbb{R}^n) \xrightarrow{d} \dots \xrightarrow{d} \Omega^n(\mathbb{R}^n) \longrightarrow 0,$$

because $d^2 = 0$.

2.4 Divergence, Gradient, Curl/Rotational, the Laplacian, and the Hodge $\star$ -Operator

2.4.1 The Hodge $\star$ -Operator

Definition 2.18 (Hodge star on $\mathbb{R}^n$). *Fix the standard Euclidean inner product and orientation on $\mathbb{R}^n$. For a basis k -form $dx^{i_1} \wedge \cdots \wedge dx^{i_k}$, define the $(n-k)$ -form $\star(dx^{i_1} \wedge \cdots \wedge dx^{i_k})$ to be the unique basis $(n-k)$ -form $\pm dx^{j_1} \wedge \cdots \wedge dx^{j_{n-k}}$ such that*

$$(dx^{i_1} \wedge \cdots \wedge dx^{i_k}) \wedge \star(dx^{i_1} \wedge \cdots \wedge dx^{i_k}) = dx^1 \wedge \cdots \wedge dx^n.$$

Equivalently,

$$\star(dx^{i_1} \wedge \cdots \wedge dx^{i_k}) = \text{sgn}(\sigma) dx^{j_1} \wedge \cdots \wedge dx^{j_{n-k}},$$

where σ is the permutation sending $(1, \dots, n)$ to $(i_1, \dots, i_k, j_1, \dots, j_{n-k})$. Extend linearly to all k -forms.

Remark 2.19. The Hodge star gives an isomorphism $\Omega^k(\mathbb{R}^n) \cong \Omega^{n-k}(\mathbb{R}^n)$, and $\dim \bigwedge^k(\mathbb{R}^n)^* = \binom{n}{k} = \binom{n}{n-k}$ reflects this.

Example 2.20. Let $\omega = a_{12} dx^1 \wedge dx^2 + a_{13} dx^1 \wedge dx^3 + a_{23} dx^2 \wedge dx^3 \in \Omega^2(\mathbb{R}^3)$. Then

$$\star\omega = a_{12} dx^3 - a_{13} dx^2 + a_{23} dx^1.$$

Similarly, if $\omega = a_1 dx^1 + a_2 dx^2 \in \Omega^1(\mathbb{R}^2)$ then

$$\star\omega = a_1 dx^2 - a_2 dx^1.$$

2.4.2 The Divergence Operator

Definition 2.21 (Divergence). *A smooth vector field $X : \mathbb{R}^n \rightarrow \mathbb{R}^n$ may be written $X = \sum_{i=1}^n X^i \frac{\partial}{\partial x^i}$. Its divergence is the function*

$$\text{div}(X) = \sum_{i=1}^n \frac{\partial X^i}{\partial x^i}.$$

Equivalently, $\text{div}(X)(p) = \text{Tr}((dX)_p)$.

Definition 2.22 (Volume form). *The standard volume form on $\mathbb{R}^n$ is the n -form*

$$\nu := dx^1 \wedge \cdots \wedge dx^n.$$

It satisfies $\nu(e_1, \dots, e_n) = 1$ and $\nu(v_1, \dots, v_n) = \det(a_{ij})$ when $v_i = \sum_j a_{ij} e_j$.

Proposition 2.23 (Divergence via d and $\star$). *Let X be a smooth vector field on $\mathbb{R}^n$ and let ω be the 1-form metrically dual to X , i.e. $\omega(\cdot) = \langle \cdot, X \rangle$. Then*

$$d(\star\omega) = (\text{div } X) \nu.$$

Proof. Write $\omega = \sum_i X^i dx^i$. Then $\star\omega = \sum_i (-1)^{i-1} X^i dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n$. Taking d and using that only the dx^i -term in dX^i survives when wedged against the omitted basis form yields $d(\star\omega) = (\sum_i \partial_i X^i) \nu$. $\square$

2.4.3 The Gradient

Definition 2.24 (Gradient). *For a smooth function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the gradient is the vector field $\text{grad}(f)$ uniquely characterized by*

$$\langle \text{grad}(f), \cdot \rangle = df.$$

In coordinates,

$$\text{grad}(f) = \sum_{i=1}^n \frac{\partial f}{\partial x^i} \frac{\partial}{\partial x^i}.$$

2.4.4 The Laplacian

Definition 2.25 (Laplacian). *For a smooth function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the Laplacian is*

$$\Delta f := \text{div}(\text{grad}(f)) = \sum_{i=1}^n \frac{\partial^2 f}{\partial (x^i)^2}.$$

In terms of d and $\star$ (with the Euclidean metric),

$$\Delta f = \star d \star df.$$

2.5 Identifying d with ∇ , div , and curl (in $\mathbb{R}^3$)

The exterior derivative generalizes to all dimensions, but in $\mathbb{R}^3$ one can match degrees with classical vector calculus.

Remark 2.26 (0-forms and 1-forms). For a function f ,

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i \quad \longleftrightarrow \quad \text{grad}(f) = \begin{pmatrix} \partial_1 f \\ \vdots \\ \partial_n f \end{pmatrix}.$$

Likewise, a 1-form $\sum_i a_i dx^i$ corresponds (via the Euclidean metric) to the vector field $(a_1, \dots, a_n)$.

Remark 2.27 (Divergence from $(n-1)$ -forms). Using $\star$, $(n-1)$ -forms correspond to vector fields and d corresponds to divergence as in Proposition 2.23.

2.5.1 The Rotational

Definition 2.28 (Rotational). *Let X be a smooth vector field on $\mathbb{R}^n$ and let ω be its metrically dual 1-form. Define the rotational of X to be the $(n-2)$ -form*

$$\text{rot}(X) := \star(d\omega).$$

Remark 2.29 (Relationship to curl in $\mathbb{R}^3$). When $n = 3$, $\text{rot}(X)$ is a 1-form; identifying 1-forms with vector fields gives the usual curl:

$$\text{rot}(X) \longleftrightarrow \text{curl}(X).$$

Proposition 2.30 (Vector calculus identities). *For any smooth function f and vector field X on $\mathbb{R}^n$,*

$$\text{rot}(\text{grad } f) = 0, \quad \text{div}(\text{rot } X) = 0.$$

In $\mathbb{R}^3$ these become $\text{curl}(\nabla f) = 0$ and $\text{div}(\text{curl } X) = 0$.

Proof. Both follow immediately from $d^2 = 0$ and the definitions: $\text{rot}(\text{grad } f) = \star(d(df)) = 0$ and $\text{div}(\text{rot } X) \nu = d(\star(\star d\omega)) = \pm d(d\omega) = 0$. $\square$

3 Differentiable Manifolds

We extend differential forms from $\mathbb{R}^n$ to smooth manifolds.

3.1 Topological Manifolds and Differential Structures

Definition 3.1 (Topological manifolds). *A topological manifold M^n is a Hausdorff, second-countable topological space that is locally Euclidean of dimension n : every point $p \in M$ has an open neighborhood U_p homeomorphic to an open subset of $\mathbb{R}^n$. A homeomorphism $\varphi : U \rightarrow \varphi(U) \subseteq \mathbb{R}^n$ is called a chart and U a coordinate neighborhood.*

Definition 3.2 (Smooth compatibility and transition maps). *Charts $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) are C^∞ -compatible if the transition maps*

$$\varphi_\alpha \circ \varphi_\beta^{-1} : \varphi_\beta(U_\alpha \cap U_\beta) \rightarrow \varphi_\alpha(U_\alpha \cap U_\beta), \quad \varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \rightarrow \varphi_\beta(U_\alpha \cap U_\beta)$$

are smooth.

Definition 3.3 (Atlas and smooth manifold). *A (smooth) atlas is a collection of pairwise compatible charts that cover M . A smooth manifold is a topological manifold equipped with a maximal smooth atlas.*

Example 3.4 (An atlas on S^1). Let $S^1 = \{e^{it} \in \mathbb{C} : 0 \leq t \leq 2\pi\}$. Set

$$U_1 = \{e^{it} : -\pi < t < \pi\}, \quad U_2 = \{e^{it} : 0 < t < 2\pi\},$$

and define $\varphi_1(e^{it}) = t$ on U_1 and $\varphi_2(e^{it}) = t$ on U_2 . On overlaps, the transition functions are smooth, giving an atlas on S^1 .

Example 3.5 (Real projective space). Real projective space $\mathbb{RP}^n$ is the set of lines through the origin in $\mathbb{R}^{n+1}$:

$$\mathbb{RP}^n = (\mathbb{R}^{n+1} \setminus \{0\}) / \sim, \quad x \sim \lambda x, \quad \lambda \in \mathbb{R}^\times.$$

Write equivalence classes as homogeneous coordinates $[x_0 : \cdots : x_n]$. For each i let

$$U_i = \{[x_0 : \cdots : x_n] \in \mathbb{RP}^n : x_i \neq 0\},$$

and define the chart

$$\varphi_i([x_0 : \cdots : x_n]) = \left(\frac{x_0}{x_i}, \dots, \frac{\widehat{x_i}}{x_i}, \dots, \frac{x_n}{x_i} \right) \in \mathbb{R}^n.$$

Transition maps are rational (hence smooth) on overlaps, so these charts define a smooth structure.

Theorem 3.6 (Product manifolds). *If M^m and N^n are smooth manifolds, then $M \times N$ is a smooth manifold of dimension $m + n$.*

Proof. Second countability and Hausdorffness are preserved by products of such spaces. Given charts $(U_\alpha, \varphi_\alpha)$ on M and (V_β, ψ_β) on N , the product chart

$$(U_\alpha \times V_\beta, \varphi_\alpha \times \psi_\beta) : U_\alpha \times V_\beta \rightarrow \mathbb{R}^m \times \mathbb{R}^n \cong \mathbb{R}^{m+n}$$

gives an atlas whose transition maps are products of smooth maps, hence smooth. $\square$

3.2 Smooth Maps Between Manifolds

Definition 3.7 (Smooth maps and diffeomorphisms). *Let M^m and N^n be smooth manifolds. A map $f : M \rightarrow N$ is smooth at $p \in M$ if for charts (U, φ) around p and (V, ψ) around $f(p)$ the coordinate expression $\psi \circ f \circ \varphi^{-1}$ is smooth near $\varphi(p)$. A diffeomorphism is a smooth bijection whose inverse is smooth.*

3.3 Differential Forms on Manifolds

Definition 3.8 (Tangent space). *Let M be a smooth manifold and $p \in M$. A tangent vector at p can be defined as the velocity of a smooth curve $\alpha : (-\epsilon, \epsilon) \rightarrow M$ with $\alpha(0) = p$:*

$$\alpha'(0)[\phi] := \left. \frac{d}{dt} \right|_{t=0} (\phi \circ \alpha)(t), \quad \phi \in C^\infty(M).$$

The set of such derivations forms a vector space denoted $T_p M$.

Definition 3.9 (Tangent bundle). *The tangent bundle of M is*

$$TM := \bigsqcup_{p \in M} T_p M = \{(p, v) : p \in M, v \in T_p M\}.$$

*Similarly, the cotangent bundle is $T^*M = \bigsqcup_{p \in M} T_p^* M$.*

Definition 3.10 (Differential forms on a manifold). *An exterior k -form on M is a smooth assignment $\omega(p) \in \bigwedge^k T_p^* M$. The space of smooth k -forms is denoted $\Omega^k(M)$.*

Remark 3.11 (Local expressions). *In a chart (U, x) with coordinates $(x^1, \dots, x^n)$, a k -form has the local expression*

$$\omega|_U = \sum_{i_1 < \dots < i_k} a_{i_1 \dots i_k}(x) dx^{i_1} \wedge \dots \wedge dx^{i_k}.$$

Smoothness means the coefficient functions are smooth in every chart and the expressions transform by pullback under transition functions.

Definition 3.12 (Exterior derivative on manifolds). *The exterior derivative $d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$ is defined by requiring:*

1. *On functions it agrees with the differential.*
2. *It satisfies the graded Leibniz rule $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^{\deg \alpha} \alpha \wedge d\beta$.*
3. *It is local and agrees with the Euclidean definition in charts.*

It satisfies $d^2 = 0$.

3.4 Orientation of Manifolds

Definition 3.13 (Orientation of a vector space). *An orientation on an n -dimensional real vector space V is a choice of one of the two equivalence classes of ordered bases, where two bases are equivalent if the change-of-basis matrix has positive determinant.*

Remark 3.14 (Orientation via top forms). *If $\beta \in \bigwedge^n V^*$ is nonzero, then β determines an orientation by declaring a basis $(v_1, \dots, v_n)$ to be positive if $\beta(v_1, \dots, v_n) > 0$. Two nonzero n -forms determine the same orientation iff they differ by a positive scalar.*

Example 3.15. Let $\left(\frac{\partial}{\partial x}\Big|_p, \frac{\partial}{\partial y}\Big|_p\right)$ be the standard basis of $T_p\mathbb{R}^2$. Then $(dx \wedge dy)_p$ evaluates to 1 on this ordered basis, so it gives the usual (counterclockwise) orientation.

Theorem 3.16 (Orientability criterion). *An n -manifold M is orientable if and only if there exists a smooth nowhere-vanishing n -form $\omega \in \Omega^n(M)$.*

Definition 3.17 (Orientable atlas). *An n -manifold M is orientable if it admits an atlas $\{(U_\alpha, \varphi_\alpha)\}$ such that for all overlapping charts the Jacobian determinant of the transition map is positive:*

$$\det(d(\varphi_\beta \circ \varphi_\alpha^{-1})) > 0.$$

Such an atlas is called positively oriented.

Example 3.18 (Orientation of S^n via stereographic projection). Let $S^n = \{x \in \mathbb{R}^{n+1} : \|x\| = 1\}$. Stereographic projection from the north and south poles gives charts $h : S^n \setminus \{N\} \rightarrow \mathbb{R}^n$ and $h' : S^n \setminus \{S\} \rightarrow \mathbb{R}^n$. On the overlap, the transition map is

$$h' \circ h^{-1}(y) = \frac{1}{\|y\|^2} y,$$

whose Jacobian determinant is nonzero on $\mathbb{R}^n \setminus \{0\}$. Since the overlap is connected, we may choose a consistent sign and thereby orient S^n .

4 Integrating Differential Forms on Manifolds

Differential forms provide a coordinate-free framework for integration. We first treat line integrals of 1-forms, then integration of top-degree forms on oriented manifolds, and finally Stokes' theorem and the Poincaré lemma.

4.1 Line Integrals

Definition 4.1 (Pullback of a 1-form along a curve). *Let $\omega \in \Omega^1(U)$ for an open set $U \subseteq \mathbb{R}^n$, and let $c : [a, b] \rightarrow U$ be a piecewise smooth curve. The pullback $c^*\omega$ is a 1-form on $[a, b] \subseteq \mathbb{R}$, hence of the form $f(t) dt$.*

Definition 4.2 (Line integral of a 1-form). *With notation as above, define the line integral of ω along c by*

$$\int_c \omega := \int_a^b c^*\omega.$$

Equivalently, if $\omega = \sum_{i=1}^n a_i(x) dx^i$ and $c(t) = (c^1(t), \dots, c^n(t))$, then

$$\int_c \omega = \int_a^b \sum_{i=1}^n a_i(c(t)) \frac{dc^i}{dt}(t) dt.$$

Example 4.3 (Fundamental theorem for exact forms). If $\omega = df$ is exact and $c : [a, b] \rightarrow U$ is a piecewise smooth curve, then

$$\int_c df = f(c(b)) - f(c(a)).$$

Proof. By the chain rule, $c^*(df) = d(f \circ c) = (f \circ c)'(t) dt$ and integrate. □

4.2 Integration of Differential k -Forms

Integration over parameterized domains

Definition 4.4 (Integral over a parameterization). *Let $U \subseteq \mathbb{R}^k$ be open and let $\varphi : U \rightarrow \mathbb{R}^n$ be smooth and one-to-one onto its image. Let $\omega \in \Omega^k(\mathbb{R}^n)$ have compact support in $\varphi(U)$. Define the integral of ω over the parameterized k -dimensional domain $\varphi(U)$ by*

$$\int_{\varphi(U)} \omega := \int_U \varphi^* \omega.$$

Remark 4.5 (Change of variables). If $\psi : U' \rightarrow U$ is an orientation-preserving diffeomorphism, then

$$\int_U \varphi^* \omega = \int_{U'} (\varphi \circ \psi)^* \omega,$$

so the definition is independent of the choice of oriented coordinates.

Integration of top forms on oriented manifolds

Definition 4.6 (Integration on oriented manifolds). *Let M^n be an oriented smooth manifold and let $\omega \in \Omega^n(M)$ have compact support. Choose a partition of unity $\{\rho_\alpha\}$ subordinate to an oriented atlas $\{(U_\alpha, \varphi_\alpha)\}$. Define*

$$\int_M \omega := \sum_\alpha \int_{\varphi_\alpha(U_\alpha)} (\varphi_\alpha^{-1})^* (\rho_\alpha \omega),$$

where each term is an integral of an n -form on $\mathbb{R}^n$ (hence a usual multiple integral).

Remark 4.7. One checks that the value is independent of the partition of unity and the choice of oriented atlas. This construction is the standard way to define integration of top-degree forms on manifolds.

4.3 Stokes' Theorem

Definition 4.8 (Manifolds with boundary). *A smooth n -manifold with boundary is a space locally modeled on the half-space $\mathbb{R}_{\geq 0}^n = \{(x^1, \dots, x^n) \in \mathbb{R}^n : x^n \geq 0\}$ with smooth transition functions. Its boundary ∂M is the set of points mapping to $x^n = 0$ in boundary charts.*

Definition 4.9 (Boundary orientation). *If M^n is oriented, then ∂M inherits an orientation characterized by: a basis $(v_1, \dots, v_{n-1})$ of $T_p(\partial M)$ is positive if $(\nu, v_1, \dots, v_{n-1})$ is a positive basis of $T_p M$, where ν is an outward-pointing normal vector.*

Theorem 4.10 (Stokes' theorem). *Let M be an oriented smooth n -manifold with (possibly empty) boundary ∂M . If $\omega \in \Omega^{n-1}(M)$ has compact support, then*

$$\int_M d\omega = \int_{\partial M} \omega.$$

Remark 4.11 (Classical theorems as special cases). In $\mathbb{R}^3$, Stokes' theorem contains the classical Green, Gauss, and Stokes theorems of vector calculus once one identifies vector fields with differential forms via the Euclidean metric and $\star$.

4.4 Poincaré's Lemma

Theorem 4.12 (Poincaré lemma). *Let $U \subseteq \mathbb{R}^n$ be a star-shaped open set (in particular, contractible). Then every closed k -form on U is exact for $k \geq 1$:*

$$d\omega = 0 \implies \exists \eta \in \Omega^{k-1}(U) \text{ with } \omega = d\eta.$$

Hence $H_{\text{dR}}^k(U) = 0$ for $k \geq 1$ and $H_{\text{dR}}^0(U) \cong \mathbb{R}$.

Idea of proof. Let $H_t : U \rightarrow U$ be the straight-line homotopy $H_t(x) = tx$ to the origin (after translating so the star center is at 0). Define the homotopy operator

$$K(\omega) := \int_0^1 H_t^*(\iota_{\partial_t} \omega) dt,$$

where ι denotes interior product. One shows $dK + Kd = \text{id} - H_0^*$. If $d\omega = 0$ and $k \geq 1$, then $H_0^* \omega = 0$ (because it pulls back to a form at a point), so $\omega = d(K\omega)$. $\square$

5 De Rham Cohomology

Definition 5.1 (De Rham complex). *Let M^n be a smooth manifold. The sequence*

$$0 \longrightarrow \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \dots \xrightarrow{d} \Omega^{n-1}(M) \xrightarrow{d} \Omega^n(M) \longrightarrow 0$$

is called the de Rham complex.

Definition 5.2 (Closed and exact forms). *A k -form $\omega \in \Omega^k(M)$ is closed if $d\omega = 0$ and exact if $\omega = d\eta$ for some $\eta \in \Omega^{k-1}(M)$. Let $Z^k(M)$ denote closed forms and $B^k(M)$ exact forms; then $B^k(M) \subseteq Z^k(M) \subseteq \Omega^k(M)$.*

Definition 5.3 (De Rham cohomology). *The k th de Rham cohomology group is the quotient*

$$H_{\text{dR}}^k(M) := Z^k(M) / B^k(M).$$

Two closed forms are cohomologous if they differ by an exact form.

Definition 5.4 (Euler characteristic). *If $H_{\text{dR}}^k(M)$ are finite-dimensional (e.g. if M is compact), the Euler characteristic is*

$$\chi(M) := \sum_{k=0}^n (-1)^k \dim H_{\text{dR}}^k(M).$$

Theorem 5.5 (Poincaré duality (statement)). *If M is a compact oriented n -manifold, the pairing*

$$H_{\text{dR}}^k(M) \times H_{\text{dR}}^{n-k}(M) \longrightarrow \mathbb{R}, \quad ([\omega], [\mu]) \longmapsto \int_M \omega \wedge \mu$$

is nondegenerate (a perfect pairing). In particular $\dim H_{\text{dR}}^k(M) = \dim H_{\text{dR}}^{n-k}(M)$ and $\chi(M) = 0$ for odd n .

Additional examples and computations

Example 5.6 (De Rham cohomology of $\mathbb{R}^n$). By the Poincaré lemma, $H_{\text{dR}}^0(\mathbb{R}^n) \cong \mathbb{R}$ and $H_{\text{dR}}^k(\mathbb{R}^n) = 0$ for $k \geq 1$.

Example 5.7 (The circle). Let S^1 be the unit circle. One may show

$$H_{\text{dR}}^0(S^1) \cong \mathbb{R}, \quad H_{\text{dR}}^1(S^1) \cong \mathbb{R}, \quad H_{\text{dR}}^k(S^1) = 0 \quad (k \geq 2).$$

A generator of $H_{\text{dR}}^1(S^1)$ is represented by the angular form $d\theta$ on S^1 .

Remark 5.8. The computation of $H_{\text{dR}}^*(S^1)$ can be done using a cover by two arcs and the Mayer–Vietoris sequence; see Section 8.8 for the homological algebra behind this.

6 Worked Examples in de Rham Cohomology

The abstract definitions of $Z^k(M)$, $B^k(M)$, and $H_{\text{dR}}^k(M)$ become much more concrete once we compute a few examples carefully. The goal of this section is twofold:

- (1) to practice the basic tools (Poincaré lemma, Mayer–Vietoris, and partitions of unity),
- (2) and to build geometric intuition for what cohomology classes of forms “measure” (winding number, flux, volume, etc.).

6.1 The Poincaré lemma in $\mathbb{R}^n$

We stated earlier that “closed implies locally exact.” We now prove this on star-shaped domains.

Definition 6.1 (Star-shaped domain). *An open set $U \subseteq \mathbb{R}^n$ is star-shaped (with respect to 0) if for every $x \in U$ the line segment $t \mapsto tx$ lies in U for all $t \in [0, 1]$.*

Theorem 6.2 (Poincaré lemma on star-shaped sets). *If $U \subseteq \mathbb{R}^n$ is star-shaped, then every closed k -form on U is exact for $k \geq 1$:*

$$d\omega = 0 \implies \exists \eta \in \Omega^{k-1}(U) \text{ with } \omega = d\eta.$$

Proof. Let $R = \sum_{i=1}^n x_i \frac{\partial}{\partial x_i}$ be the radial vector field on U . Define a homotopy $H : [0, 1] \times U \rightarrow U$ by $H(t, x) = tx$. For a form $\omega \in \Omega^k(U)$ define the operator

$$K\omega := \int_0^1 \iota_R(H_t^* \omega) \frac{dt}{t}, \quad \text{where } H_t(x) = H(t, x) = tx.$$

A standard computation using Cartan’s formula and differentiation under the integral sign shows

$$d(K\omega) + K(d\omega) = \omega - \omega_0,$$

where ω_0 is the constant form obtained by evaluating ω at 0 (and pulling back by the inclusion of the point). If $k \geq 1$, then $\omega_0 = 0$, hence $d(K\omega) + K(d\omega) = \omega$. If $d\omega = 0$, this gives $\omega = d(K\omega)$. $\square$

Remark 6.3. The proof exhibits an explicit *chain homotopy* between the identity and the “evaluation at 0” map. This is an early appearance of the “homotopy operator” idea that returns in both de Rham theory and homological algebra.

6.2 Mayer–Vietoris for de Rham cohomology

We already stated Mayer–Vietoris informally; we now spell it out and explain the connecting homomorphism.

Theorem 6.4 (Mayer–Vietoris for de Rham). *Let $M = U \cup V$ be an open cover. Then there is a natural long exact sequence*

$$\cdots \rightarrow H_{\text{dR}}^{k-1}(U \cap V) \xrightarrow{\delta} H_{\text{dR}}^k(M) \xrightarrow{(r_U, r_V)} H_{\text{dR}}^k(U) \oplus H_{\text{dR}}^k(V) \xrightarrow{r_U - r_V} H_{\text{dR}}^k(U \cap V) \xrightarrow{\delta} \cdots$$

where r_U, r_V denote restriction maps.

Construction of δ . Choose a partition of unity $\{\rho_U, \rho_V\}$ subordinate to $\{U, V\}$, so $\rho_U + \rho_V = 1$. Given a closed form $\alpha \in \Omega^{k-1}(U \cap V)$, extend it to U and V by multiplying with cutoffs: define $\beta_U = \rho_V \alpha$ (viewed as a form on U) and $\beta_V = -\rho_U \alpha$ (a form on V). Then $\beta_U + \beta_V = 0$ on $U \cap V$, so β_U and β_V glue to a global $(k-1)$ -form β on M . Define

$$\delta([\alpha]) := [d\beta] \in H_{\text{dR}}^k(M).$$

A check shows this is well-defined (independent of choices and respects cohomology classes), and exactness follows from a standard diagram chase. $\square$

6.3 Computation of $H_{\text{dR}}^*(S^1)$

We compute the de Rham cohomology of the circle in a way that makes the generator explicit.

Proposition 6.5.

$$H_{\text{dR}}^k(S^1) \cong \begin{cases} \mathbb{R}, & k = 0, \\ \mathbb{R}, & k = 1, \\ 0, & k \geq 2. \end{cases}$$

Proof. Connectedness gives $H_{\text{dR}}^0(S^1) \cong \mathbb{R}$. Let θ be the angular coordinate on S^1 (defined locally). The 1-form $d\theta$ is closed. It is not exact: if $d\theta = df$, then integrating around the circle gives

$$\int_{S^1} d\theta = 2\pi \neq 0 = \int_{S^1} df$$

by Stokes' theorem. Hence $[d\theta] \neq 0$. To show H^1 is one-dimensional, cover S^1 by two arcs U and V with contractible intersection $U \cap V$. Mayer–Vietoris for the cover shows that $H^1(S^1)$ fits into an exact sequence

$$0 \rightarrow H^0(S^1) \rightarrow H^0(U) \oplus H^0(V) \rightarrow H^0(U \cap V) \xrightarrow{\delta} H^1(S^1) \rightarrow 0.$$

All H^0 groups here are $\mathbb{R}$ and the map $\mathbb{R} \oplus \mathbb{R} \rightarrow \mathbb{R}$ is subtraction, so δ is an isomorphism. Thus $H^1(S^1) \cong \mathbb{R}$. $\square$

Remark 6.6 (Winding number). If $\gamma : S^1 \rightarrow S^1$ is smooth, then $\deg(\gamma) = \frac{1}{2\pi} \int_{S^1} \gamma^*(d\theta)$. Thus the generator $[d\theta]$ detects winding.

6.4 Computation of $H_{\text{dR}}^*(S^n)$

We now compute the cohomology of spheres in full detail.

Proposition 6.7. *For $n \geq 1$,*

$$H_{\text{dR}}^k(S^n) \cong \begin{cases} \mathbb{R}, & k = 0 \text{ or } k = n, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. We argue by induction on n . The case $n = 1$ is Proposition 6.5. For $n \geq 2$, cover S^n by open neighborhoods U and V of the north and south hemispheres. Then U and V are contractible and $U \cap V$ deformation retracts to S^{n-1} . By induction, $H^k(U \cap V) \cong H^k(S^{n-1})$ is $\mathbb{R}$ in degrees 0 and $n-1$ and 0 otherwise.

Mayer–Vietoris yields a long exact sequence. In degrees $1 \leq k \leq n-2$, we have $H^k(U) = H^k(V) = 0$ and $H^{k-1}(U \cap V) = 0$, so exactness forces $H^k(S^n) = 0$.

In degree $k = n$, exactness gives

$$0 = H^{n-1}(U) \oplus H^{n-1}(V) \rightarrow H^{n-1}(U \cap V) \xrightarrow{\delta} H^n(S^n) \rightarrow H^n(U) \oplus H^n(V) = 0,$$

so $\delta : H^{n-1}(U \cap V) \rightarrow H^n(S^n)$ is an isomorphism. Since $H^{n-1}(U \cap V) \cong H^{n-1}(S^{n-1}) \cong \mathbb{R}$, we get $H^n(S^n) \cong \mathbb{R}$.

Finally $H^0(S^n) \cong \mathbb{R}$ because S^n is connected. $\square$

Remark 6.8 (A generator of $H^n(S^n)$). Let ν be the standard volume form on S^n associated to the induced orientation and Riemannian metric. Then $[\nu]$ generates $H_{\text{dR}}^n(S^n)$, and $\int_{S^n} \nu = \text{Vol}(S^n) \neq 0$ shows ν is not exact.

6.5 The torus and products

We now compute the cohomology of T^n by an explicit differential-forms argument (compatible with Künneth).

Proposition 6.9. *Let $T^n = \mathbb{R}^n / \mathbb{Z}^n$ with coordinates $\theta^1, \dots, \theta^n \pmod{2\pi}$. Then $H_{\text{dR}}^*(T^n)$ is the exterior algebra on the classes of $d\theta^1, \dots, d\theta^n$:*

$$H_{\text{dR}}^*(T^n) \cong \Lambda([d\theta^1], \dots, [d\theta^n]).$$

Sketch. The forms $d\theta^i$ descend from constant 1-forms on $\mathbb{R}^n$ and are closed. They are not exact on T^n because integrating $d\theta^i$ over the i th coordinate circle gives 2π . Wedge products of these classes are likewise detected by integration over the corresponding subtori. A dimension-counting argument using Künneth (or Fourier series) shows these generate all classes. $\square$

6.6 A first look at surfaces

For a closed connected oriented surface Σ_g of genus g ,

$$\dim H_{\text{dR}}^0(\Sigma_g) = 1, \quad \dim H_{\text{dR}}^1(\Sigma_g) = 2g, \quad \dim H_{\text{dR}}^2(\Sigma_g) = 1,$$

so $\chi(\Sigma_g) = 2 - 2g$. One can prove this using a cellular decomposition and the de Rham theorem, or by Mayer–Vietoris. In this course we often take the surface computation as motivation: H^1 measures the number of independent “handles”.

6.7 Exercises

Exercise 6.10. Let $\omega = \frac{-y dx + x dy}{x^2 + y^2}$ on $\mathbb{R}^2 \setminus \{0\}$. Show $d\omega = 0$ but ω is not exact. Interpret $\int_{S^1} \omega$.

Exercise 6.11. Show directly that $H_{\text{dR}}^1(\mathbb{R}^2 \setminus \{0\}) \cong \mathbb{R}$ by exhibiting an explicit generator and using Stokes' theorem.

Exercise 6.12. Compute $H_{\text{dR}}^*(S^1 \times S^1)$ directly using the forms $d\theta^1, d\theta^2$ and integration over cycles.

7 Exercises

This section contains the original exercises plus additional ones to reinforce the main concepts.

7.1 Exact Sequences

Exercise 7.1. Consider the sequence of linear maps

$$0 \longrightarrow \mathbb{R}^2 \xrightarrow{A} \mathbb{R}^3 \xrightarrow{B} \mathbb{R}^3 \xrightarrow{C} \mathbb{R}^2 \longrightarrow 0,$$

where

$$A = \begin{pmatrix} 2 & 3 \\ 0 & 1 \\ 2 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -3 & -1 \\ 1 & -3 & -1 \\ 1 & -3 & -1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Is the sequence exact? If not, where does exactness fail? Could such a sequence occur as the (co)homology groups obtained from the de Rham complex of some oriented manifold M ? Why or why not?

7.2 The Exterior Derivative

Exercise 7.2. Take the exterior derivative of the following form:

$$\omega = a_1 dx \in \Omega^1(\mathbb{R}^4),$$

where a_1 is a smooth function of (x, y, z, w) . Would you get a different answer if we considered ω as an element of $\Omega^1(\mathbb{R}^5)$? Explain.

Additional exercises

Exercise 7.3. Show that for a smooth map $f : M \rightarrow N$ between manifolds, f^* commutes with d :

$$f^*(d\omega) = d(f^*\omega).$$

Exercise 7.4. Compute $H_{\text{dR}}^*(S^n)$ using Mayer-Vietoris and a cover by two hemispheres.

8 Homology Functors

The last part of the original notes introduces basic homological algebra tools used for diagram chasing and for constructing long exact sequences. This expanded version fills in missing definitions and standard proofs.

8.1 Exact sequences, kernels, cokernels

We begin with a few warm-up facts that are repeatedly used in diagram chases.

Definition 8.1 (Kernel, cokernel). *Let $f : M \rightarrow N$ be a morphism in an abelian category (or a linear map of vector spaces). The kernel $\ker(f)$ is the universal object mapping into M whose composite with f is 0. The cokernel $\operatorname{coker}(f)$ is the universal object receiving a map from N that kills $\operatorname{im}(f)$. In $\mathbf{Ab}$ or $R\text{-Mod}$ one has $\operatorname{coker}(f) = N/\operatorname{im}(f)$.*

Lemma 8.2 (Surjectivity and cokernels). *A morphism $f : M \rightarrow N$ is surjective (an epimorphism) if and only if $\operatorname{coker}(f) = 0$.*

Proof. In modules (and hence in vector spaces), $\operatorname{coker}(f) = N/\operatorname{im}(f)$. If f is surjective then $\operatorname{im}(f) = N$, so $N/\operatorname{im}(f) = 0$. Conversely, if $\operatorname{coker}(f) = 0$ then $N/\operatorname{im}(f) = 0$, hence $\operatorname{im}(f) = N$ and f is surjective. The same argument holds abstractly in an abelian category where f is epi iff $\operatorname{coker}(f) = 0$ by definition. $\square$

Lemma 8.3 (The “standard” exact sequence). *For any morphism $f : M \rightarrow N$ there is a canonical exact sequence*

$$0 \longrightarrow \ker(f) \longrightarrow M \xrightarrow{f} N \longrightarrow \operatorname{coker}(f) \longrightarrow 0.$$

Proof. The map $0 \rightarrow \ker(f)$ is monic by definition. The inclusion $\ker(f) \rightarrow M$ has image $\ker(f)$, so exactness at M is precisely $\operatorname{im}(\ker(f) \rightarrow M) = \ker(f) = \ker(f : M \rightarrow N)$. By definition $\operatorname{coker}(f) = N/\operatorname{im}(f)$, so $N \rightarrow \operatorname{coker}(f)$ is epic with kernel $\operatorname{im}(f)$; exactness at N follows. Finally $\operatorname{coker}(f) \rightarrow 0$ is obviously exact. $\square$

Lemma 8.4. *The sequence $0 \rightarrow M \rightarrow 0$ is exact if and only if $M = 0$.*

Proof. Exactness at M forces $\operatorname{im}(0) = 0 = \ker(M \rightarrow 0) = M$, hence $M = 0$. The converse is immediate. $\square$

Proposition 8.5 (A small exactness test). *Assume*

$$A \xrightarrow{f'} B \xrightarrow{f} C \xrightarrow{f''} D$$

is exact. Then the following are equivalent:

- (a) f' is surjective;
- (b) $f = 0$ (the zero morphism);
- (c) f'' is injective.

Proof. If f' is surjective then $\operatorname{im}(f') = B = \ker(f)$ by exactness, so $\ker(f) = B$ and hence $f = 0$. If $f = 0$ then $\operatorname{im}(f) = 0$, hence $\ker(f'') = 0$ by exactness; thus f'' is injective. If f'' is injective then $\ker(f'') = 0 = \operatorname{im}(f)$, so $f = 0$; exactness then gives $\ker(f) = B = \operatorname{im}(f')$, so f' is surjective. $\square$

Proposition 8.6. *Assume*

$$A \xrightarrow{f'} B \xrightarrow{f} C \xrightarrow{f''} D \xrightarrow{f'''} E$$

is exact. If f' and f''' are isomorphisms, then $C = 0$.

Proof. If f' is an isomorphism then $\operatorname{im}(f') = B$, hence $\ker(f) = B$ and $f = 0$. If f''' is an isomorphism then $\ker(f''') = 0$, hence $\operatorname{im}(f'') = 0$ and $f'' = 0$. Thus the middle portion $B \xrightarrow{0} C \xrightarrow{0} D$ consists of zero maps. Exactness at C forces $\ker(0 : C \rightarrow D) = \operatorname{im}(0 : B \rightarrow C) = 0$, but $\ker(0) = C$, so $C = 0$. $\square$

8.2 A first diagram chase: the 3×3 lemma

The original notes include a detailed element chase in a commutative 3×3 diagram. We record a standard formulation (sometimes called the *Nine Lemma*).

Theorem 8.7 (Nine Lemma / 3×3 lemma). *Consider a commutative diagram in an abelian category with exact columns:*

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & A' & \longrightarrow & A & \longrightarrow & A'' \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & B' & \longrightarrow & B & \longrightarrow & B'' \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & C' & \longrightarrow & C & \longrightarrow & C'' \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & 0 & & 0 & & 0
 \end{array}$$

(1) *If the bottom two rows are exact, then the top row is exact.*

(2) *If the top two rows are exact, then the bottom row is exact.*

Proof. We sketch the chases in the spirit of the notes.

(1) Assume the bottom two rows are exact. First we show $A \rightarrow A''$ is surjective. Let $a'' \in A''$. Its image in B'' (under the vertical map) maps to 0 in C'' by exactness of the third column, hence lies in $\ker(B'' \rightarrow C'') = \text{im}(B \rightarrow B'')$. Choose $b \in B$ mapping to that element of B'' . Let $c \in C$ be the image of b . Commutativity implies the image of c in C'' is 0, so $c \in \ker(C \rightarrow C'') = \text{im}(C' \rightarrow C)$. Choose $c' \in C'$ mapping to c . By surjectivity of $B' \rightarrow C'$ there exists $b' \in B'$ mapping to c' . Let $\tilde{b} \in B$ be the image of b' . Then $b - \tilde{b}$ maps to 0 in C (since both map to c), so by exactness of the middle column $b - \tilde{b}$ lies in $\text{im}(A \rightarrow B)$. Choose $a \in A$ mapping to $b - \tilde{b}$. Tracing to A'' shows a maps to a'' (injectivity of $A'' \rightarrow B''$ is used here), proving surjectivity.

Next, injectivity of $A' \rightarrow A$ follows from injectivity of $A' \rightarrow B'$ and $B' \rightarrow B$ together with commutativity. Finally, exactness at A (i.e. $\text{im}(A' \rightarrow A) = \ker(A \rightarrow A'')$) is proved by a standard chase: if $a \in A$ maps to 0 in A'' then it maps to 0 in B'' ; exactness of the middle row gives that a comes from some $b' \in B'$, and then exactness of the left column identifies b' as coming from a unique $a' \in A'$.

(2) Assume the top two rows are exact. Surjectivity of $C \rightarrow C''$ is shown by lifting $c'' \in C''$ to some $b'' \in B''$, then to $b \in B$, and then mapping to $c \in C$. Injectivity of $C' \rightarrow C$ is a chase dual to the previous one: if $c' \in C'$ maps to 0 in C , lift c' to $b' \in B'$, compare its image in B , and use exactness of columns to show $c' = 0$. Exactness at C (image equals kernel) follows similarly: if $c \in C$ maps to 0 in C'' , lift c to $b \in B$ and then to $b' \in B'$, and use exactness of the upper rows to find a preimage $c' \in C'$. $\square$

8.3 More diagram chasing: the Four Lemma and the Five Lemma

The Nine Lemma is one of several “standard” diagram-chasing results. Two others that appear constantly in homological algebra and algebraic topology are the *Four Lemma* and the *Five*

Lemma. They are often used to show that a morphism between long exact sequences is injective/surjective/an isomorphism.

Theorem 8.8 (Four Lemma). *Consider a commutative diagram of abelian groups (or modules) with exact rows*

$$\begin{array}{ccccccc} A_1 & \longrightarrow & A_2 & \longrightarrow & A_3 & \longrightarrow & A_4 \\ \downarrow f_1 & & \downarrow f_2 & & \downarrow f_3 & & \downarrow f_4 \\ B_1 & \longrightarrow & B_2 & \longrightarrow & B_3 & \longrightarrow & B_4 \end{array}$$

(a) *If f_1 is surjective and f_2 is injective, then f_3 is injective.*

(b) *If f_3 is surjective and f_4 is injective, then f_2 is surjective.*

Proof of (a). Let $a_3 \in A_3$ with $f_3(a_3) = 0$. Choose $a_2 \in A_2$ mapping to a_3 by exactness. Then $f_2(a_2)$ maps to $f_3(a_3) = 0$ in B_3 , hence $f_2(a_2) \in \ker(B_2 \rightarrow B_3) = \text{im}(B_1 \rightarrow B_2)$. Choose $b_1 \in B_1$ with image $f_2(a_2)$. Since f_1 is surjective, pick $a_1 \in A_1$ with $f_1(a_1) = b_1$. Commutativity shows that $f_2(a_2 - \text{im}(a_1)) = 0$, and injectivity of f_2 implies a_2 lies in $\text{im}(A_1 \rightarrow A_2)$. Therefore $a_3 = 0$ in A_3 , proving f_3 is injective. The proof of (b) is similar (a dual chase). $\square$

Theorem 8.9 (Five Lemma). *Consider a commutative diagram with exact rows*

$$\begin{array}{ccccccccc} A_1 & \longrightarrow & A_2 & \longrightarrow & A_3 & \longrightarrow & A_4 & \longrightarrow & A_5 \\ \downarrow f_1 & & \downarrow f_2 & & \downarrow f_3 & & \downarrow f_4 & & \downarrow f_5 \\ B_1 & \longrightarrow & B_2 & \longrightarrow & B_3 & \longrightarrow & B_4 & \longrightarrow & B_5 \end{array}$$

If f_1 and f_5 are isomorphisms, f_2 is surjective, f_4 is injective, and f_2, f_4 are also isomorphisms (or at least f_2 injective and f_4 surjective), then f_3 is an isomorphism. A common useful special case: if f_1, f_2, f_4, f_5 are isomorphisms, then f_3 is an isomorphism.

Sketch. Injectivity of f_3 follows from the Four Lemma applied to the left part of the diagram; surjectivity follows from the Four Lemma applied to the right part. $\square$

8.4 The snake lemma: an element chase

We proved the snake lemma abstractly earlier, but it is instructive to see the explicit element chase (the style used in the original notes).

Consider a commutative diagram with exact rows:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A' & \xrightarrow{i} & A & \xrightarrow{p} & A'' & \longrightarrow & 0 \\ & & \downarrow f' & & \downarrow f & & \downarrow f'' & & \\ 0 & \longrightarrow & B' & \xrightarrow{j} & B & \xrightarrow{q} & B'' & \longrightarrow & 0 \\ & & \downarrow g' & & \downarrow g & & \downarrow g'' & & \\ 0 & \longrightarrow & C' & \xrightarrow{k} & C & \xrightarrow{r} & C'' & \longrightarrow & 0 \end{array}$$

The snake lemma produces a long exact sequence

$$0 \rightarrow \ker(f') \rightarrow \ker(f) \rightarrow \ker(f'') \xrightarrow{\delta} \text{coker}(f') \rightarrow \text{coker}(f) \rightarrow \text{coker}(f'') \rightarrow 0.$$

We now construct the connecting morphism δ concretely.

Construction 8.10 (Connecting morphism). Take $a'' \in \ker(f'') \subseteq A''$. Choose $a \in A$ with $p(a) = a''$ (possible because p is surjective). Then $0 = f''(a'') = f''(p(a)) = q(f(a))$ by commutativity, so $f(a) \in \ker(q) = \operatorname{im}(j)$. Choose $b' \in B'$ with $j(b') = f(a)$. Now consider $g'(b') \in C'$. One checks that $g'(b') \in \operatorname{coker}(f')$ depends only on the class of a'' . Define

$$\delta(a'') := [g'(b')] \in \operatorname{coker}(f') = C'/\operatorname{im}(f').$$

Why this lands in $\operatorname{coker}(f')$ and is well-defined. If we change the lift a by an element of $\operatorname{im}(i)$, say $a \mapsto a + i(a')$, then $f(a)$ changes by $j(f'(a'))$, so b' changes by $f'(a')$. Thus $g'(b')$ changes by $g'(f'(a')) = f'(?)$ (commutativity) and hence by an element of $\operatorname{im}(f')$, i.e. the class in $\operatorname{coker}(f')$ does not change. A similar check shows independence of the choice of b' , and additivity follows from choosing lifts for sums. Exactness of the resulting sequence is another (slightly longer) chase. $\square$

8.5 Projective Resolutions

Definition 8.11 (Chain complexes in an abelian category). Let $\mathcal{A}$ be an abelian category (e.g. $\mathbf{Ab}$ or $R\text{-Mod}$). A chain complex $(C_\bullet, d_\bullet)$ in $\mathcal{A}$ is a sequence

$$\cdots \xrightarrow{d_{n+2}} C_{n+1} \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \xrightarrow{d_{n-1}} \cdots$$

such that $d_n \circ d_{n+1} = 0$ for all n .

Definition 8.12 (Cycles, boundaries, homology). In a chain complex $(C_\bullet, d)$, define

$$Z_n(C) := \ker(d_n) \subseteq C_n, \quad B_n(C) := \operatorname{im}(d_{n+1}) \subseteq C_n.$$

Since $d_n d_{n+1} = 0$, we have $B_n(C) \subseteq Z_n(C)$. The n th homology object is

$$H_n(C) := Z_n(C)/B_n(C).$$

Definition 8.13 (Projective object). An object P in an abelian category $\mathcal{A}$ is projective if for every epimorphism $e : X \twoheadrightarrow Y$ and every morphism $f : P \rightarrow Y$, there exists a lift $\tilde{f} : P \rightarrow X$ with $e \circ \tilde{f} = f$.

Definition 8.14 (Projective resolution). A projective resolution of an object M in $\mathcal{A}$ is an exact sequence

$$\cdots \rightarrow P_2 \rightarrow P_1 \rightarrow P_0 \rightarrow M \rightarrow 0$$

where each P_i is projective.

Remark 8.15. Projective resolutions are used to define derived functors such as Ext and Tor . While we do not develop the full theory here, the diagram-chasing tools below are precisely those used to prove functorial properties and long exact sequences in homology.

8.6 The Snake Lemma

We now construct the connecting homomorphism and derive the long exact sequence in homology. The original notes present the construction in the case of a short exact sequence of chain complexes.

Lemma 8.16 (Connecting homomorphism). *Let*

$$0 \rightarrow C'_\bullet \xrightarrow{i} C_\bullet \xrightarrow{p} C''_\bullet \rightarrow 0$$

be a short exact sequence of chain complexes in an abelian category. Then there is a natural homomorphism (the connecting homomorphism)

$$\partial_n : H_n(C'') \rightarrow H_{n-1}(C')$$

defined as follows: for a cycle $z''_n \in Z_n(C'')$, choose $c_n \in C_n$ with $p(c_n) = z''_n$ and set

$$\partial_n([z''_n]) := [i^{-1}(d_n c_n)] \in H_{n-1}(C').$$

Proof. Well-definedness. If $p(c_n) = z''_n$ and $p(\hat{c}_n) = z''_n$, then $c_n - \hat{c}_n \in \ker(p) = \text{im}(i)$, so $c_n - \hat{c}_n = i(c'_n)$. Applying d and using commutativity gives $d(c_n - \hat{c}_n) = i(dc'_n)$. Thus $i^{-1}(dc_n) - i^{-1}(d\hat{c}_n) = dc'_n$ is a boundary in C' , so the homology class is independent of the chosen lift.

Cycle condition. Since z''_n is a cycle, $0 = dz''_n = d(p(c_n)) = p(dc_n)$, so $dc_n \in \ker(p) = \text{im}(i)$ and $i^{-1}(dc_n)$ is defined.

Homomorphism. Choosing lifts compatibly shows $\partial_n([z''_n + \hat{z}''_n]) = \partial_n([z''_n]) + \partial_n([\hat{z}''_n])$. $\square$

Theorem 8.17 (Long exact sequence in homology). *Given a short exact sequence of chain complexes*

$$0 \rightarrow C'_\bullet \xrightarrow{i} C_\bullet \xrightarrow{p} C''_\bullet \rightarrow 0,$$

there is a natural long exact sequence

$$\cdots \rightarrow H_n(C') \xrightarrow{i_*} H_n(C) \xrightarrow{p_*} H_n(C'') \xrightarrow{\partial_n} H_{n-1}(C') \rightarrow \cdots$$

Proof sketch. Exactness at $H_n(C)$ follows from exactness of $0 \rightarrow C'_n \rightarrow C_n \rightarrow C''_n \rightarrow 0$ and the fact that cycles/boundaries are functorial. Exactness at $H_n(C'')$ is the heart of the snake lemma construction: $\ker(\partial_n) = \text{im}(p_*)$ and $\text{im}(\partial_n) = \ker(i_*)$ are shown by chasing representatives and using exactness in each degree. $\square$

Lemma 8.18 (Snake lemma). *Let*

$$\begin{array}{ccccccc} 0 & \longrightarrow & A' & \longrightarrow & A & \longrightarrow & A'' \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & B' & \longrightarrow & B & \longrightarrow & B'' \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & C' & \longrightarrow & C & \longrightarrow & C'' \longrightarrow 0 \end{array}$$

be a commutative diagram with exact rows. Then there is a connecting morphism $\delta : \ker(B'' \rightarrow C'') \rightarrow \text{coker}(A' \rightarrow B')$ yielding an exact sequence

$$\ker(A \rightarrow B) \rightarrow \ker(A'' \rightarrow B'') \xrightarrow{\delta} \text{coker}(A' \rightarrow B') \rightarrow \text{coker}(A \rightarrow B) \rightarrow \text{coker}(A'' \rightarrow B'').$$

Remark 8.19. The long exact sequence in homology is a special case of the snake lemma applied to the diagram of cycles and boundaries in a short exact sequence of chain complexes.

8.7 Chain Complexes and Contracting Maps

Definition 8.20 (Degree p maps). *Let $C_\bullet$ and $D_\bullet$ be chain complexes. A degree p map $s : C_\bullet \rightarrow D_\bullet$ is a collection of maps $s_n : C_n \rightarrow D_{n+p}$.*

Definition 8.21 (Chain homotopy). *Two chain maps $f, g : (C, d) \rightarrow (C', d')$ are chain homotopic if there exists a degree $+1$ map $s : C \rightarrow C'$ such that*

$$f - g = d's + sd.$$

Definition 8.22 (Null-homotopic and contracting homotopy). *A chain map is null-homotopic if it is chain homotopic to 0. A chain complex (C, d) is contractible if id_C is null-homotopic, i.e. there exists s with*

$$\text{id}_C = ds + sd.$$

Such an s is a contracting homotopy.

Theorem 8.23 (Contractible complexes are acyclic). *If (C, d) is contractible, then $H_n(C) = 0$ for all n .*

Proof. Let $z \in Z_n(C)$, so $dz = 0$. If $\text{id} = ds + sd$, then $z = ds(z) + s(dz) = ds(z)$, so z is a boundary. Thus $Z_n(C) = B_n(C)$. $\square$

8.8 The Mayer–Vietoris Sequence

We briefly recall the Mayer–Vietoris sequence in de Rham cohomology.

Theorem 8.24 (Mayer–Vietoris for de Rham cohomology). *Let M be a smooth manifold covered by open sets U and V . There is a long exact sequence*

$$\cdots \rightarrow H_{\text{dR}}^k(M) \xrightarrow{(r_U, r_V)} H_{\text{dR}}^k(U) \oplus H_{\text{dR}}^k(V) \xrightarrow{r_U - r_V} H_{\text{dR}}^k(U \cap V) \xrightarrow{\delta} H_{\text{dR}}^{k+1}(M) \rightarrow \cdots$$

Remark 8.25 (Idea). The sequence comes from the short exact sequence of complexes

$$0 \rightarrow \Omega^\bullet(M) \xrightarrow{(|_U, |_V)} \Omega^\bullet(U) \oplus \Omega^\bullet(V) \xrightarrow{-} \Omega^\bullet(U \cap V) \rightarrow 0$$

and the long exact sequence in cohomology associated to it.

8.9 Computations for the Circle and Spheres

Proposition 8.26 (Cohomology of spheres). *For $n \geq 1$,*

$$H_{\text{dR}}^k(S^n) \cong \begin{cases} \mathbb{R}, & k = 0 \text{ or } k = n, \\ 0, & \text{otherwise.} \end{cases}$$

Sketch. Cover S^n by two contractible open sets U and V (slightly thickened hemispheres). Then $U \cap V$ deformation retracts to S^{n-1} . Apply Mayer–Vietoris and induct on n , using the base case S^1 . $\square$

9 Differential Topology Essentials

The earlier parts of these notes develop the algebra of differential forms and the beginnings of de Rham cohomology. Differential topology studies *smooth maps* and the geometric information one can read off from their differentials. In this section we collect several foundational results (rank theorem, regular value theorem, Sard's theorem, transversality, and degree) that constitute a standard “toolbox” for first courses in differential topology.

9.1 The differential and the rank

Let $f : M \rightarrow N$ be a smooth map between smooth manifolds of dimensions m and n .

Definition 9.1 (Differential). *For $p \in M$, the differential (or derivative) of f at p is the linear map*

$$df_p : T_p M \longrightarrow T_{f(p)} N$$

defined as follows: for $v \in T_p M$ and for any smooth function φ defined near $f(p)$,

$$(df_p(v))[\varphi] = v[\varphi \circ f].$$

In local coordinates, df_p is represented by the Jacobian matrix of the coordinate expression of f .

Definition 9.2 (Rank). *The rank of f at p is the rank of the linear map df_p :*

$$\text{rank}_p(f) := \dim \text{im}(df_p) \leq \min\{m, n\}.$$

We say f has constant rank r on a set $U \subseteq M$ if $\text{rank}_p(f) = r$ for all $p \in U$.

Definition 9.3 (Immersion, submersion). *A smooth map $f : M^m \rightarrow N^n$ is*

- *an immersion at p if df_p is injective (equivalently $\text{rank}_p(f) = m$);*
- *a submersion at p if df_p is surjective (equivalently $\text{rank}_p(f) = n$).*

If the condition holds at every point of M , we say f is an immersion (resp. submersion).

Example 9.4. (a) The inclusion $S^n \hookrightarrow \mathbb{R}^{n+1}$ is an immersion and an embedding.

(b) The projection $\pi : M \times N \rightarrow M$ is a submersion.

(c) A smooth curve $\gamma : I \rightarrow M$ is an immersion exactly when $\gamma'(t) \neq 0$ for all t .

9.2 The constant rank theorem

The following theorem is the local normal form of a smooth map whose derivative has constant rank.

Theorem 9.5 (Constant Rank Theorem). *Let $f : M^m \rightarrow N^n$ be smooth and assume $\text{rank}_p(f) = r$ in a neighborhood of $p \in M$. Then there exist charts (U, ϕ) about p and (V, ψ) about $f(p)$ such that in these coordinates*

$$(\psi \circ f \circ \phi^{-1})(x_1, \dots, x_m) = (x_1, \dots, x_r, 0, \dots, 0) \in \mathbb{R}^n.$$

Proof sketch. In coordinates one reduces to a map $F : \mathbb{R}^m \rightarrow \mathbb{R}^n$ with constant rank r near the origin. Choose an $r \times r$ minor of the Jacobian with nonzero determinant at 0. The implicit function theorem produces local changes of coordinates on domain and target that straighten the map to the stated normal form. See, for example, [lee2013](#) or [guilleminpollack](#) for a full proof. $\square$

Remark 9.6. Two important special cases:

- (a) If $r = m \leq n$ (an immersion), then locally f looks like an inclusion $(x_1, \dots, x_m) \mapsto (x_1, \dots, x_m, 0, \dots, 0)$.
- (b) If $r = n \leq m$ (a submersion), then locally f looks like a projection $(x_1, \dots, x_m) \mapsto (x_1, \dots, x_n)$.

9.3 Regular values and submanifolds

Definition 9.7 (Regular and critical points/values). *Let $f : M^m \rightarrow N^n$ be smooth. A point $p \in M$ is a critical point if df_p is not surjective (i.e. $\text{rank}_p(f) < n$). Otherwise p is regular. A point $q \in N$ is a regular value if every $p \in f^{-1}(q)$ is regular; otherwise q is a critical value.*

Theorem 9.8 (Regular Value Theorem / Preimage Theorem). *Let $f : M^m \rightarrow N^n$ be smooth and let $q \in N$ be a regular value of f . Then $f^{-1}(q)$ is either empty or a smooth embedded submanifold of M of dimension $m - n$. Moreover, in local coordinates near any $p \in f^{-1}(q)$, the set $f^{-1}(q)$ is given by equations $x_1 = \dots = x_n = 0$.*

Idea. At $p \in f^{-1}(q)$, surjectivity of df_p means f is a submersion at p . By the constant rank theorem, in suitable coordinates f becomes a projection $(x_1, \dots, x_m) \mapsto (x_1, \dots, x_n)$. Then $f^{-1}(q)$ is locally the set $\{x_1 = \dots = x_n = 0\}$, an embedded $\mathbb{R}^{m-n}$. $\square$

Example 9.9 (Spheres as regular level sets). Consider $F : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ given by $F(x) = \|x\|^2 = x_1^2 + \dots + x_{n+1}^2$. Then $dF_x(v) = 2\langle x, v \rangle$, so $dF_x \neq 0$ for $x \neq 0$. Hence 1 is a regular value and

$$S^n = F^{-1}(1)$$

is a smooth n -dimensional submanifold of $\mathbb{R}^{n+1}$.

Example 9.10 (Graphs). Let $g : U \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^n$ be smooth. The graph $\Gamma_g = \{(x, g(x)) : x \in U\} \subseteq \mathbb{R}^{m+n}$ is a smooth embedded submanifold of dimension m . One proof uses the map $F(x, y) = y - g(x)$ and the fact that 0 is a regular value of F .

9.4 Orientations of regular level sets

If M is oriented and q is a regular value of $f : M^m \rightarrow N^n$, then $f^{-1}(q)$ inherits a natural orientation once we choose an orientation on N . The cleanest formulation uses differential forms.

Proposition 9.11 (Induced orientation via pullback). *Let $f : M^m \rightarrow N^n$ be a submersion and assume M and N are oriented. Let ω_N be a nowhere-vanishing n -form on N representing the orientation. Then $f^*\omega_N$ is a nowhere-vanishing n -form on M whose kernel defines a splitting of TM into vertical and horizontal directions. In particular, each fiber $f^{-1}(q)$ is an oriented $(m - n)$ -manifold.*

Remark 9.12 (Orientation of boundaries). A particularly important case is $f = \rho : M \rightarrow \mathbb{R}$ a defining function for a boundary: if $\partial M = \rho^{-1}(0)$ with $d\rho \neq 0$ along ∂M , then ∂M is oriented by the *outward normal first* convention.

9.5 Sard's theorem

Critical values of a smooth map can be complicated, but they are “small” in a precise measure-theoretic sense.

Theorem 9.13 (Sard's Theorem). *Let $f : M^m \rightarrow N^n$ be smooth. The set of critical values of f has measure zero in N . In particular, regular values are dense in N (and, when N is connected, nonempty unless f is very degenerate).*

Remark 9.14. Sard's theorem is one of the engines behind transversality: it implies that “most” parameters give regular values.

9.6 Transversality

Transversality is the correct replacement for “general position” in the smooth category.

Definition 9.15 (Transverse maps). *Let $f : M \rightarrow N$ be smooth and let $S \subseteq N$ be an embedded submanifold. We say f is transverse to S , written $f \pitchfork S$, if for every $p \in M$ with $f(p) \in S$,*

$$df_p(T_p M) + T_{f(p)} S = T_{f(p)} N.$$

Equivalently, the induced map $T_p M \rightarrow T_{f(p)} N / T_{f(p)} S$ is surjective.

Theorem 9.16 (Preimage theorem, transverse form). *If $f : M \rightarrow N$ is transverse to an embedded submanifold $S \subseteq N$, then $f^{-1}(S)$ is an embedded submanifold of M of codimension $\text{codim}(S)$.*

Idea. Choose local coordinates on N in which S is a coordinate subspace; then transversality becomes a surjectivity condition, so the implicit function theorem applies. $\square$

Definition 9.17 (Transverse submanifolds). *Two embedded submanifolds $A, B \subseteq M$ are transverse, written $A \pitchfork B$, if for every $p \in A \cap B$,*

$$T_p A + T_p B = T_p M.$$

In this case $A \cap B$ is an embedded submanifold of dimension $\dim A + \dim B - \dim M$.

Example 9.18. In $\mathbb{R}^2$, two smooth curves intersect transversely at a point if their tangent lines are not parallel. A tangential intersection (same tangent line) is not transverse and is not stable under perturbation.

Theorem 9.19 (Transversality theorem, informal). *Transversality is generic: after a small perturbation, a smooth map can be made transverse to a given submanifold. More precisely, for a large class of parameterized families $F : M \times P \rightarrow N$, the set of parameters $p \in P$ for which F_p is transverse to S is dense (and often residual) in P .*

9.7 Degree of a smooth map

Degree theory is a basic global invariant of smooth maps between oriented manifolds of the same dimension.

Definition 9.20 (Local degree). *Let $f : M^n \rightarrow N^n$ be smooth between oriented n -manifolds and let $q \in N$ be a regular value. For each $p \in f^{-1}(q)$, the differential $df_p : T_p M \rightarrow T_q N$ is an isomorphism. Define the sign of p (relative to q) by*

$$\text{sgn}_p(f) := \begin{cases} +1, & df_p \text{ preserves orientation,} \\ -1, & df_p \text{ reverses orientation.} \end{cases}$$

Definition 9.21 (Degree). Assume M is compact and N is connected and oriented. For a regular value $q \in N$, the degree of f is

$$\deg(f) := \sum_{p \in f^{-1}(q)} \operatorname{sgn}_p(f) \in \mathbb{Z}.$$

Theorem 9.22 (Well-definedness and homotopy invariance). For $f : M^n \rightarrow N^n$ as above, $\deg(f)$ is independent of the chosen regular value q . Moreover, $\deg(f)$ is invariant under smooth homotopies of f .

Idea. Independence of q follows by analyzing what happens when q crosses a critical value; pairs of preimages are created/annihilated with opposite signs. Homotopy invariance follows by applying the regular value theorem to the map $F : M \times [0, 1] \rightarrow N$ given by a homotopy and considering preimages of a regular value. $\square$

Proposition 9.23 (Degree via integration). Let $f : M^n \rightarrow N^n$ be smooth between compact oriented n -manifolds. For any n -form ω on N ,

$$\int_M f^* \omega = \deg(f) \int_N \omega.$$

Sketch. Choose ω supported in a small neighborhood of a regular value q and use change of variables in local charts around points of $f^{-1}(q)$. Summing the signed Jacobians gives the degree. A partition of unity then extends the identity to all forms. $\square$

Example 9.24 (Antipodal map on S^n). The antipodal map $A : S^n \rightarrow S^n$, $A(x) = -x$, has degree $\deg(A) = (-1)^{n+1}$: it preserves orientation when n is odd and reverses when n is even.

9.8 Exercises

Exercise 9.25. Show that if $f : M \rightarrow N$ is a submersion, then f is an open map.

Exercise 9.26. Let $f : S^1 \rightarrow S^1$ be given by $f(e^{it}) = e^{ikt}$. Compute $\deg(f)$ and use Proposition 9.23 with $\omega = d\theta$ to verify your answer.

Exercise 9.27. Let $A, B \subseteq M$ be embedded submanifolds with $\dim A + \dim B = \dim M$. Assuming $A \pitchfork B$ and M oriented, define the *intersection number* $\#(A \cap B)$. Compute the intersection number of the equator and a meridian in S^2 .

10 Embeddings, Tubular Neighborhoods, and Vector Fields

This section continues the differential-topology toolbox with three closely related themes:

- (1) embedding theorems (especially Whitney's),
- (2) tubular neighborhoods of submanifolds,
- (3) and global information encoded by vector fields (Poincaré–Hopf).

10.1 Embeddings and proper maps

Definition 10.1 (Embedding). *A smooth map $f : M \rightarrow N$ is an embedding if it is an immersion and a homeomorphism onto its image (with the subspace topology). Equivalently, f is an immersion and is a topological embedding.*

Remark 10.2 (Proper immersions). A useful criterion: if M is a manifold and $f : M \rightarrow N$ is a proper immersion (preimages of compact sets are compact), then f is an embedding. The proof uses that proper maps are closed and that immersions are locally embeddings by the constant rank theorem.

Example 10.3. The map $\gamma : \mathbb{R} \rightarrow \mathbb{R}^2$, $\gamma(t) = (\cos t, \sin t)$ is an immersion but not an embedding (it is not injective). Restricting to $[0, 2\pi)$ gives an embedding of an interval whose image is S^1 with a point removed.

10.2 Whitney embedding theorem

The following is one of the most fundamental existence results in differential topology.

Theorem 10.4 (Whitney Embedding Theorem). *Every smooth n -manifold admits a smooth embedding into $\mathbb{R}^{2n}$. Moreover, every smooth n -manifold admits an immersion into $\mathbb{R}^{2n-1}$.*

Very rough idea. First embed M into some $\mathbb{R}^N$ by choosing a countable atlas and a partition of unity to build a global map with injective differential. Then use transversality to perturb the map so that self-intersections disappear when the target dimension is large enough. Whitney’s original argument is a masterful application of the implicit function theorem and general position. $\square$

Remark 10.5. Whitney’s theorem implies that, for many purposes, one may think of manifolds as submanifolds of Euclidean space. This is especially convenient when discussing differential forms, orientations, and integration.

10.3 Tubular neighborhoods

If $S \subseteq M$ is an embedded submanifold, it has a neighborhood that looks like its normal bundle.

Definition 10.6 (Normal bundle). *Let $S \subseteq M$ be an embedded submanifold and choose a Riemannian metric on M . The normal space at $p \in S$ is the orthogonal complement*

$$\nu_p(S) := (T_p S)^\perp \subseteq T_p M.$$

The collection $\nu(S) = \bigsqcup_{p \in S} \nu_p(S)$ is a vector bundle over S , called the normal bundle.

Theorem 10.7 (Tubular Neighborhood Theorem). *Let $S \subseteq M$ be an embedded submanifold. There exists an open neighborhood U of the zero section in the normal bundle $\nu(S)$ and an open neighborhood W of S in M together with a diffeomorphism*

$$\Phi : U \longrightarrow W$$

that restricts to the identity on S (identified with the zero section).

Remark 10.8. In $\mathbb{R}^n$, the tubular neighborhood theorem says that a submanifold has a neighborhood diffeomorphic to a neighborhood in its normal bundle, i.e. “thickening” the submanifold in the normal directions produces a product-like neighborhood. This theorem underlies the definition of intersection numbers and the construction of Thom classes.

10.4 Vector fields and the Euler characteristic

We close with one of the most beautiful global theorems of differential topology.

Definition 10.9 (Index of an isolated zero). *Let X be a smooth vector field on an oriented n -manifold M , and assume $p \in M$ is an isolated zero of X . Choose a small coordinate ball B around p with boundary sphere $S = \partial B$. The map $S \rightarrow S^{n-1}$ given by $x \mapsto \frac{X(x)}{\|X(x)\|}$ has a well-defined degree. This integer is called the index of X at p , denoted $\text{ind}_p(X)$.*

Theorem 10.10 (Poincaré–Hopf Theorem). *Let M be a compact oriented manifold and let X be a smooth vector field on M with isolated zeros. Then*

$$\sum_{p \in Z(X)} \text{ind}_p(X) = \chi(M),$$

where $Z(X)$ is the (finite) set of zeros of X and $\chi(M)$ is the Euler characteristic.

Example 10.11 (Hairy ball theorem). On S^2 , $\chi(S^2) = 2$. Therefore every vector field on S^2 must have a zero (otherwise the sum of indices would be 0). This is the classical “hairy ball theorem.”

Remark 10.12. Poincaré–Hopf is a far-reaching generalization of the fundamental theorem of algebra: for a complex polynomial $p(z)$, the sum of multiplicities of its roots equals the degree of p .

10.5 Exercises

Exercise 10.13. Show that if $f : M \rightarrow N$ is a proper immersion, then f is an embedding.

Exercise 10.14. Let M be compact and connected. Prove that every smooth function $f : M \rightarrow \mathbb{R}$ attains a maximum and minimum, and explain why this implies f has at least two critical points.

Exercise 10.15. Use Poincaré–Hopf to show that the torus T^2 admits a nowhere-vanishing vector field.

11 An Introduction to Morse Theory

Morse theory is one of the most powerful bridges between smooth analysis and topology: it relates the topology of a manifold to the critical points of a smooth function on it. Even a short introduction fits well with the themes of these notes: differentials, Hessians, local normal forms, and homology.

11.1 Critical points and the Hessian

Let M^n be a smooth manifold and $f : M \rightarrow \mathbb{R}$ a smooth function.

Definition 11.1 (Critical point). *A point $p \in M$ is a critical point of f if $df_p = 0$ (i.e. $df_p(v) = 0$ for all $v \in T_p M$). The value $f(p) \in \mathbb{R}$ is a critical value. A point $c \in \mathbb{R}$ that is not a critical value is a regular value.*

If we choose local coordinates $x = (x_1, \dots, x_n)$ near p with $x(p) = 0$, then p is critical iff all first partial derivatives vanish at 0. The second derivative information is captured by the Hessian.

Definition 11.2 (Hessian). Let p be a critical point of f . In local coordinates, the Hessian of f at p is the symmetric matrix

$$\text{Hess}_p(f) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(0) \right)_{i,j}.$$

Although the matrix depends on coordinates, its nondegeneracy is coordinate independent.

Definition 11.3 (Morse function). A critical point p of f is nondegenerate if $\det(\text{Hess}_p(f)) \neq 0$. A smooth function $f : M \rightarrow \mathbb{R}$ is a Morse function if all of its critical points are nondegenerate.

Definition 11.4 (Index). If p is a nondegenerate critical point, the index $\lambda(p)$ is the number of negative eigenvalues of $\text{Hess}_p(f)$ (counted with multiplicity).

Example 11.5 (Height on the sphere). Let $S^n \subset \mathbb{R}^{n+1}$ and let $f(x) = x_{n+1}$ be the height function. Then f has exactly two critical points (north and south poles), both nondegenerate. The minimum has index 0 and the maximum has index n .

11.2 The Morse lemma

The Morse lemma is the analogue of the constant rank theorem for functions near a nondegenerate critical point.

Theorem 11.6 (Morse Lemma). Let $f : M^n \rightarrow \mathbb{R}$ be smooth and let $p \in M$ be a nondegenerate critical point of index λ . Then there exist local coordinates $(x_1, \dots, x_n)$ centered at p such that

$$f(x) = f(p) - x_1^2 - \dots - x_\lambda^2 + x_{\lambda+1}^2 + \dots + x_n^2.$$

Idea. In local coordinates, Taylor expand f at the critical point: $f(x) = f(0) + \frac{1}{2}x^T Hx + (\text{higher order})$. Nondegeneracy of H allows one to eliminate the higher order terms by a smooth change of coordinates, reducing f to the quadratic form determined by the signature of H . $\square$

Remark 11.7. A key moral: *nondegenerate critical points are isolated*. Thus a Morse function on a compact manifold has finitely many critical points.

11.3 Handle attachments and topology change

For $a \in \mathbb{R}$, define the sublevel set $M_a = \{p \in M : f(p) \leq a\}$. As a increases, the topology of M_a changes only when a crosses a critical value.

Theorem 11.8 (Topology does not change at regular values). If $[a, b]$ contains no critical values of f , then M_a is diffeomorphic to M_b .

Idea. On the region $f^{-1}([a, b])$ the gradient ∇f is nonzero, so one can flow along a suitably chosen vector field that increases f at unit speed. This produces a diffeomorphism between level sets and hence between sublevel sets. $\square$

Theorem 11.9 (Handle attachment at a critical point). Suppose f is Morse and c is a critical value such that $f^{-1}(c)$ contains a single critical point p of index λ . Then for $\varepsilon > 0$ small, $M_{c+\varepsilon}$ is obtained from $M_{c-\varepsilon}$ by attaching a λ -handle $D^\lambda \times D^{n-\lambda}$ along $\partial D^\lambda \times D^{n-\lambda}$.

Remark 11.10. Handle attachment implies a CW-structure on M with one λ -cell for each index- λ critical point. This leads to the Morse inequalities comparing numbers of critical points to Betti numbers.

11.4 Morse inequalities and Euler characteristic

Let m_λ be the number of critical points of index λ . Let $b_\lambda = \dim H_\lambda(M; \mathbb{R})$ be the λ th Betti number.

Theorem 11.11 (Weak Morse inequalities). *For each λ one has $m_\lambda \geq b_\lambda$.*

Theorem 11.12 (Euler characteristic from Morse data). *If M is compact and f is Morse, then*

$$\chi(M) = \sum_{\lambda=0}^n (-1)^\lambda m_\lambda.$$

Idea. The handle decomposition gives a cellular chain complex whose ranks in degree λ are m_λ . Taking the alternating sum of ranks yields the Euler characteristic, while homology computes the Betti numbers. $\square$

Example 11.13 (Euler characteristic of S^n). In Example 11.5, there is one critical point of index 0 and one of index n , so $\chi(S^n) = 1 + (-1)^n$.

11.5 Exercises

Exercise 11.14. Show that if $f : M \rightarrow \mathbb{R}$ is Morse and M is compact, then f has finitely many critical points.

Exercise 11.15. Let $T^2 \subset \mathbb{R}^3$ be the standard embedded torus and let f be the height function. How many critical points does f have, what are their indices, and what Euler characteristic does Theorem 11.12 give?

Exercise 11.16. (Prove a special case of Theorem 11.8.) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be smooth and assume $\nabla f \neq 0$ on $f^{-1}([a, b])$. Show that $f^{-1}(a)$ and $f^{-1}(b)$ are diffeomorphic hypersurfaces.

12 Partitions of Unity and the de Rham Theorem

Many constructions in differential topology are *local*: we prove something in coordinates on $\mathbb{R}^n$ and then want to “patch” the result on a manifold. The standard tool for patching is a partition of unity. We also use partitions of unity to explain (at a high level) why de Rham cohomology agrees with singular cohomology with real coefficients.

12.1 Bump functions

Definition 12.1 (Support). *If $f : M \rightarrow \mathbb{R}$ is a function, its support is the closed set*

$$\text{supp}(f) := \overline{\{p \in M : f(p) \neq 0\}}.$$

Definition 12.2 (Bump function). *A bump function on a manifold M is a smooth function $\rho : M \rightarrow [0, 1]$ with compact support. In $\mathbb{R}^n$ one can build bump functions by composing a one-variable bump with a norm.*

Lemma 12.3 (Existence of bumps). *Let M be a smooth manifold and let $U \subseteq M$ be open with $K \subseteq U$ compact. There exists a bump function ρ such that $\rho \equiv 1$ on K and $\text{supp}(\rho) \subseteq U$.*

Idea. Using charts and compactness of K , reduce to the standard bump functions on $\mathbb{R}^n$ and patch with a locally finite sum. $\square$

12.2 Partitions of unity

Definition 12.4 (Partition of unity). *Let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of M . A family of smooth functions $\{\rho_\alpha : M \rightarrow [0, 1]\}_{\alpha \in A}$ is a partition of unity subordinate to the cover if:*

- (1) $\text{supp}(\rho_\alpha) \subseteq U_\alpha$ for each α ;
- (2) the family is locally finite (each point has a neighborhood meeting only finitely many supports);
- (3) $\sum_{\alpha \in A} \rho_\alpha(p) = 1$ for all $p \in M$.

Theorem 12.5 (Existence of partitions of unity). *Every smooth manifold admits partitions of unity subordinate to any open cover.*

Sketch. Second countability implies paracompactness for manifolds; paracompactness yields locally finite refinements of covers. Using Lemma 12.3, choose bump functions supported in sets of the refinement and normalize the sum to equal 1. $\square$

Remark 12.6. Partitions of unity are the reason many analytic constructions on $\mathbb{R}^n$ extend to manifolds: connections, Riemannian metrics, embeddings into Euclidean space, and the definition of integration of top forms are all globalized by patching local data with partitions of unity.

12.3 The de Rham theorem

We already defined the de Rham cohomology groups $H_{\text{dR}}^k(M) = Z^k(M)/B^k(M)$. A fundamental result is that these groups are purely topological invariants.

Theorem 12.7 (de Rham Theorem). *For every smooth manifold M , there is a natural isomorphism*

$$H_{\text{dR}}^k(M) \cong H_{\text{sing}}^k(M; \mathbb{R})$$

between de Rham cohomology and singular cohomology with real coefficients.

Overview. There is a natural map from forms to singular cochains given by integration: a k -form ω defines a singular cochain $I(\omega)$ by

$$I(\omega)(\sigma) := \int_{\Delta^k} \sigma^* \omega$$

for each singular simplex $\sigma : \Delta^k \rightarrow M$. Stokes' theorem shows $I(d\omega) = \delta I(\omega)$, so I is a cochain map and induces a homomorphism $H_{\text{dR}}^k(M) \rightarrow H_{\text{sing}}^k(M; \mathbb{R})$. The core of the theorem is to show this map is an isomorphism. One approach uses a good cover $\{U_\alpha\}$ of M (all finite intersections contractible), the Poincaré lemma (exactness on contractible sets), and a double-complex argument comparing de Rham cohomology and de Rham cohomology. Partitions of unity enter in constructing homotopies and comparing cochains. See **bott-tu** or **tu2011** for a complete proof. $\square$

12.4 Consequences and examples

Corollary 12.8 (Homotopy invariance). *If M and N are homotopy equivalent, then $H_{\text{dR}}^*(M) \cong H_{\text{dR}}^*(N)$.*

Proof. Singular cohomology with real coefficients is homotopy invariant, so the claim follows from Theorem 12.7. $\square$

Example 12.9 (The circle again). Since S^1 deformation retracts onto itself and has singular cohomology $H^0(S^1; \mathbb{R}) \cong \mathbb{R}$ and $H^1(S^1; \mathbb{R}) \cong \mathbb{R}$, all higher groups vanish. Thus $H_{\text{dR}}^k(S^1)$ agrees with this computation (as already shown in Section 12).

12.5 Products and the Künneth formula

For de Rham cohomology with real coefficients, the Künneth formula takes a particularly clean form.

Theorem 12.10 (Künneth for de Rham cohomology). *Let M and N be smooth manifolds. Then there is a natural graded-algebra isomorphism*

$$H_{\text{dR}}^*(M \times N) \cong H_{\text{dR}}^*(M) \otimes_{\mathbb{R}} H_{\text{dR}}^*(N),$$

where the product on the left is induced by the wedge product and the product on the right is the graded tensor product.

Idea. Use the de Rham theorem to identify both sides with singular cohomology with real coefficients and apply the usual Künneth theorem (torsion issues disappear over $\mathbb{R}$). $\square$

Example 12.11 (The n -torus). Let $T^n = (S^1)^n$. By Künneth and $H^*(S^1; \mathbb{R}) \cong \Lambda(\alpha)$ with $\deg(\alpha) = 1$, we obtain

$$H_{\text{dR}}^*(T^n) \cong \Lambda(\alpha_1, \dots, \alpha_n),$$

the exterior algebra on n degree-1 generators. Concretely, on $T^n = \mathbb{R}^n/\mathbb{Z}^n$ the constant 1-forms $d\theta^1, \dots, d\theta^n$ are closed and represent these generators.

Example 12.12 (Real projective space with real coefficients). For $\mathbb{RP}^n$, all intermediate cohomology groups are 2-torsion over $\mathbb{Z}$, hence vanish over $\mathbb{R}$. Thus

$$H_{\text{dR}}^k(\mathbb{RP}^n) \cong \begin{cases} \mathbb{R}, & k = 0, \\ \mathbb{R}, & k = n \text{ and } n \text{ odd (orientable case)}, \\ 0, & \text{otherwise.} \end{cases}$$

12.6 Exercises

Exercise 12.13. Construct an explicit bump function on $\mathbb{R}$ that equals 1 on $[-1, 1]$ and is supported in $[-2, 2]$.

Exercise 12.14. Let M be a compact oriented manifold and let $\omega \in \Omega^n(M)$ be a nowhere-vanishing n -form. Show that $\int_M \omega \neq 0$ if and only if $[\omega] \neq 0$ in $H_{\text{dR}}^n(M)$.

Exercise 12.15. Compute $H_{\text{dR}}^*(S^1 \times S^2)$ using the Künneth formula and interpret the generators via explicit differential forms.

13 Vector Bundles as the Organizing Principle

Up to this point we have treated differential forms as objects that can be written in coordinates (e.g. $\omega = \sum_I a_I dx^I$) and that transform well under smooth maps (via pullback). A powerful way to unify this viewpoint is to recognize that *most geometric fields are sections of vector bundles*.

13.1 Definition and first examples

Definition 13.1 (Vector bundle). A (real) *vector bundle of rank k* over a smooth manifold M is a smooth manifold E together with a smooth surjection $\pi : E \rightarrow M$ such that

- (1) each fiber $E_p := \pi^{-1}(p)$ is a k -dimensional real vector space;
- (2) E is *locally trivial*: every $p \in M$ has a neighborhood U and a diffeomorphism

$$\Phi : \pi^{-1}(U) \longrightarrow U \times \mathbb{R}^k$$

that restricts to a linear isomorphism $E_q \cong \{q\} \times \mathbb{R}^k$ on each fiber.

A *section* is a smooth map $s : M \rightarrow E$ with $\pi \circ s = \text{id}_M$.

Example 13.2 (Trivial bundle). The projection $\pi : M \times \mathbb{R}^k \rightarrow M$ is a rank- k vector bundle. Sections are exactly smooth maps $M \rightarrow \mathbb{R}^k$ (i.e. k -tuples of smooth functions).

Example 13.3 (Tangent and cotangent bundles). The tangent bundle $\pi : TM \rightarrow M$ has fiber $T_p M$. Similarly, the cotangent bundle $\pi : T^*M \rightarrow M$ has fiber T_p^*M . A smooth vector field is a section of TM ; a smooth 1-form is a section of T^*M . More generally, k -forms are sections of $\bigwedge^k T^*M$.

Remark 13.4 (Transition functions and cocycles). Local trivializations (U_α, Φ_α) define *transition maps* on overlaps $U_\alpha \cap U_\beta$:

$$g_{\alpha\beta} : U_\alpha \cap U_\beta \rightarrow GL_k(\mathbb{R}), \quad \Phi_\alpha \circ \Phi_\beta^{-1}(p, v) = (p, g_{\alpha\beta}(p)v).$$

The bundle is encoded by the collection $\{g_{\alpha\beta}\}$ satisfying the cocycle identities $g_{\alpha\alpha} = \text{id}$ and $g_{\alpha\beta}g_{\beta\gamma}g_{\gamma\alpha} = \text{id}$ on triple overlaps.

13.2 Why bundles matter in differential topology

There are three recurring themes:

- (a) **Global obstructions.** A global section may not exist even if local sections do (e.g. the Hairy Ball phenomenon for TS^2).
- (b) **Natural constructions.** Pullback bundles, quotient bundles, and the normal bundle to an embedding are best expressed in this language.
- (c) **Characteristic classes.** Certain cohomology classes are intrinsic invariants of a bundle (and thus of M).

We will only sketch the first characteristic class relevant for differential topology: the *Euler class*.

14 Euler Class and Poincaré–Hopf Revisited

Part I stated the Poincaré–Hopf theorem (Theorem 10.10) and used it to motivate degree and index. Here we add a complementary viewpoint that emphasizes *vector bundles* and *intersection theory*. The moral is:

“The Euler characteristic is the obstruction to finding a nowhere-vanishing section of the tangent bundle.”

14.1 Euler class via Thom classes (conceptual picture)

Let $\pi : E \rightarrow M$ be an oriented real vector bundle of rank n over a paracompact smooth manifold M . There is a canonical cohomology class

$$u_E \in H_c^n(E; \mathbb{Z}),$$

called the *Thom class* (compactly supported cohomology in the total space). The Thom class is characterized by the property that its restriction to each fiber $E_p \cong \mathbb{R}^n$ integrates to 1.

The Thom isomorphism says that cup product with u_E induces an isomorphism

$$H^k(M; \mathbb{Z}) \xrightarrow{\cong} H_c^{k+n}(E; \mathbb{Z}).$$

Now consider the *zero section* $s_0 : M \rightarrow E$. Pulling back the Thom class along s_0 defines the *Euler class*:

$$e(E) := s_0^*(u_E) \in H^n(M; \mathbb{Z}).$$

This definition packages a lot:

- the Euler class is functorial under pullback of bundles;
- $e(E) = 0$ if E admits a nowhere-vanishing section (there is then a splitting $E \cong \mathbb{R} \oplus E'$);
- if M is closed and oriented of dimension n , the integer $\langle e(E), [M] \rangle$ is an intersection number.

14.2 Zeros of sections and local degree

Let $s : M \rightarrow E$ be a smooth section. If s is transverse to the zero section, then its zero set $Z(s)$ is a submanifold of codimension n . In particular, if $\dim M = n$ (the tangent bundle case), then $Z(s)$ is a finite set of points.

Near an isolated zero $p \in Z(s)$, choose a local trivialization $E|_U \cong U \times \mathbb{R}^n$. The section becomes a map $f : U \rightarrow \mathbb{R}^n$ with $f(p) = 0$. One defines a local degree $\deg_p(f)$ (equivalently the degree of the normalized map on a small sphere). In the tangent-bundle setting this local degree is exactly the index $\text{ind}_p(X)$ from Definition 10.9.

Remark 14.1 (Intersection number viewpoint). Transversality ensures that at each zero p the section intersects the zero section cleanly. The sign of the intersection is determined by whether the derivative df_p preserves orientation. This is the same sign that appears in the degree/index definition.

14.3 Recovering Poincaré–Hopf from Euler classes

When $E = TM$ for a closed oriented n -manifold, there is a fundamental identity

$$\langle e(TM), [M] \rangle = \chi(M).$$

Combining this with the “zeros count as Euler class” interpretation gives Poincaré–Hopf: for a vector field X with isolated zeros,

$$\sum_{p: X(p)=0} \text{ind}_p(X) = \langle e(TM), [M] \rangle = \chi(M).$$

Thus degree theory, transversality, and characteristic classes fit together into a single picture:

- **Degree** measures how maps wrap spheres.
- **Index** measures how a section wraps a small sphere around a zero.
- **Euler class** is the global cohomology class detected by these local degrees.

14.4 Examples and sanity checks

Example 14.2 (Circle and torus). The tangent bundle of S^1 is trivial, so $e(TS^1) = 0$ and $\chi(S^1) = 0$. Indeed, the unit tangent vector field has no zeros.

Similarly, $T\mathbb{T}^2$ is trivial, so $\chi(\mathbb{T}^2) = 0$. This matches the existence of a nowhere-vanishing vector field on the torus (in fact, many).

Example 14.3 (Sphere). For S^2 , $\chi(S^2) = 2$, so every vector field has zeros and the sum of indices is 2. A familiar example is the gradient field of the height function: it has a north-pole source (index +1) and a south-pole sink (index +1).

Exercise 14.4. Let M be a closed oriented surface of genus g . Use a cell decomposition or a Mayer–Vietoris argument to show $\chi(M) = 2 - 2g$. Interpret this as a constraint on the number and type of zeros of a vector field on M .

15 Hodge Theory: Harmonic Representatives of Cohomology

Part I introduced the Hodge star on $\mathbb{R}^n$ and used it to package operations like divergence and curl. On a compact oriented *Riemannian* manifold, the Hodge star upgrades de Rham cohomology from a purely topological invariant to something that also has a canonical analytic representative: *harmonic forms*. This is one of the most useful “bridges” between analysis and topology.

15.1 The Hodge star and the codifferential

Let (M^n, g) be an oriented Riemannian manifold. The metric induces an inner product on k -forms:

$$\langle \alpha, \beta \rangle = \int_M \alpha \wedge \star \beta,$$

where $\star : \Omega^k(M) \rightarrow \Omega^{n-k}(M)$ is the Hodge star (defined pointwise via g and the orientation).

The adjoint of d with respect to this inner product is the *codifferential* $\delta : \Omega^k(M) \rightarrow \Omega^{k-1}(M)$, defined (up to a sign convention) by

$$\delta = (-1)^{n(k+1)+1} \star d \star.$$

It satisfies the formal integration-by-parts identity

$$\langle d\alpha, \beta \rangle = \langle \alpha, \delta\beta \rangle$$

for compactly supported forms (and for all smooth forms if M is closed).

15.2 The Hodge Laplacian and harmonic forms

Define the (Hodge) Laplacian on k -forms by

$$\Delta := d\delta + \delta d : \Omega^k(M) \rightarrow \Omega^k(M).$$

A form ω is *harmonic* if $\Delta\omega = 0$. We denote the finite-dimensional space of harmonic k -forms by $\mathcal{H}^k(M)$.

Remark 15.1 (Closed and coclosed). If ω is harmonic, then $d\omega = 0$ and $\delta\omega = 0$. Indeed,

$$0 = \langle \Delta\omega, \omega \rangle = \langle d\omega, d\omega \rangle + \langle \delta\omega, \delta\omega \rangle,$$

so both terms vanish. Thus harmonic forms are simultaneously closed and coclosed.

15.3 Hodge decomposition and the Hodge theorem

When M is closed (compact without boundary), elliptic theory implies the following fundamental structure theorem.

Theorem 15.2 (Hodge decomposition (informal statement)). *Let M be a closed oriented Riemannian manifold. Then every k -form decomposes orthogonally as*

$$\Omega^k(M) = d\Omega^{k-1}(M) \oplus \delta\Omega^{k+1}(M) \oplus \mathcal{H}^k(M).$$

Moreover, each de Rham cohomology class in $H_{\text{dR}}^k(M)$ has a unique harmonic representative. Equivalently,

$$H_{\text{dR}}^k(M) \cong \mathcal{H}^k(M).$$

Remark 15.3 (What is nontrivial here?). The novelty is not the formal algebra $\Delta = d\delta + \delta d$, but the analytic fact that Δ is an elliptic operator on a compact manifold. Ellipticity gives:

- finite-dimensionality of $\mathcal{H}^k(M)$,
- existence of Green’s operators solving $\Delta\eta = \alpha$ modulo harmonic parts,
- and the orthogonal splitting above.

A full proof belongs to elliptic PDE and global analysis; we treat it as optional background.

15.4 Examples

Example 15.4 (The circle). Let $M = S^1$ with coordinate $\theta \in [0, 2\pi)$ and metric $d\theta^2$. Then $\mathcal{H}^0(S^1)$ consists of constants and $\mathcal{H}^1(S^1)$ is spanned by $d\theta$. Thus $H_{\text{dR}}^0(S^1) \cong \mathbb{R}$ and $H_{\text{dR}}^1(S^1) \cong \mathbb{R}$ with canonical representatives.

Example 15.5 (Flat torus). On the flat torus $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ with its standard metric, the harmonic forms are precisely the constant-coefficient forms. Hence

$$\dim H_{\text{dR}}^k(\mathbb{T}^n) = \binom{n}{k}.$$

15.5 Connection to transport and diffusion

Appendix A expressed deterministic transport of densities using Lie derivatives. Diffusion terms (as in Fokker–Planck) involve Laplacians. Hodge theory is one clean way to see why Laplacians are natural on manifolds: they arise as the canonical elliptic operators built from d and its adjoint δ . This perspective becomes essential when one wants to formulate stochastic dynamics or diffusion-based models intrinsically on curved state spaces.

Exercise 15.6 (Harmonic forms and Poincaré duality). Assume M is closed and oriented. Using the inner product $\int_M \alpha \wedge \star\beta$, show that the Hodge star maps harmonic k -forms to harmonic $(n-k)$ -forms. Deduce Poincaré duality in de Rham cohomology from Theorem 15.2.

16 More Diagram Chasing

Part I included the Snake Lemma and Mayer–Vietoris as core tools. Here we record two additional lemmas that are used constantly in homological arguments, and we work out one “complete chase” as a model.

16.1 Five Lemma

Lemma 16.1 (Five Lemma). *Consider a commutative diagram in an abelian category (or in vector spaces/abelian groups) with exact rows:*

$$\begin{array}{ccccccccc} A_1 & \longrightarrow & A_2 & \longrightarrow & A_3 & \longrightarrow & A_4 & \longrightarrow & A_5 \\ \downarrow f_1 & & \downarrow f_2 & & \downarrow f_3 & & \downarrow f_4 & & \downarrow f_5 \\ B_1 & \longrightarrow & B_2 & \longrightarrow & B_3 & \longrightarrow & B_4 & \longrightarrow & B_5 \end{array}$$

If f_1 is surjective, f_2 is injective, f_4 is surjective, and f_5 is injective, and if f_2, f_4 are isomorphisms, then f_3 is an isomorphism. (There are several standard variants; this is one common form.)

Idea. Injectivity of f_3 is proved by chasing an element $a_3 \in A_3$ with $f_3(a_3) = 0$ leftwards to show it must come from A_2 and then be zero by injectivity. Surjectivity is a dual chase: start with $b_3 \in B_3$ and lift it right-to-left using surjectivity and exactness. $\square$

16.2 Nine Lemma (a.k.a. the 3×3 Lemma)

Lemma 16.2 (Nine Lemma). *Given a commutative 3×3 diagram with exact columns and exact first two rows, the third row is exact, provided the category is abelian and the connecting morphisms exist.*

Remark 16.3. A detailed proof is a good exercise in diagram chasing. It can be organized as three applications of the Snake Lemma.

16.3 A complete chase: long exact sequence in cohomology

A standard construction is: a short exact sequence of cochain complexes induces a long exact sequence in cohomology. The Snake Lemma is the engine.

Theorem 16.4 (LES from a short exact sequence of cochain complexes). *Let*

$$0 \rightarrow (A^\bullet, d) \xrightarrow{i} (B^\bullet, d) \xrightarrow{p} (C^\bullet, d) \rightarrow 0$$

be a short exact sequence of cochain complexes (exact in each degree, and commuting with differentials). Then there is a natural long exact sequence

$$\cdots \rightarrow H^k(A) \xrightarrow{i^*} H^k(B) \xrightarrow{p^*} H^k(C) \xrightarrow{\delta} H^{k+1}(A) \rightarrow \cdots$$

where δ is the connecting homomorphism.

Sketch with the essential chase. Fix k . Consider the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A^k & \xrightarrow{i} & B^k & \xrightarrow{p} & C^k & \longrightarrow & 0 \\ & & \downarrow d & & \downarrow d & & \downarrow d & & \\ 0 & \longrightarrow & A^{k+1} & \xrightarrow{i} & B^{k+1} & \xrightarrow{p} & C^{k+1} & \longrightarrow & 0 \end{array}$$

and apply the Snake Lemma to obtain a connecting homomorphism $\delta : \ker(d : C^k \rightarrow C^{k+1}) \rightarrow \operatorname{coker}(d : A^k \rightarrow A^{k+1})$. One checks that

$$\ker(d : C^k \rightarrow C^{k+1}) = Z^k(C), \quad \operatorname{coker}(d : A^k \rightarrow A^{k+1}) = A^{k+1}/B^{k+1}(A),$$

and that δ vanishes on coboundaries so it descends to cohomology. Exactness at each term is then verified by straightforward chases using exactness of rows and the definition of δ . $\square$

Exercise 16.5. Work out the definition of δ explicitly in the setting of differential forms: take $A^\bullet = \Omega^\bullet(M)$, $B^\bullet = \Omega^\bullet(U) \oplus \Omega^\bullet(V)$, and $C^\bullet = \Omega^\bullet(U \cap V)$ for $M = U \cup V$. Show that Theorem 16.4 recovers the Mayer–Vietoris long exact sequence.

Part II

Applications: Densities as Forms and Energy-Based Flow Matching

17 Densities as Top Forms and Transport by Flows

Part I developed differential k -forms as smooth sections of the bundle $\bigwedge^k T^*M$. In applications to dynamics and generative modeling, the most important special case is $k = n = \dim M$: a *density* on an oriented n -manifold can be encoded as a nowhere-negative n -form.

17.1 Densities, volume forms, and change of variables

Let M be an oriented smooth n -manifold. An orientation can be represented by a nowhere-vanishing n -form $\nu \in \Omega^n(M)$ (a *volume form*). Any other n -form can then be written uniquely as

$$\omega = \rho \nu, \quad \rho \in C^\infty(M).$$

If $\rho \geq 0$ and $\int_M \rho \nu = 1$, we can interpret $\rho \nu$ as a probability measure on M with density ρ relative to ν .

Remark 17.1 (Densities as coordinate-free objects). In $\mathbb{R}^n$ with the standard orientation, $\nu = dx^1 \wedge \cdots \wedge dx^n$, so $\rho \nu$ is exactly the familiar density $\rho(x) dx$ from multivariable calculus. On manifolds, writing densities as top forms has two advantages:

1. It packages the Jacobian determinant into the pullback of a single tensorial object.
2. It makes the continuity equation and change-of-variables formula coordinate-free.

Let $F : M \rightarrow M$ be an orientation-preserving diffeomorphism. The pullback acts on top forms by

$$F^*(\rho \nu) = (\rho \circ F) F^* \nu.$$

In a local coordinate chart, if F is represented by $y = F(x)$, then

$$F^*(dy^1 \wedge \cdots \wedge dy^n) = \det(dF_x) dx^1 \wedge \cdots \wedge dx^n.$$

Hence (locally)

$$F^*(\rho(y) dy) = \rho(F(x)) \det(dF_x) dx.$$

This is precisely the analytic change-of-variables rule. Equivalently, for integrals,

$$\int_{F(U)} \rho \nu = \int_U F^*(\rho \nu).$$

We will use this viewpoint repeatedly in Part II.

17.2 Flows and the continuity equation via Lie derivatives

Let v_t be a time-dependent vector field on M , and let Φ_t be its flow (when it exists):

$$\frac{d}{dt}\Phi_t(p) = v_t(\Phi_t(p)), \quad \Phi_0(p) = p.$$

If μ_0 is a measure (or density form) on M , we can transport it by the flow to obtain $\mu_t := (\Phi_t)_*\mu_0$. When μ_t is represented by a time-dependent top form $\omega_t \in \Omega^n(M)$, there is a clean differential equation for ω_t .

Theorem 17.2 (Continuity equation in differential-forms form). *Let $\omega_t \in \Omega^n(M)$ be a smooth time-dependent top form and let v_t be a smooth time-dependent vector field. The condition that ω_t is transported by the flow of v_t (i.e. $\omega_t = (\Phi_t)_*\omega_0$) is equivalent to*

$$\partial_t \omega_t + \mathcal{L}_{v_t} \omega_t = 0,$$

where $\mathcal{L}_{v_t}$ is the Lie derivative along v_t .

Proof sketch. The statement is a standard “differentiate the pullback” identity:

$$\frac{d}{dt}(\Phi_t^* \omega_t) = \Phi_t^*(\partial_t \omega_t + \mathcal{L}_{v_t} \omega_t).$$

If $\omega_t = (\Phi_t)_*\omega_0$ then $\Phi_t^* \omega_t = \omega_0$ is constant in t , and the right-hand side must vanish. Conversely, if the right-hand side vanishes, then $\Phi_t^* \omega_t$ is constant, so ω_t must be the pushforward of ω_0 . $\square$

To connect this to the familiar PDE $\partial_t \rho + \operatorname{div}(\rho v) = 0$, write $\omega_t = \rho_t \nu$. Cartan’s formula gives

$$\mathcal{L}_v = d \iota_v + \iota_v d.$$

Since $d\omega_t = 0$ for top forms (degree $n+1$ forms vanish), we get

$$\mathcal{L}_{v_t} \omega_t = d \iota_{v_t} \omega_t.$$

In a coordinate chart on $\mathbb{R}^n$ with $\nu = dx^1 \wedge \cdots \wedge dx^n$, one checks explicitly that

$$\mathcal{L}_v(\rho \nu) = (\nabla \rho \cdot v + \rho \operatorname{div} v) \nu = \operatorname{div}(\rho v) \nu.$$

Thus Theorem 17.2 reduces to

$$\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0,$$

the standard continuity equation.

Remark 17.3 (What changes on a general manifold?). On a general oriented manifold, “div” depends on a choice of volume form ν . Given ν , divergence is defined by $\mathcal{L}_v \nu = (\operatorname{div}_\nu v) \nu$. This matches the construction in Part I where divergence is extracted from $d(\star \omega)$ when ω corresponds to v via the metric.

17.3 Continuous normalizing flows (CNFs) as transported densities

In $\mathbb{R}^d$, a *continuous normalizing flow* is a diffeomorphism obtained as the time-1 flow map Φ_1 of an ODE

$$\dot{x}_t = v_t(x_t).$$

If $x_0 \sim p_0$, then $x_1 = \Phi_1(x_0)$ has density $p_1 = (\Phi_1)_* p_0$. The continuity equation describes the entire path p_t .

An equivalent (and practically important) identity is the *instantaneous change-of-variables formula*: if x_t follows the ODE, then along trajectories

$$\frac{d}{dt} \log p_t(x_t) = -\operatorname{div} v_t(x_t).$$

One derivation is to combine the continuity equation with the method of characteristics. In differential-forms language, it is the statement that the Jacobian determinant of Φ_t evolves by the trace of dv_t .

Exercise 17.4. Derive the identity $\frac{d}{dt} \log \det(d\Phi_t) = \operatorname{div} v_t \circ \Phi_t$ and use it to prove $\frac{d}{dt} \log p_t(x_t) = -\operatorname{div} v_t(x_t)$.

17.4 Why flow matching is a natural “inverse problem” for the continuity equation

The continuity equation

$$\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0$$

is an underdetermined PDE for the pair (ρ_t, v_t) . In generative modeling we often specify (or approximate) a family ρ_t and then seek a vector field v_t that realizes it, or we specify endpoint distributions ρ_0, ρ_1 and seek a suitable interpolating pair (ρ_t, v_t) .

Flow matching (as introduced by **lipman2022flowmatching**) can be viewed as a supervised learning proxy for this inverse problem: one chooses a probabilistic coupling between endpoints (or other construction) that yields *conditional* velocities $u_t(x \mid \text{context})$, and then trains a model $v_\theta(t, x)$ to match the conditional expectation

$$v^*(t, x) = \mathbb{E}[u_t(X \mid \text{context}) \mid X_t = x].$$

Under mild conditions, this v^* is precisely the vector field that drives the marginal density ρ_t .

Remark 17.5 (On manifolds). Because all of the above can be phrased in terms of top forms, pullbacks, and Lie derivatives, the picture extends essentially verbatim to flows on manifolds as long as we have a notion of density (a top form). This is one reason differential topology is a natural language for modern flow-based modeling.

17.5 Energy-based targets and iEFM in words

A key obstacle in scientific applications is that the target density may be known only up to a normalizing constant:

$$p^*(x) \propto \exp(-E(x)),$$

where E is an energy function (possibly defined by a physics simulator or a learned surrogate). Sampling from p^* directly can be hard.

Iterated energy-based flow matching (iEFM) provides a simulation-free, off-policy training strategy for CNFs in this setting (see Part II and **woo2024iefm**):

- One maintains a current flow model p_θ that can be sampled.
- One constructs (implicitly or explicitly) intermediate distributions connecting p_θ to p^* using the energy E .
- One trains a velocity field v_θ so that its induced continuity equation progressively reduces the energy mismatch.

The case study that follows combines this training idea with a learned quantum-accuracy potential (UMA) to target enzyme catalysis design objectives.

18 Temperature-Conditioned Iterated Energy-Based Flow Matching with a Learned Quantum Potential for Simulation-Free Enzyme Path and Sequence Design

Anonymous Authors (from the original draft; notation aligned for this unified edition)

Abstract

This paper describes a practical adaptation of *iterated energy-based flow matching* (iEFM)—a simulation-free, off-policy training framework for continuous normalizing flows from unnormalized densities—to the enzyme catalysis setting where the energy function is provided by a large learned *quantum-accuracy surrogate* interatomic potential. The method replaces expensive quantum chemistry and molecular simulation pipelines with direct calls to a pretrained Universal Model for Atoms (UMA), which predicts DFT-level energies and forces across broad chemical space. The main contribution is a *three-step cyclic training objective* tailored to de novo enzyme design with minimal supervision: (i) a temperature-conditioned equilibrium (EQ) objective that learns Boltzmann samplers using iEFM with a discrete temperature ladder; (ii) a transition-path sampling objective (TPS) formulated as a *path-space Boltzmann density* with endpoint constraints given only by reactant-bound and product-bound structures (e.g. `.cif` files), augmented with controlled inpainting masks; and (iii) an enzyme sequence design objective (SEQ) using *categorical discrete flow matching* conditioned on structural embeddings from endpoints and a sampled mid-trajectory state. Each cycle through EQ→TPS→SEQ constitutes one training step and produces interpretable per-objective losses and monitoring metrics. The exposition emphasizes implementable design choices, pseudocode, and monitoring/validation protocols (including ProteinConformers and DynamicPDB benchmarks). A mathematical appendix (in the language of differential forms) summarizes the background theory and the restriction to this special case.

Reader’s note on notation and placement

This case study is written in a research-paper style, but it is intentionally embedded here as a continuation of the foundational narrative. We keep the original sectioning and claims, while (i) aligning notation with Part I (e.g. d for differentials, Ω^k for k -forms, and consistent pullback/pushforward conventions), and (ii) adding brief clarifications when they improve readability for a first pass.

19 Introduction

Many problems in enzyme catalysis can be expressed as sampling and design problems on an energy landscape: (i) sampling thermally weighted conformational ensembles of an enzyme–ligand complex at temperature T ; (ii) sampling plausible *reactive transition paths* between reactant-bound

and product-bound states; and (iii) designing an amino-acid sequence that stabilizes a desired mechanistic pathway and binding geometry. In principle, these tasks can be addressed by molecular dynamics (MD), enhanced sampling, and quantum mechanical (QM) calculations. In practice, such pipelines are expensive and difficult to scale across large enzyme families, long timescales, and repeated design iterations.

This paper proposes a simulation-free alternative built around two recent ingredients. First, *iterated energy-based flow matching* (iEFM) trains a continuous normalizing flow (CNF) generator from evaluations of an energy function $E(x)$ defining an unnormalized Boltzmann density, without requiring target samples and without generating training trajectories from the model (**woo2024iefm**). Second, UMA is a family of large pretrained interatomic potentials trained on $\sim 5 \times 10^8$ DFT-labeled 3D atomic structures and designed to act as a fast DFT surrogate, returning total energy and forces (negative energy gradients) (**wood2025uma**). By using UMA as a fixed energy backend, iEFM-style training becomes feasible for all-atom enzyme systems where classical force fields may be insufficient and QM is prohibitively slow.

Scope. The focus is restricted to *enzyme catalysis* and *all-atom* modeling. Training constraints consists of endpoint structures for enzyme-reactant and enzyme-product complexes (e.g. `.cif` files), along with optional metadata (charge/spin/task identifiers compatible with UMA). No intermediate trajectories or simulations are assumed for training; all intermediate conformations and paths are learned via the cyclic objectives, with the categorical discrete sequence flow-matching network conditioned on initial and middle frame embeddings. The method is designed for conditioning on masked enzyme atoms and residues, enabling *trajectory inpainting* and *sequence design* conditioned on reactant and product ligand endpoints.

Contributions.

- **Temperature-conditioned iEFM with a learned quantum potential.** The equilibrium objective trains a CNF to sample from $\mu_T(x) \propto \exp(-E_{\text{UMA}}(x)/(k_B T))$ using iEFM’s simulation-free, off-policy estimator, and conditions the vector field on a discrete temperature ladder $T_{\text{ladder}} = [755, 500, 400, 350, 300]\text{K}$.
- **Path-space Boltzmann TPS with endpoint constraints and inpainting.** Transition-path sampling is formulated as sampling from an unnormalized density on *trajectory space* with fixed endpoints, enabling training with only reactant/product structures generated as follows:
 1. generate high quality conformers for each reactant and product ligand utilizing ETFlow **hassan2024etflow**, an equivariant flow-matching model trained to generate high quality conformers (initial tests with many ligands utilizing RDKit and openbabel returned many failed attempts at generating embeddings despite high quality settings),
 2. filter conformers based on QA measures such as clashes, energy, and (optional) bond statistics,
 3. utilize the generated conformers as templates in a new custom version of Boltz-2 we are calling *Boltz-TS* that docks multi-fragment ligands utilizing `.sdf` templates, with a RMSD threshold of $< 1.5\text{\AA}$, with pocket residue constraints pulled from UniProt on January 30th, 2026, and a pocket threshold of $7.0 - 9.0\text{\AA}$, with 300 diffusion steps, 10 recycles, and an MSA generated from the latest version of UniRef30 database pulled on January 30th, 2026 with 4096 sequences for each enzyme MSA.

We *do not* provide the transition state in any form to the model during training, expecting that the quantum accuracy of UMA will be a strong signal for the model to learn accurate physics of transition paths, informed in part by its capability to find high quality, accurate transition states when run inside of ColabReaction [karasawa2025colabreaction](#).

Two and only two masking modes are supported for TPS inpainting: (a) *residue masking* (mask all atoms of selected residues), and (b) *side-chain masking* (mask side-chain atoms for selected residues while keeping the backbone unmasked). Masking difficulty increases linearly over training steps.

- **Sequence design via categorical discrete flow matching.** A discrete flow-matching model generates enzyme sequences conditioned on structural embeddings from an endpoint and a sampled mid-trajectory state; an ablation additionally conditions on the product-bound endpoint. Masking in the discrete objective follows a linear schedule compatible with the continuous objectives. Monitoring includes *sequence recovery* as a key metric with the sequences from the initial frame constraints.
- **Cyclic training as a single optimization primitive.** EQ, TPS, and SEQ are executed in a fixed cycle; each cycle yields $(L_{\text{eq}}, L_{\text{tps}}, L_{\text{seq}}, L_{\text{cycle}})$ and auxiliary metrics for stable training and reproducible diagnostics.
- **Implementation and validation plan.** Practical integration details are given for the Mora-iFEM-1 codebase (Hydra entrypoint `dem/train.py`) ([ifem1code](#)). Validation protocols are specified using ProteinConformers ([proteinconformers2025](#)) and DynamicPDB ([triki2024dynamicpdb](#)).

20 Background and Problem Setup

20.1 Boltzmann densities with temperature

Let $x \in \mathbb{R}^{3N}$ denote all-atom Cartesian coordinates (for an enzyme–ligand complex with N atoms). Given an energy function $E(x)$ (in eV), the Boltzmann density at temperature T is

$$\mu_T(x) = \frac{\exp(-\beta(T) E(x))}{Z(T)}, \quad \beta(T) := \frac{1}{k_B T}, \quad (1)$$

$$Z(T) = \int \exp(-\beta(T) E(x)) \, dx. \quad (2)$$

The normalizing constant $Z(T)$ is intractable for realistic systems. The learning objective is to train a sampler that produces $x \sim \mu_T$ and, additionally, to learn conditional models over paths and sequences as described later.

20.2 Simulation-free flow matching from unnormalized densities

A continuous normalizing flow (CNF) defines an ODE

$$\frac{dx}{dt} = u_\theta(x, t, c), \quad t \in [0, 1], \quad (3)$$

where u_θ is a time-dependent vector field (a neural network) and c is optional conditioning information. Solving (3) pushes a simple base distribution p_0 (e.g. Gaussian) to a model distribution p_1^θ . Flow matching trains u_θ by regressing to a target marginal vector field $v_t(x)$ associated with a chosen probability path, via the objective $\mathbb{E}_{t, x \sim p_t} \|u_\theta(x, t) - v_t(x)\|^2$ ([lipman2022flowmatching](#)).

iEFM introduces a *Monte Carlo estimator* for $v_t(x)$ when the endpoint density $p_1(x) \propto \exp(-E(x))$ is only known up to normalization (**woo2024iefm**). Crucially, the iEFM training objective is (i) *off-policy* (samples can come from a replay buffer) and (ii) *simulation-free* (no trajectories from the current model are required to form training targets). Section 21.3 summarizes the specialization needed for enzyme systems and temperature conditioning.

20.3 UMA as a learned quantum-accuracy energy and force model

UMA is trained to emulate density functional theory (DFT) energy and force labels across multiple chemical domains (materials, molecules, catalysts, molecular crystals, and MOFs) and multiple DFT settings, returning total energy and per-atom forces (**wood2025uma**). DFT is a quantum mechanical method that approximates ground-state electronic structure; UMA is thus a *quantum-accuracy surrogate* at the DFT level (not a classical empirical force field). UMA models are designed to be both accurate and efficient at inference, supporting simulation workloads and large atom counts on modern GPUs (**wood2025uma**).

In this work, UMA is used as a fixed differentiable energy backend:

$$E(x) \equiv E_{\text{UMA}}(x; \text{task}, q, s), \quad (4)$$

$$F(x) \equiv -\nabla_x E_{\text{UMA}}(x; \text{task}, q, s), \quad (5)$$

optionally conditioned on total charge q , spin multiplicity s , and a task identifier encoding the relevant DFT settings (**wood2025uma**). The availability of forces $F(x)$ enables gradient-based variance reduction in iEFM estimators and direct gradients for path-space objectives.

20.4 Enzyme catalysis setting and supervision

Training constraints are assumed to consist of endpoint structure pairs:

$$(x_{\text{full}}^{\text{R}}, x_{\text{full}}^{\text{P}}),$$

where $x_{\text{full}}^{\text{R}}$ is an all-atom enzyme–reactant complex and $x_{\text{full}}^{\text{P}}$ is an all-atom enzyme–product complex for the same enzyme scaffold, parsed from `.cif` (or equivalent) files. No intermediate structures or trajectory data are required for training. During training, masking is applied to create additional *conditioning* $x_{\text{obs}}^{\text{R}}, x_{\text{obs}}^{\text{P}}$ and masked variables $x_{\text{mask}}^{\text{R}}, x_{\text{mask}}^{\text{P}}$ depending on the objective. The overall goal is *de novo design* conditioned on ligand endpoints (reactants and products) with substantial or complete enzyme masking, so that the model learns to generate missing enzyme structure, transition paths and masked trajectory data that are plausible in terms of physics, and compatible sequences.

21 Method: Mora-iFEM

21.1 Overview: three-step cyclic training

Mora-iFEM consists of three models trained jointly (with optional parameter sharing):

1. **EQ model** (equilibrium sampler): a temperature-conditioned CNF vector field $u_{\theta_{\text{eq}}}(x, t, T, \cdot)$ trained by iEFM to sample from μ_T at temperatures in $\mathcal{T}_{\text{ladder}}$.
2. **TPS model** (trajectory sampler): a CNF on *path space* that samples discrete-time trajectories $\mathbf{x} = (x_0, \dots, x_M)$ consistent with fixed endpoints and a path-space Boltzmann density.

Algorithm 1: Cyclic training loop for Mora-iFEM. Each cycle is one training step.

Input: UMA energy/force oracle $E_{\text{UMA}}, F_{\text{UMA}}$;
temperature ladder $\mathcal{T}_{\text{ladder}} = [755, 500, 400, 350, 300]\text{K}$;
models $\theta_{\text{eq}}, \theta_{\text{tps}}, \theta_{\text{seq}}$;
buffers $\mathcal{B}_{\text{eq}}, \mathcal{B}_{\text{tps}}$;
cycle step counter $s = 0, 1, 2, \dots$
Output: Updated parameters $(\theta_{\text{eq}}, \theta_{\text{tps}}, \theta_{\text{seq}})$ and logged metrics.

for $s = 0, 1, \dots$ **do**
 Select temperature $T_s \in \mathcal{T}_{\text{ladder}}$ using the annealing schedule in Sec. 21.3.5
 Compute masking rates $\rho_{\text{tps}}(s), \rho_{\text{seq}}(s)$ via linear schedules (Sec. 21.4.5)
 // (1) EQ: temperature-conditioned iEFM update
 $L_{\text{eq}} \leftarrow \text{EQ_Step}(\theta_{\text{eq}}, \mathcal{B}_{\text{eq}}, T_s; E_{\text{UMA}}, F_{\text{UMA}})$
 // (2) TPS: endpoint-conditioned path-space sampling with inpainting
 masks
 $L_{\text{tps}} \leftarrow \text{TPS_Step}(\theta_{\text{tps}}, \mathcal{B}_{\text{tps}}, T_s, \rho_{\text{tps}}(s); E_{\text{UMA}}, F_{\text{UMA}})$
 // (3) SEQ: categorical DFM sequence design conditioned on embeddings for
 initial and middle frames
 $L_{\text{seq}} \leftarrow \text{SEQ_Step}(\theta_{\text{seq}}, \theta_{\text{tps}}, T_s, \rho_{\text{seq}}(s))$
 // Composite cycle loss (logged each cycle)
 $L_{\text{cycle}} \leftarrow w_{\text{eq}}L_{\text{eq}} + w_{\text{tps}}L_{\text{tps}} + w_{\text{seq}}L_{\text{seq}}$
 Update $(\theta_{\text{eq}}, \theta_{\text{tps}}, \theta_{\text{seq}})$ using ∇L_{cycle}
 Log per-objective losses and diagnostics (Sec. 23)

3. **SEQ model** (sequence generator): a categorical discrete flow-matching (DFM) model that generates amino-acid sequences conditioned on structural embeddings from an endpoint and a sampled mid-trajectory state.

Training proceeds in *cycles*. Each cycle executes one EQ step, one TPS step, and one SEQ step, and then computes a composite loss used for an optimizer update. This strict cycling makes it straightforward to monitor stability and identify failure modes by objective.

21.2 Shared representation and conditioning

The three objectives require a representation of atomic structure and, for SEQ, a representation of sequence. There is no restriction to iEFM-style architectures, however, in this work the EGNN $E(n)$ -equivariant neural network is utilized. Although SE(3)-equivariance is often the most appropriate architecture when training on molecular data involving chiral ligands, our trajectory-free training paradigm eliminates the need for SE(3)-equivariance or data augmentation, as explained in Appendix B.

Concretely, let $x \in \mathbb{R}^{3N}$ be centered coordinates and let $\phi_\psi(x) \in \mathbb{R}^{d_\phi}$ be a learned structural embedding (e.g. obtained by a message-passing network on a neighbor graph built from distance cut-offs, while enforcing E(3) symmetry constraints). The EQ and TPS vector fields can optionally take $\phi_\psi(x)$ as additional input; the SEQ model conditions on $\phi_\psi(x)$ from endpoint and mid-trajectory frames.

21.3 Objective (1): EQ — temperature-conditioned iEFM

21.3.1 Target density and temperature conditioning

The equilibrium target at temperature T is

$$p_1^{(T)}(x) \propto \exp(-\beta(T)E_{\text{UMA}}(x)). \quad (6)$$

Temperature enters only through $\beta(T)$ and via an explicit temperature embedding provided to $u_{\theta_{\text{eq}}}$. A practical implementation embeds T (in Kelvin) via a small MLP on $\log T$ concatenated to the usual time embedding.

21.3.2 Simulation-free EFM target via Monte Carlo reweighting

iEFM builds an estimator for the marginal vector field $v_t(x)$ by sampling K candidates $x_1^{(i)}$ from an auxiliary distribution $q(x_1; x, t) \propto p_t(x|x_1)$ and computing a weighted average of conditional vector fields (**woo2024iefm**). For each sample (x, t) ,

$$\hat{v}_t(x; T) \equiv U_K(x, t; T) = \sum_{i=1}^K w_i(x, t; T) v_t(x | x_1^{(i)}), \quad (7)$$

$$w_i(x, t; T) = \frac{\exp(-\beta(T)E_{\text{UMA}}(x_1^{(i)}))}{\sum_{j=1}^K \exp(-\beta(T)E_{\text{UMA}}(x_1^{(j)}))}. \quad (8)$$

The conditional vector field $v_t(x | x_1)$ is determined by the chosen conditional probability path (e.g. variance-exploding or optimal-transport Gaussian paths). The EQ loss is then

$$L_{\text{eq}}(\theta_{\text{eq}}; T) = \mathbb{E}_{t \sim \mathcal{U}[0,1], x \sim r_t} \left[\left\| u_{\theta_{\text{eq}}}(x, t, T) - U_K(x, t; T) \right\|^2 \right], \quad (9)$$

For later reference (and to match the notation used in the appendix), define the temperature-conditioned iEFM flow-matching objective at temperature T as

$$\mathcal{L}_T(\theta_{\text{eq}}) := \mathbb{E}_{t \sim \mathcal{U}[0,1], x \sim r_t} \left[\left\| u_{\theta_{\text{eq}}}(x, t, T) - U_K(x, t; T) \right\|^2 \right], \quad (10)$$

which coincides with L_{eq} in (9) under the same sampling scheme.

where r_t is an off-policy reference distribution. Following iEFM, r_t is formed by sampling x_1 from a replay buffer $\mathcal{B}_{\text{eq}}$ and sampling $x \sim p_t(x | x_1)$.

21.3.3 Using UMA forces for variance reduction (optional but recommended)

The estimator (8) can suffer from weight degeneracy at low temperature because $\exp(-\beta E)$ becomes sharply peaked. Forces from UMA enable gradient-based refinements of the proposal samples $x_1^{(i)}$ toward the *posterior* proportional to $p_t(x | x_1) p_1^{(T)}(x_1)$. Starting from $x_1^{(i,0)} \sim q(\cdot; x, t)$, perform L steps of overdamped Langevin refinement:

$$\begin{aligned} x_1^{(i,\ell+1)} &= x_1^{(i,\ell)} + \\ &+ \epsilon \left(\nabla_{x_1} \log p_t(x | x_1^{(i,\ell)}) - \beta(T) \nabla E_{\text{UMA}}(x_1^{(i,\ell)}) \right) + \\ &+ \sqrt{2\epsilon} \xi^{(i,\ell)}, \end{aligned} \quad (11)$$

with $\xi^{(i,\ell)} \sim \mathcal{N}(0, I)$ and $\nabla E_{\text{UMA}} = -F_{\text{UMA}}$. The Gaussian term $\nabla_{x_1} \log p_t(x | x_1)$ is analytic for the Gaussian conditional paths. In practice, $L \in \{1, 2, 4\}$ is sufficient to reduce variance while maintaining the “simulation-free” spirit (no long trajectories or model rollouts are generated). When refinement is used, $x_1^{(i)}$ in (8) is replaced by $x_1^{(i,L)}$.

21.3.4 Replay buffer and off-policy sampling

As in iEFM, Mora-iFEM maintains a replay buffer $\mathcal{B}_{\text{eq}}$ storing recent samples x_1 generated by integrating the EQ CNF from the base distribution. The EQ update reuses buffer samples for many inner-loop steps, which is crucial for all-atom systems where ODE solves are expensive.

21.3.5 Temperature ladder annealing schedule

Training is conditioned on a discrete ladder

$$\mathcal{T}_{\text{ladder}} = [755, 500, 400, 350, 300]\text{K}.$$

A default *monotone annealing* schedule assigns temperatures by epoch index e (or by steps):

$$T(e) = \begin{cases} 755\text{K}, & e = 0, \\ 500\text{K}, & e = 1, \\ 400\text{K}, & e = 2, \\ 350\text{K}, & e = 3, \\ 300\text{K}, & e \geq 4. \end{cases} \quad (12)$$

This mirrors standard simulated annealing curricula. Implementation-wise, e can be the number of full passes through the endpoint dataset or through the replay buffer. Alternative schedules (e.g. sampling temperatures with a fixed categorical distribution) are straightforward, but (12) is the ablation method described here, with a step based default schedule (one epoch).

21.3.6 EQ diagnostics

Useful monitoring metrics for EQ include:

- L_{eq} in (9).
- **Weight degeneracy:** entropy $H(w)$ and effective sample size $\text{ESS} = (\sum_i w_i)^2 / (\sum_i w_i^2)$.
- **Energy statistics:** running mean/variance of $E_{\text{UMA}}(x_1)$ for generated samples at each T .
- **ODE cost:** number of function evaluations (NFE) per CNF solve and solver divergence rates.

21.4 Objective (2): TPS — endpoint-conditioned path-space sampling with inpainting

21.4.1 Path representation and endpoint constraints

Let $M \in \mathbb{N}$ be the number of path segments and $\mathbf{x} = (x_0, x_1, \dots, x_M) \in \mathbb{R}^{3N(M+1)}$ a discrete trajectory. TPS conditions on fixed endpoints:

$$x_0 = x^{\text{R}}, \quad x_M = x^{\text{P}}.$$

In training, endpoints may be partially masked (Sec. 21.4.4) to enable inpainting; unmasked atoms are treated as *hard constraints*.

21.4.2 A simulation-free path-space Boltzmann density

A simple and practical path distribution is a Boltzmann density on trajectory space:

$$\pi_T(\mathbf{x}) \propto e^{(-\beta(T) \sum_{m=0}^M E_{\text{UMA}}(x_m))} e^{(-\lambda \sum_{m=0}^{M-1} \|x_{m+1} - x_m\|^2)}, \quad (13)$$

restricted to trajectories satisfying the endpoint constraints. The first factor favors low-energy conformations along the path; the second is a smoothness prior (a discretized kinetic regularizer) that prevents unrealistic jumps. Other priors (e.g. torsion-based smoothness) can be added, but (13) is the baseline due to simplicity and differentiability.

Because (13) is an unnormalized density over $\mathbf{x}$, it can be sampled with the same iEFM machinery by treating $\mathbf{x}$ as a single high-dimensional variable and providing an energy functional

$$\mathcal{E}_T(\mathbf{x}) = \beta(T) \sum_{m=0}^M E_{\text{UMA}}(x_m) + \lambda \sum_{m=0}^{M-1} \|x_{m+1} - x_m\|^2, \quad (14)$$

plus hard constraints for endpoints and unmasked atoms. UMA forces provide $\nabla_{x_m} E_{\text{UMA}}(x_m)$ needed for gradient-based diagnostics and optional proposal refinement as in Sec. 21.3.3.

21.4.3 TPS model and training loss

Let $u_{\theta_{\text{tps}}}(\mathbf{x}, t, T, \cdot)$ denote the TPS CNF vector field on trajectory space. The TPS loss mirrors (9):

$$L_{\text{tps}}(\theta_{\text{tps}}; T) = \mathbb{E}_{t, \mathbf{x} \sim \tau_t^{\text{tps}}} \left[\left\| u_{\theta_{\text{tps}}}(\mathbf{x}, t, T) - U_K^{\text{tps}}(\mathbf{x}, t; T) \right\|^2 \right] + \gamma L_{\text{geom}}(\mathbf{x}), \quad (15)$$

where U_K^{tps} is the iEFM estimator computed using the path energy $\mathcal{E}_T(\cdot)$, and L_{geom} is a geometry validity penalty (e.g. bond-length and steric-clash violations) computed on generated trajectories. Importantly, no molecular simulation trajectories are required: training targets are computed from energy evaluations (and optionally forces) on sampled path candidates.

21.4.4 TPS inpainting masks: exactly two modes

TPS inpainting is parameterized by a residue-level mask set $\mathcal{M} \subset \{1, \dots, R\}$, where R is the number of residues. Two and only two masking modes are used:

1. **Residue masking (backbone+side-chain):** for each residue $r \in \mathcal{M}$, mask *all atoms* of residue r across all frames where inpainting is applied. This supports missing loops or segments and forces the model to reconstruct both backbone geometry and side chains.
2. **Side-chain masking (backbone unmasked):** for each residue $r \in \mathcal{M}$, mask only side-chain atoms (including hydrogens if present), while keeping backbone atoms (N, CA, C, O) unmasked. This supports active-site redesign where the backbone is scaffolded but side-chain packing must be generated.

The ligand atoms can be treated as always unmasked constraints in TPS (default) or partially masked in ablations; the primary design goal is enzyme completion or complete *de novo* design, conditioned on ligand endpoints.

21.4.5 Linear masking curriculum over training steps

Let s be the cycle index (Algorithm 1). The masking rate for TPS is a linear schedule:

$$\rho_{\text{tps}}(s) = \rho_{\min} + (\rho_{\max} - \rho_{\min}) \min\left(\frac{s}{S_{\text{ramp}}}, 1\right), \quad (16)$$

$$(\rho_{\min}, \rho_{\max}) = (0.25, 0.50), \quad (17)$$

where $\rho_{\text{tps}}(s)$ denotes the fraction of residues selected into $\mathcal{M}$. The ramp length S_{ramp} is a hyperparameter (e.g. $S_{\text{ramp}} = 50,000$ cycles). This curriculum starts with moderate masking and increases difficulty gradually, which is especially important because TPS is trained without ground-truth trajectories. As an alternative strategy, we reverse the schedule and mask out 50% or residues in the beginning for exploration purposes, and follow the linear schedule to 30%.

21.4.6 TPS diagnostics

Beyond L_{tps} , monitor:

- **Endpoint constraint satisfaction:** RMSD of generated x_0, x_M to the constrained endpoints on unmasked atoms (should be near zero by construction).
- **Path smoothness:** $\sum_m \|x_{m+1} - x_m\|^2$ and its distribution.
- **Energy barrier proxy:** $\max_m E_{\text{UMA}}(x_m) - \min(E_{\text{UMA}}(x_0), E_{\text{UMA}}(x_M))$.
- **Inpainting quality:** geometry violation rates and, when reference structures exist, masked-atom RMSD.

21.5 Objective (3): SEQ — sequence design via categorical discrete flow matching

21.5.1 Sequence as a categorical variable

Let $a \in \{1, \dots, 20\}^R$ denote the amino-acid sequence over R residues (20 standard amino acids). The SEQ model generates a conditioned on structural *embeddings* extracted from the TPS model.

21.5.2 Conditioning on endpoints and mid-trajectory embeddings

Let $\phi_\psi(x)$ be a residue-level embedding (one vector per residue). For each training cycle, draw a mid-trajectory frame index $m^* \sim \text{Unif}([0.3M], [0.7M])$, or m^* drawn from a normal distribution over the interval $[0.3M], [0.7M]$, and compute embeddings

$$h^{\text{R}} = \phi_\psi(x_0), \quad h^{\text{mid}} = \phi_\psi(x_{m^*}).$$

The SEQ model is conditioned on $(h^{\text{R}}, h^{\text{mid}})$. An ablation additionally conditions on $h^{\text{P}} = \phi_\psi(x_M)$ to quantify the incremental value of product-state conditioning.

21.5.3 Categorical discrete flow matching

We adopt categorical discrete flow matching (DFM) as a training principle for inverse folding and sequence generation (**yi2025adflip**). DFM defines a stochastic corruption process over sequences (e.g. masking and resampling) and learns a *flow* that maps noise to data in discrete time. Let

$s_d \in [0, 1]$ denote discrete-flow time (to distinguish it from continuous CNF time t). Let $a^{(s_d)}$ denote a corrupted sequence at time s_d . The model predicts per-position categorical probabilities $p_\theta(\cdot \mid a^{(s_d)}, s_d, h^R, h^{\text{mid}})$, trained to match the DFM target distribution. A practical implementation uses a transformer over residue positions with conditioning via a projection layer into a joint embedding space, or cross-attention to the structural embeddings.

21.5.4 Masking curriculum for SEQ

The discrete corruption process includes a residue masking rate $\rho_{\text{seq}}(s)$ scheduled linearly with cycle index s :

$$\rho_{\text{seq}}(s) = \rho_{\min} + (\rho_{\max} - \rho_{\min}) \min\left(\frac{s}{S_{\text{ramp}}}, 1\right),$$

$$(\rho_{\min}, \rho_{\max}) = (0.25, 0.50), \quad (18)$$

mirroring (17) to keep difficulty aligned across continuous and discrete objectives. Here $\rho_{\text{seq}}(s)$ is the fraction of residues whose amino-acid identities are masked and must be predicted, with the alternative inverse schedule annealing from 50% down to 30%.

21.5.5 SEQ loss and monitoring

Let L_{seq} denote the DFM training loss (typically a cross-entropy between predicted and target categorical distributions at sampled s_d). Recommended monitoring metrics include:

- L_{seq} (DFM loss) and per-residue negative log-likelihood.
- **Sequence recovery:** with the reference sequence a^{ref} obtained from the initial frame constraint file, report $\text{Rec}_{\text{all}} = \frac{1}{R} \sum_r \mathbb{1}[a_r = a_r^{\text{ref}}]$ and $\text{Rec}_{\text{mask}} = \frac{1}{|\mathcal{M}_{\text{seq}}|} \sum_{r \in \mathcal{M}_{\text{seq}}} \mathbb{1}[a_r = a_r^{\text{ref}}]$, where $\mathcal{M}_{\text{seq}}$ is the masked residue set.
- **Diversity/novelty:** pairwise Hamming distance among sampled sequences at fixed conditioning (often called *sequence similarity*).

21.6 Composite loss and optimization

Each training cycle produces a tuple of losses. The composite loss

$$L_{\text{cycle}} = w_{\text{eq}} L_{\text{eq}} + w_{\text{tps}} L_{\text{tps}} + w_{\text{seq}} L_{\text{seq}} \quad (19)$$

is computed *after* running all three sub-objectives and is the only loss used for parameter updates. This design encourages synchronized progress and provides clean diagnostics. Weights $w_{\text{eq}}, w_{\text{tps}}, w_{\text{seq}}$ are treated as hyperparameters and can be tuned to prioritize stability (e.g. up-weighting EQ early to stabilize samplers before intensive TPS/SEQ training).

22 Practical Implementation in the iFEM-1 Codebase

This section describes concrete integration steps for implementing Mora-iFEM in the public iFEM-1 repository (**ifem1code**). The repository is Hydra-based and uses an entrypoint `python dem/train.py experiment=...` (as shown in its README), with separate experiment configurations for different energies and probability paths. Mora-iFEM can be implemented as a new experiment with additional modules for UMA interfacing, endpoint datasets, masking, and cyclic training.

22.1 Code organization and new components

A minimal integration introduces the following components:

UMA wrapper (`dem/energies/uma_oracle.py`). A thin module that loads UMA weights (via `fairchem` and/or `HuggingFace`) and exposes:

$$\begin{aligned} & (E_{\text{UMA}}(x), F_{\text{UMA}}(x)) \\ & = \text{UMAOOracle}(\text{atoms}, \text{positions}, \text{charge}, \text{spin}, \text{task}). \end{aligned} \tag{20}$$

Unit conversions are handled here (Sec. 22.2).

Endpoint constraints (`dem/data/enzyme_endpoints.py`). A dataset class that reads `.cif` endpoint pairs, constructs atom types, residue indices, and ligand masks, and returns aligned pairs $(x^{\text{R}}, x^{\text{P}})$ with metadata.

Masking utilities (`dem/masking/residue_masks.py`). Implements the two TPS masking modes (residue mask and side-chain-only mask) and sequence masking sets. The utilities return boolean masks over atoms (TPS) or residues (SEQ) and can be scheduled by s .

TPS path-space model (`dem/models/path_cnf.py`). Implements a CNF whose state is a concatenation of $M+1$ frames. Endpoint constraints are enforced by masking and clamping coordinates during ODE integration and during the EFM target construction.

SEQ discrete model (`dem/models/seq_dfm.py`). A discrete-flow model (e.g. a transformer) that predicts amino acids from masked sequences conditioned on structural embeddings.

Cyclic trainer (`dem/training/cycle_trainer.py`). Encapsulates Algorithm 1 and maintains separate optimizers or a joint optimizer.

22.2 Numerical details and units

UMA outputs energies in eV and forces in eV/Å (as typical for ML interatomic potentials). Temperatures are specified in Kelvin and k_{B} must be consistent, e.g. $k_{\text{B}} \approx 8.617333262145 \times 10^{-5}$ eV/K. All Boltzmann factors use $\beta(T) = 1/(k_{\text{B}}T)$ in inverse-eV.

For large systems, numerical stability requires:

- force/energy clipping for out-of-distribution conformations (log and reject extreme energies);
- gradient clipping in the optimizer (e.g. norm 1–10 for CNF parameters);
- mixed precision for model networks while keeping UMA inference in a stable mode (as recommended in the UMA paper) (`wood2025uma`).

22.3 Practical pseudocode for each step

Below, high-level pseudocode sketches each objective implementation within the cyclic trainer. Detailed derivations of the EFM target estimator are deferred to the appendix.

Algorithm 2: EQ_Step: temperature-conditioned iEFM update with UMA.

Input: Replay buffer $\mathcal{B}_{\text{eq}}$, temperature T , proposal count K , optional Langevin steps L .

Output: L_{eq} and diagnostics.

▷ **Outer:** add fresh samples to buffer (infrequent) Sample $x_0 \sim p_0$; integrate $\dot{x} = u_{\theta_{\text{eq}}}(x, t, T)$ to obtain x_1 ; push x_1 into $\mathcal{B}_{\text{eq}}$
 ▷ **Inner:** compute simulation-free EFM targets and update Sample $x_1 \sim \mathcal{B}_{\text{eq}}$; sample $t \sim \text{U}[0, 1]$; sample $x \sim p_t(\cdot | x_1)$
 Sample proposals $\{x_1^{(i)}\}_{i=1}^K \sim q(\cdot; x, t)$
if $L > 0$ **then**
 | Refine $\{x_1^{(i)}\}$ via (11) using UMA forces
 Compute weights $w_i \propto \exp(-\beta(T)E_{\text{UMA}}(x_1^{(i)}))$
 Compute $U_K(x, t; T) = \sum_i w_i v_t(x | x_1^{(i)})$
 Return $L_{\text{eq}} = \|u_{\theta_{\text{eq}}}(x, t, T) - U_K(x, t; T)\|^2$ and diagnostics (ESS, weight entropy)

Algorithm 3: TPS_Step: path-space iEFM update with endpoint constraints and in-painting masks.

Input: Endpoint pair $(x^{\text{R}}, x^{\text{P}})$, mask mode, mask rate ρ_{tps} , temperature T .

Output: L_{tps} and diagnostics.

Sample residue mask set $\mathcal{M}$ with fraction ρ_{tps}
 Construct atom-level mask (residue mask or side-chain-only mask)
 Clamp unmasked atoms in endpoints; initialize masked atoms from a prior (e.g. Gaussian around local neighborhoods)
 Form initial trajectory $\mathbf{x}$ by linear interpolation in Cartesian space on masked atoms, with endpoints fixed
 Sample $t \sim \text{U}[0, 1]$ and draw $\mathbf{x}^{(t)} \sim p_t(\cdot | \mathbf{x}_1)$ from the TPS conditional path
 Compute iEFM target $U_K^{\text{tps}}(\mathbf{x}^{(t)}, t; T)$ using path energy $\mathcal{E}_T$ from (14)
 Return $L_{\text{tps}} = \|u_{\theta_{\text{tps}}}(\mathbf{x}^{(t)}, t, T) - U_K^{\text{tps}}(\mathbf{x}^{(t)}, t; T)\|^2 + \gamma L_{\text{geom}}$

23 Monitoring and Diagnostics

Cyclic training produces an interpretable set of losses and diagnostics at every step. This section summarizes recommended logged quantities for stable training and reproducibility.

23.1 Per-objective losses

Log:

$$L_{\text{eq}}, \quad L_{\text{tps}}, \quad L_{\text{seq}}, \quad L_{\text{cycle}}.$$

It is beneficial to also log moving averages and per-temperature slices of L_{eq} and TPS energy statistics.

23.2 Flow-matching diagnostics

For EQ and TPS, log:

Algorithm 4: SEQ_Step: categorical discrete flow matching for sequence design conditioned on structures.

Input: Structural embeddings h^R, h^{mid} (and optional h^P), mask rate ρ_{seq} .

Output: L_{seq} and diagnostics.

Sample a residue mask set $\mathcal{M}_{\text{seq}}$ with fraction ρ_{seq}

Corrupt sequence a by masking residues in $\mathcal{M}_{\text{seq}}$ and applying discrete-time noise

Sample discrete-flow time $s_d \sim \text{U}[0, 1]$

Predict categorical probabilities $p_{\theta_{\text{seq}}}(\cdot \mid a^{(s_d)}, s_d, h^R, h^{\text{mid}})$

Compute DFM loss L_{seq} (cross-entropy to flow-matching targets) and sequence recovery metrics if reference available

- **Estimator health:** ESS and $H(w)$ for iEFM weights.
- **Vector-field scale:** $\mathbb{E} \|u_\theta\|$, $\mathbb{E} \|U_K\|$, and their cosine similarity.
- **ODE solver metrics:** NFE, step rejection count, and final integration error estimates.

23.3 Geometry and chemistry validity

For generated conformations and trajectories, log:

- bond-length and bond-angle violation rates (requires topology);
- steric clash count (distance-based);
- Ramachandran region violations for backbone torsions (optional).

These diagnostics are essential because the method trains without trajectory supervision and can otherwise drift into unphysical regions.

23.4 Sequence diagnostics

Log:

- sequence recovery (overall and masked positions) when a reference sequence exists;
- per-residue perplexity or NLL;
- diversity metrics (entropy per position, average pairwise Hamming distance).

24 Validation and Evaluation Plan

This work does not report empirical results. Instead, this section specifies a rigorous evaluation plan and metrics suitable for enzyme catalysis systems trained with only endpoint structures.

24.1 Equilibrium ensemble validation

Small-system sanity checks. On small peptides or small enzyme fragments where reference sampling is feasible, compare EQ-generated ensembles to long MD or MCMC accelerated by UMA or other learned energy/force-field models with comparable accuracy, using: energy histograms, torsion distributions, TICA plots, and structural RMSD distributions.

ProteinConformers. ProteinConformers provides benchmark tasks and evaluation protocols for protein conformer generation (**proteinconformers2025**). Evaluate generated ensembles (conditioned on a single structure) using the benchmark’s recommended coverage and quality metrics (e.g. RMSD-based diversity/coverage, physical plausibility checks).

Temperature stratification. Because Mora-iFEM is temperature-conditioned, evaluate ensembles separately at each ladder temperature and check for physically plausible trends (e.g. increased flexibility at higher T), with optional additional temperatures not seen during training in the same range as the annealing ladder.

24.2 Transition-path validation

Endpoint satisfaction and committor-like proxies. TPS trajectories must satisfy endpoint constraints exactly on unmasked atoms. Beyond endpoint correctness, evaluate whether intermediate frames interpolate through chemically plausible regions by examining: (i) the energy barrier proxy; (ii) smoothness; and (iii) monotonic progress along chosen reaction coordinates (e.g. distances/angles defining bond formation/breaking).

DynamicPDB. DynamicPDB provides curated MD trajectories for proteins and proposes evaluation protocols for structure-based generative models (**triki2024dynamicpdb**). Use it as an out-of-distribution benchmark to test whether TPS trajectories capture realistic conformational transitions, using the dataset’s metrics (trajectory coverage, conformational diversity, and physical stability proxies).

Inpainting evaluation. For TPS inpainting, create held-out masks on known structures and measure masked-atom dynamics reconstruction error by utilizing the Jensen-Shannon divergence of ENCORE inside of MDAnalysis **mdanalysis2011**, **tiberti2015encore**, and violation rates, separated by masking mode (residue vs side-chain-only) and by mask rate ρ_{tps} .

ColabReaction Transition State-Matching As an additional validation step, we generate transition paths for reactant-product ligand pairs only, leaving the enzyme out, and compare the discovered transition states to those generated by the model utilizing ColabReaction **karasawa2025colabreaction**.

24.3 Sequence design validation

Sequence recovery. When ground-truth sequences are known for endpoint structures, measure sequence recovery at all residues and at masked residues (Sec. 21.5). Report results stratified by whether conditioning uses (h^R, h^{mid}) only or also h^P (ablation).

Structure-conditioned consistency. Sample sequences, build side chains (or run fast relaxation), and evaluate UMA energy and force magnitudes to detect gross incompatibilities. This is not a definitive functional assessment, but it is a useful early filter.

Functional and mechanistic proxies. For select systems, validate designed enzymes using: docking scores for reactant/product ligands, binding affinity from Boltz-TS, QM/MM single-point calculations near the active site, and, where feasible, short MD relaxations to assess stability, accelerated by a learned potential. These validations are computationally heavier but can be applied to a small subset of candidates.

25 Limitations and Discussion

Mora-iFEM makes several approximations that should be explicitly acknowledged.

Energy model mismatch. UMA targets DFT settings across a broad set of tasks ([wood2025uma](#)), but enzyme catalysis may require specialized treatment (e.g. explicit solvent, protonation states, electronic reorganization). The method is only as accurate as the energy backend and the chosen metadata (charge/spin/task), however, conditioning on dynamics with a quantum accurate learned energy and force-field function like UMA is already above and beyond current SOTA models such as RFDiffusion3, that only condition on static structures with no Boltzmann density conditioning, or reactant-bound enzyme complex and product-bound enzyme complex structures as initial and final constraint conditions (forcing the model to learn transition paths and states, rather than conditioning on them).

Path-space density is a modeling choice. The TPS objective uses a tractable path-space Boltzmann density ([13](#)) with a smoothness prior. This is an approximation to true dynamical transition-path ensembles. Nevertheless, it provides a principled simulation-free training target that can be refined (e.g. by more expressive priors or reaction-coordinate constraints) as needed.

Scaling. All-atom enzyme complexes can be large. While UMA is designed for efficient inference on large systems ([wood2025uma](#)), CNF-based flow matching on $\mathbb{R}^{3N}$ remains expensive. Practical training will require careful batching, neighbor-graph caching, and possibly coarse-to-fine strategies (outside the present scope).

26 Conclusion

This paper presented Mora-iFEM, a temperature-conditioned, simulation-free adaptation of iFEM to enzyme catalysis using a learned DFT-accuracy energy and force model. The core idea is a three-step cyclic training objective that unifies equilibrium sampling, endpoint-conditioned transition-path generation with inpainting, and enzyme sequence design via categorical discrete flow matching. The method is designed to operate with minimal supervision (only endpoint structures) and includes detailed monitoring metrics and validation protocols. Mathematical background in the language of differential forms is provided in the appendix, restricted to the special case relevant here.

27 Didactic Interlude: Energies, Boltzmann Densities, and Temperatures

Part II targets distributions of the form

$$p^*(x) \propto \exp(-E(x)),$$

possibly with an explicit temperature parameter. Because the applied sections mix ideas from statistical mechanics, molecular simulation, and generative modeling, it helps to pause and fix the basic conventions.

27.1 Units and the role of $k_B T$

In equilibrium statistical mechanics, one often writes

$$p_T(x) = \frac{1}{Z(T)} \exp\left(-\frac{U(x)}{k_B T}\right),$$

where $U(x)$ is a physical potential energy (e.g. in kcal/mol) and T is an absolute temperature in Kelvin. The normalizing constant

$$Z(T) = \int \exp\left(-\frac{U(x)}{k_B T}\right) dx$$

is the *partition function*; its logarithm encodes thermodynamic free energies.

In machine-learning energy-based modeling, it is common to absorb $k_B T$ into the definition of the energy: set

$$E_T(x) := \frac{U(x)}{k_B T}, \quad \text{so that} \quad p_T(x) \propto e^{-E_T(x)}.$$

In this convention, *changing T rescales the energy landscape*. Lower temperature sharpens modes and increases stiffness of dynamics; higher temperature flattens the landscape and promotes exploration.

Remark 27.1 (Why temperature ladders help). A temperature ladder $\mathcal{T}_{\text{ladder}} = \{T_1 > T_2 > \dots > T_m\}$ provides a principled curriculum: learn to sample a broad, high-temperature distribution first, then gradually refine to the sharper low-temperature target. In Part II this ladder is used not only to help mixing, but also to stabilize training of the flow.

27.2 Marginals, conditionals, and reaction coordinates

In enzyme design one often cares about *functionals* of the conformation x :

$$\xi(x) \in \mathbb{R},$$

for example a reaction coordinate or an active-site distance. Even if the full density $p_T(x)$ is hard to sample, one may want:

- the marginal density of ξ ,
- the conditional density of x given $\xi(x) \approx \xi_0$,
- or *transition paths* that move from one metastable region to another.

Normalizing-flow models are attractive here because once a flow is trained, it can generate large numbers of independent samples and can be conditioned (with additional machinery) on constraints or observables.

28 Deterministic Transport vs Diffusion: Continuity and Fokker–Planck

Part II focuses on deterministic flows (ODEs) that transport a base density to a target. In molecular simulation, however, equilibrium sampling is frequently performed using stochastic dynamics (e.g. Langevin). It is therefore useful to see the PDEs side-by-side.

28.1 Deterministic transport

For a deterministic flow $\dot{x}_t = v_t(x_t)$, the density evolves by the continuity equation

$$\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0.$$

In differential-forms language this is $\partial_t(\rho_t \nu) + \mathcal{L}_{v_t}(\rho_t \nu) = 0$ (Appendix A).

28.2 Overdamped Langevin and the Fokker–Planck equation

The overdamped Langevin SDE is

$$dX_t = -\nabla U(X_t) dt + \sqrt{2k_B T} dW_t,$$

where W_t is standard Brownian motion. The density ρ_t of X_t solves the Fokker–Planck equation

$$\partial_t \rho_t = \operatorname{div}(\rho_t \nabla U) + k_B T \Delta \rho_t.$$

The first term looks like deterministic transport with velocity $v = -\nabla U$, and the second is a diffusion term.

Remark 28.1 (Why this matters for generative modeling). Diffusion models can be interpreted as (time-reversed) solutions to SDEs with a diffusion term. Flow models (CNFs) remove the diffusion term and instead represent transformations by invertible maps. Both paradigms are governed by transport PDEs; the difference is whether diffusion is present.

28.3 A geometric aside: diffusion and the Laplacian

On a Riemannian manifold, the Laplacian Δ can be defined intrinsically. For functions it coincides with $\operatorname{div} \circ \operatorname{grad}$. For differential forms, one uses the Hodge Laplacian $\Delta = d\delta + \delta d$. Thus the Fokker–Planck equation on a manifold can be phrased in terms of divergence and intrinsic Laplacians.

29 A Toy Example: Flow Matching for a One-Dimensional Double Well

To make the flow-matching objective concrete, consider a 1D energy function with two metastable basins:

$$U(x) = x^4 - 2x^2.$$

At temperature T , the target density is

$$p_T(x) \propto \exp\left(-\frac{x^4 - 2x^2}{k_B T}\right).$$

For small T this distribution has two sharp modes near $x = \pm 1$.

29.1 Why sampling is hard at low temperature

The energy barrier at $x = 0$ suppresses transitions between wells. A local sampler (e.g. Langevin with small noise) can mix slowly: it may stay in one well for a long time before crossing.

A flow model attempts something different: learn a deterministic map that *directly* transforms a simple base distribution into p_T . If the map is expressive enough, it can move probability mass into both wells without requiring rare barrier crossings during sampling.

29.2 A simple flow matching path

A basic flow-matching construction chooses a random endpoint pair (x_0, x_1) where x_0 comes from a base distribution (say a standard normal) and x_1 comes from the target. For each pair, define an interpolation

$$x_t = (1 - t)x_0 + tx_1, \quad t \in [0, 1],$$

and a “ground truth” velocity $u_t(x_t | x_0, x_1) = x_1 - x_0$. Flow matching trains $v_\theta(t, x)$ to match the conditional expectation of u_t given x_t . The resulting v_θ drives a continuity equation whose marginals are (ideally) the desired ρ_t .

Remark 29.1 (Energy-based setting). In the energy-based setting, we do not have direct samples $x_1 \sim p_T$. Instead, iEFM uses the current model and energy gradients to create effective training signals that move the model toward the energy-based target. The toy example is still useful as intuition: we are learning a family of velocities that push mass into low-energy regions.

30 Practical Notes for Neural ODE Flows in Scientific Settings

This section collects pragmatic considerations that frequently arise when implementing CNFs in high-dimensional scientific problems (such as molecular conformations).

30.1 Divergence estimation

The instantaneous change-of-variables formula requires $\text{div } v_t(x)$. For a neural network vector field $v_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^d$, the divergence is the trace of the Jacobian:

$$\text{div } v_\theta(x) = \text{tr}(dv_\theta(x)).$$

When d is large, computing the full Jacobian is expensive. A common alternative is Hutchinson’s trace estimator:

$$\text{tr}(J) = \mathbb{E}_\varepsilon[\varepsilon^\top J \varepsilon]$$

for random vectors ε with $\mathbb{E}[\varepsilon\varepsilon^\top] = I$. This trades exactness for stochastic approximation and is often sufficient for training and evaluation.

30.2 Stiffness and temperature schedules

Low-temperature targets can create stiffness because the learned vector field must vary rapidly near sharp modes. Symptoms include:

- unstable ODE integration (large local truncation errors),
- exploding divergence estimates,
- or training collapse to a single mode.

Temperature conditioning and ladder schedules address this by gradually increasing sharpness as the model becomes capable of representing it.

30.3 Time reparameterization

Because the ODE is autonomous in the augmented time variable, one can reparameterize time to allocate more “integration budget” to difficult regions. Concretely, one can train a model in a rescaled time variable $\tau = \sigma(t)$ where σ is monotone and concentrates steps near $t = 1$ (or elsewhere). This does not change the expressivity of the flow, but can improve numerical stability.

30.4 Geometry-aware evaluation

For molecular conformations, naive Euclidean metrics in $\mathbb{R}^{3N}$ can be misleading. It is often important to evaluate samples using symmetry-aware metrics (e.g. RMSD modulo rotation) and to enforce physical constraints (bond lengths, clashes). Equivariance (Appendix B) helps, but it does not automatically enforce all constraints.

31 Discrete Sequence Modeling and Discrete Flow Matching

Part II ultimately couples conformational sampling (continuous variables) with *sequence* design (discrete variables). Even if a project only uses discrete modeling as a downstream module, it helps to have a minimal dictionary for what “flow” means on a discrete space.

31.1 Sequences as points in a finite product space

An amino-acid sequence of length L can be represented as

$$s = (s_1, \dots, s_L) \in \mathcal{A}^L,$$

where $\mathcal{A}$ is a finite alphabet (typically 20 standard amino acids, sometimes augmented with special tokens). A probability model is a distribution $p(s)$ on this finite set.

31.2 Continuous-time Markov chains and the master equation

One way to build an expressive sequence model is via a continuous-time Markov chain (CTMC) on $\mathcal{A}^L$. A time-dependent generator $Q_t(s \rightarrow s')$ specifies infinitesimal transition rates. If p_t is the distribution over sequences at time t , it evolves by the *master equation*

$$\partial_t p_t(s) = \sum_{s' \neq s} p_t(s') Q_t(s' \rightarrow s) - p_t(s) \sum_{s' \neq s} Q_t(s \rightarrow s').$$

This equation plays the role, for discrete spaces, that the continuity equation plays in continuous spaces: it describes conservation of total probability mass under a flow of probability between states.

31.3 Discrete flow matching in one paragraph

Discrete flow matching methods (see, e.g., **yi2025adflip**) propose supervised objectives that learn the generator (or related parameterization) so that the resulting CTMC transports a simple base distribution to a target distribution over sequences. Conceptually, the training signal is built from conditional transition statistics along a chosen interpolation scheme, much like continuous flow matching learns a velocity field from conditional velocities along an interpolation path.

Remark 31.1 (Geometry behind the scenes). Even though the state space is discrete, the distribution p_t lives in a probability simplex, which is a (stratified) manifold with boundary. The master equation is an ODE on this simplex. Thus, at a structural level, discrete and continuous “flows” both amount to choosing dynamics for probability distributions.

31.4 Coupling sequences and conformations

In enzyme design, the energy landscape depends on both conformation x and sequence s . A coupled generative pipeline can be organized in multiple ways:

- **Conformation-first:** sample x (path or conformer) under a fixed sequence, then propose sequence edits conditioned on x .
- **Sequence-first:** propose sequences and then sample conformers for each candidate.
- **Joint modeling:** define a joint energy $E(x, s)$ and attempt to sample jointly (harder, but conceptually clean).

Part II follows a staged strategy, which is often more tractable in practice.

32 Glossary and Notational Concordance

This final section provides a quick translation guide between the geometric language of Part I and the modeling language of Part II.

32.1 Geometric objects

- **Top form / density:** $\omega = \rho \nu \in \Omega^n(M)$ represents a density ρ relative to a volume form ν .
- **Transport:** a diffeomorphism F transports densities by pushforward; integrals transform by pullback.
- **Continuity equation:** $\partial_t \omega_t + \mathcal{L}_{v_t} \omega_t = 0$ is the coordinate-free statement that ω_t is transported by the flow of v_t .

32.2 Modeling objects

- **CNF:** an ODE flow $\dot{x}_t = v_\theta(t, x_t)$ whose time-1 map pushes a base density to a target density.
- **EBM:** a target density specified up to a normalizing constant, $p^*(x) \propto e^{-E(x)}$.
- **Flow matching:** a supervised objective for learning v_θ by matching conditional velocities that realize a chosen marginal path.
- **iEFM:** an energy-based variant in which the training signal is constructed using the energy E and samples from the current model.

32.3 Acronyms and quick reference

Acronym	Meaning / comment
CNF	Continuous normalizing flow (invertible density model defined by an ODE flow).
EBM	Energy-based model: $p(x) \propto e^{-E(x)}$.

(continued on next page)

Acronym	Meaning / comment
iEFM	Iterated energy-based flow matching (energy-based training of flow models).
UMA	“Universal model for atoms” potential (learned energy / force field).
ODE	Ordinary differential equation; used to define deterministic flows.
SDE	Stochastic differential equation; used in Langevin and diffusion modeling.
TPS	Transition path sampling (MCMC sampling of path ensembles).
EQ / SEQ	Abbreviations used in Part II for staged training components (see that part’s notation).
$E(3)$ / $SE(3)$	Euclidean / special Euclidean group; encode rigid-motion symmetries.
RMSD	Root-mean-square deviation (typically after optimal rigid alignment).
MCMC	Markov chain Monte Carlo; a general class of sampling algorithms.
OT	Optimal transport; leads to the Benamou–Brenier formulation.
KL	Kullback–Leibler divergence, $KL(q p)$.
Fokker–Planck	PDE governing densities of diffusion processes (e.g. Langevin).
Hutchinson	Stochastic trace estimator used for divergence computations in high dimension.
CTMC	Continuous-time Markov chain (discrete-space analog of a flow).
Master equation	ODE for probabilities in a CTMC (discrete analog of continuity equation).
Score	Gradient of log-density, $\nabla \log p(x)$; for EBMs it is $-\nabla E(x)$.

Exercise 32.1 (Connecting the languages). Let $M = \mathbb{R}^d$ with $\nu = dx$. Show directly that the differential-forms continuity equation $\partial_t(\rho_t \nu) + \mathcal{L}_{v_t}(\rho_t \nu) = 0$ is equivalent to $\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0$. Then derive the instantaneous change-of-variables formula along characteristics.

33 Further Reading and Conceptual Connections

This unified document deliberately spans two communities: classical differential topology and modern generative modeling. A few conceptual “bridges” are worth keeping in mind as you continue beyond these notes.

33.1 Topology and invariance

Part I emphasizes that many constructions are coordinate-free and functorial (pullbacks, Lie derivatives, Stokes’ theorem, cohomology). In Part II, symmetry constraints (e.g. $SE(3)$ equivariance) play a similar role: they enforce that the learned model respects transformations that should not change the underlying object of interest. In both cases, the guiding principle is that *a good model should not depend on arbitrary choices of coordinates*.

33.2 Transport PDEs as the common language

Whether one works with deterministic flows (CNFs), stochastic flows (diffusions), or discrete Markov chains (sequence models), the evolution of probability is governed by a conservation law:

- continuity equation for ODE-driven transport,
- Fokker–Planck for SDE-driven transport,

- master equations for CTMCs on discrete spaces.

The mathematical structure in Part I (Lie derivatives, divergence, Stokes) is exactly what makes these conservation laws coordinate-free on manifolds.

33.3 Energy landscapes and geometry

In molecular problems, the energy function is often the best available global summary of “physical plausibility.” But an energy landscape is also a geometric object: its gradient, Hessian, and level sets encode local shape. From a differential-topological perspective, critical points and their indices are the first invariants one studies. From a generative-modeling perspective, these same quantities determine stiffness, mixing, and mode structure. Thus even classical tools like Morse theory and transversality have direct relevance when reasoning about learnability and sampling difficulty.

33.4 If you want to go deeper

Depending on your interests, natural next steps include:

- **Geometric measure theory / currents:** for treating “densities” and “surfaces” beyond smooth forms.
- **Optimal transport and Wasserstein geometry:** for variational interpretations of flows and diffusion.
- **Global analysis:** for full proofs of Hodge theory and elliptic regularity on manifolds.
- **Equivariant deep learning:** for architectural guarantees of symmetry in scientific ML.

The references cited throughout Part I provide standard entry points on the differential-topological side, while the citations in Part II provide modern ML context.

Part III

Appendices

A Mathematical Appendix: Probability Transport in the Language of Differential Forms

This appendix summarizes (and slightly extends) the background theory that connects *(i)* densities as top forms, *(ii)* transport by diffeomorphisms/flows, and *(iii)* the PDE/ODE identities used in continuous normalizing flows and flow matching. Part I develops the underlying manifold and differential-forms machinery in a more general setting; here we record the special case most relevant for Part II.

A.1 Top forms as densities

Let M be an oriented smooth n -manifold. Fix a volume form $\nu \in \Omega^n(M)$ (nonvanishing everywhere). Every top form can be written as $\omega = \rho \nu$ for a unique smooth function ρ .

Definition A.1 (Probability density form). A *probability density form* on M (relative to ν) is a top form $\omega = \rho \nu$ with $\rho \geq 0$ and $\int_M \rho \nu = 1$.

In local coordinates on $\mathbb{R}^n$, $\nu = dx^1 \wedge \cdots \wedge dx^n$ and $\omega = \rho(x) dx$.

A.2 Pullback, pushforward, and Jacobians

Let $F : M \rightarrow M$ be an orientation-preserving diffeomorphism. The pullback of a top form satisfies

$$F^*(\rho \nu) = (\rho \circ F) F^* \nu.$$

In coordinates, $F^* \nu = \det(dF) \nu$ (for the standard volume form), so

$$F^*(\rho \nu) = (\rho \circ F) \det(dF) \nu.$$

Equivalently, writing measures, the pushforward $(F)_*$ on densities corresponds to the usual change-of-variables formula: if $Y = F(X)$ and X has density ρ_X then

$$\rho_Y(y) = \rho_X(F^{-1}(y)) \left| \det(dF_y^{-1}) \right|.$$

Remark A.2 (Why pullback is often cleaner). Instead of memorizing the pushforward Jacobian factor, one can remember the universal identity

$$\int_{F(U)} \omega = \int_U F^* \omega,$$

which is exactly the differential-forms version of substitution in multivariable calculus.

A.3 Lie derivative and the continuity equation

Let v_t be a time-dependent vector field on M with flow Φ_t . Given a time-dependent top form ω_t , the transport equation is

$$\partial_t \omega_t + \mathcal{L}_{v_t} \omega_t = 0.$$

Cartan's formula gives $\mathcal{L}_v = d \iota_v + \iota_v d$. Since $d\omega_t = 0$ for top forms,

$$\mathcal{L}_{v_t} \omega_t = d \iota_{v_t} \omega_t.$$

Thus the continuity equation can also be written as

$$\partial_t \omega_t + d(\iota_{v_t} \omega_t) = 0.$$

On $\mathbb{R}^n$, writing $\omega_t = \rho_t dx$ and $v_t = \sum_i v_t^i \partial_{x^i}$, one computes

$$\iota_{v_t}(\rho_t dx^1 \wedge \cdots \wedge dx^n) = \rho_t \sum_{i=1}^n (-1)^{i-1} v_t^i dx^1 \wedge \cdots \widehat{dx^i} \cdots \wedge dx^n,$$

and then $d(\iota_{v_t} \omega_t) = \operatorname{div}(\rho_t v_t) dx$. Hence

$$\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0.$$

A.4 ODE flows and the instantaneous change-of-variables identity

Assume $M = \mathbb{R}^d$. Let Φ_t be the flow of $\dot{x}_t = v_t(x_t)$ and let ρ_t solve the continuity equation with initial density ρ_0 . Then $\rho_t = (\Phi_t)_* \rho_0$. Along a trajectory $x_t = \Phi_t(x_0)$, the density satisfies

$$\frac{d}{dt} \log \rho_t(x_t) = -\operatorname{div} v_t(x_t).$$

One standard derivation. Let $J_t = \det(d\Phi_t)$. Differentiating $d\Phi_t$ in t and using Jacobi’s formula yields

$$\frac{d}{dt} \log J_t(x_0) = \operatorname{div} v_t(\Phi_t(x_0)).$$

Because $\rho_t(\Phi_t(x_0)) J_t(x_0) = \rho_0(x_0)$ (conservation of mass), we get

$$\frac{d}{dt} \log \rho_t(\Phi_t(x_0)) = -\frac{d}{dt} \log J_t(x_0) = -\operatorname{div} v_t(x_t).$$

□

This identity is the computational backbone of CNFs: it reduces log-density tracking to evaluating a divergence (often via Hutchinson trace estimators in high dimension).

A.5 Energy-based targets and temperature scaling

Energy-based modeling specifies an unnormalized density

$$\tilde{\rho}(x) = \exp(-E(x))$$

and the normalized density $\rho(x) = \tilde{\rho}(x)/Z$ where $Z = \int \exp(-E)$ is unknown. In physics, E is often an energy in units of $k_B T$, so an explicit temperature parameter appears:

$$\rho_T(x) \propto \exp\left(-\frac{E(x)}{k_B T}\right).$$

Two remarks are useful:

- (1) Lower temperatures concentrate mass near low-energy basins, often making sampling harder.
- (2) A *temperature ladder* $\mathcal{T}_{\text{ladder}}$ can be used to stage learning: start from high T (easier exploration) and gradually decrease.

This is exactly the motivation for the temperature conditioning used in Part II.

A.6 Specialization to molecular conformations

In the enzyme setting, a “state” x is typically a concatenation of atomic coordinates in $\mathbb{R}^{3N}$. Two geometric subtleties matter:

- (a) **Rigid motions.** Physical energies are invariant under global translations and rotations; one often works on a quotient by $SE(3)$ or uses an equivariant architecture so that the learned vector field respects the symmetry.
- (b) **Constraints.** Bond lengths/angles may impose a lower-dimensional constraint manifold inside $\mathbb{R}^{3N}$; in that case the “correct” density lives on that manifold and is naturally expressed as a top form there.

For the purposes of the methods described in Part II, the flow is implemented in ambient coordinates with an $E(3)$ -equivariant network, which is compatible with treating the target as a density on $\mathbb{R}^{3N}$ that is constant along rigid-motion orbits.

Remark A.3 (A conceptual map). If one wanted to be completely intrinsic, one could:

- model the state space as a manifold M (e.g. a quotient or constraint manifold),
- choose a volume form ν (often induced by a Riemannian metric),
- define the target density as a top form $\rho \nu$,
- and learn a vector field v_t on M whose flow transports a base density to the target.

Part I provides the geometric background for this intrinsic viewpoint.

B Equivariance and Symmetry Considerations

Part II relies on the fact that molecular energies are (to excellent approximation) invariant under global Euclidean motions: translating or rotating an entire molecule in space does not change its internal energy. When learning a vector field that transports densities on coordinate space $\mathbb{R}^{3N}$, it is therefore natural to impose the same symmetry on the model.

This appendix records the basic definitions and a few consequences that are useful when thinking about flow models on molecular conformations.

B.1 Group actions, invariance, and equivariance

Let G be a Lie group acting smoothly on a manifold M . We write the action as $(g, x) \mapsto g \cdot x$. A function $f : M \rightarrow \mathbb{R}$ is *G-invariant* if

$$f(g \cdot x) = f(x) \quad \text{for all } g \in G, x \in M.$$

A map $F : M \rightarrow M$ is *G-equivariant* if

$$F(g \cdot x) = g \cdot F(x) \quad \text{for all } g \in G, x \in M.$$

A time-dependent vector field v_t on M is *G-equivariant* if

$$(d(g \cdot))_x v_t(x) = v_t(g \cdot x),$$

i.e. pushing forward the vector by the group action equals evaluating the field at the transformed point.

Remark B.1 (Flows preserve symmetry). If v_t is *G-equivariant* and Φ_t is its flow, then each Φ_t is *G-equivariant*. This follows by uniqueness of solutions to ODEs: both $\Phi_t(g \cdot x)$ and $g \cdot \Phi_t(x)$ solve the same initial value problem.

B.2 Euclidean symmetry in molecular coordinate spaces

For a molecule with N atoms, a conformation can be represented as

$$x = (r_1, \dots, r_N) \in (\mathbb{R}^3)^N \cong \mathbb{R}^{3N}.$$

The Euclidean group $E(3)$ acts by

$$(R, t) \cdot (r_1, \dots, r_N) = (Rr_1 + t, \dots, Rr_N + t),$$

where $R \in O(3)$ is an orthogonal matrix (rotation or reflection) and $t \in \mathbb{R}^3$ is a translation. The subgroup $SE(3)$ consists of *proper* rotations ($R \in SO(3)$) and translations.

Physical potential energies $E(x)$ are typically invariant under $SE(3)$, and often also invariant under reflection if chirality is not modeled explicitly. In practice:

- If the data and energy model treat mirror images as physically equivalent, $E(3)$ symmetry is appropriate.
- If chirality is essential (as in proteins, which are overwhelmingly built from L-amino acids), then one should be cautious: enforcing full $E(3)$ -equivariance may implicitly identify a configuration with its mirror image, which can be undesirable.

B.3 Gradient fields and symmetry

Suppose $E : M \rightarrow \mathbb{R}$ is G -invariant and M has a G -invariant Riemannian metric. Then the gradient vector field ∇E is G -equivariant. Intuitively, invariance of E means its level sets are preserved by the action; a G -invariant metric ensures that the notion of “steepest ascent” behaves well under the symmetry.

This observation is one reason energy-based modeling pairs naturally with equivariant architectures: if the target density is $\propto e^{-E}$, then many of the “natural” vector fields derived from E inherit the symmetry automatically.

B.4 Quotients and “intrinsic” state spaces

If the density and dynamics are invariant under a group action, it can be conceptually cleaner to work on the quotient space M/G . For example, removing global translations and rotations suggests working modulo $SE(3)$. However:

- the quotient may have singularities (e.g. when points have nontrivial stabilizers),
- implementing flows directly on the quotient can be technically involved,
- and in practice equivariant neural ODEs in ambient coordinates often perform well without explicitly constructing the quotient.

The ambient-coordinate approach can be interpreted as learning a G -equivariant vector field on M so that the induced flow descends to the quotient.

B.5 A practical checklist for equivariant flow models

When reading or implementing a model such as the one in Part II, it is useful to check:

- (1) **Energy symmetry:** is the target energy (or learned potential) invariant under translations/rotations (and reflections)?
- (2) **Architecture:** does the network guarantee equivariance of the velocity field under the same group?
- (3) **Training data:** are coordinates pre-aligned/centered, or is equivariance expected to handle it?
- (4) **Evaluation:** are metrics computed in a symmetry-invariant way (e.g. RMSD modulo rotations)?

If these are mismatched, one can accidentally learn spurious coordinate artifacts (for example, a model that behaves differently after a global rotation).

C Molecular Geometry Primer for the Applied Case Study

This appendix is a short primer on geometric conventions that repeatedly appear in Part II. It is not intended as a comprehensive introduction to structural biology; the goal is simply to make the modeling choices (state representation, symmetry, and evaluation) easy to parse on a first read.

C.1 Coordinates and degrees of freedom

A protein (or enzyme) conformation can be represented at different resolutions:

- **All-atom:** each atom has a position $r_i \in \mathbb{R}^3$, giving a state $x \in \mathbb{R}^{3N}$.
- **Backbone-only:** keep only the peptide backbone atoms (e.g. N, C $_{\alpha}$, C), reducing N .
- **Coarse-grained:** represent each residue by one or a few pseudo-atoms (e.g. C $_{\alpha}$ only).

The case study uses all-atom information for the reactive region and treats the state as an element of $\mathbb{R}^{3N}$.

Not all degrees of freedom are physically independent. Bond lengths and bond angles are relatively stiff, so a large fraction of physically relevant variability is in *dihedral* (torsion) angles. From a geometric viewpoint, this means that the physically relevant conformations often lie near a lower-dimensional submanifold of $\mathbb{R}^{3N}$. Flows that operate in ambient coordinates can still work well, but one should keep this structure in mind when interpreting results.

C.2 Rigid-motion symmetry and RMSD

If two conformations differ only by a global translation and rotation, they represent the same physical internal geometry. This motivates two standard practices:

- (1) build $E(3)$ - or $SE(3)$ -equivariant models (Appendix B),
- (2) evaluate closeness using metrics that quotient out rigid motions.

The most common such metric is the *root-mean-square deviation* (RMSD). Given two point clouds $x = (r_1, \dots, r_N)$ and $y = (s_1, \dots, s_N)$ in $\mathbb{R}^{3N}$, define the RMSD after optimal alignment by

$$\text{RMSD}(x, y) = \min_{R \in SO(3), t \in \mathbb{R}^3} \left(\frac{1}{N} \sum_{i=1}^N \|Rr_i + t - s_i\|^2 \right)^{1/2}.$$

The minimization over R can be solved in closed form by the *Kabsch algorithm*, which reduces to an SVD of a 3×3 covariance matrix. In practice, toolkits such as MDAnalysis ([mdanalysis2011](#)) provide implementations.

Remark C.1 (Why RMSD can be misleading). RMSD is sensitive to outliers and mixes all atoms into a single number. For flexible proteins, two conformations can have large RMSD even if the active site geometry is nearly identical. Conversely, small RMSD does not guarantee functional similarity. For enzyme design, it is common to measure RMSD only on a subset (e.g. active-site residues) and to add chemically motivated metrics.

C.3 Internal coordinates: dihedral angles

For a polymer-like backbone, conformations are often described by dihedral angles:

- ϕ (around the N-C $_{\alpha}$ bond),
- ψ (around the C $_{\alpha}$ -C bond),
- and ω (around the peptide bond, usually near π due to planarity).

The distribution of (ϕ, ψ) pairs is summarized by a Ramachandran plot; physically plausible conformations concentrate in specific regions due to steric constraints.

From a modeling standpoint, internal coordinates suggest an alternative state space: a product of circles (angles) together with a few bond-length/angle parameters. This is closer to a manifold (e.g. $(S^1)^k$) than to $\mathbb{R}^{3N}$. Nevertheless, many modern equivariant architectures operate directly in Cartesian coordinates because they handle symmetries and local geometry effectively.

C.4 Side chains, rotamers, and sequence dependence

Side chains introduce additional dihedral angles (often denoted $\chi_1, \chi_2, \dots$). In protein structures, these angles concentrate near discrete preferred values called *rotamers*. Sequence design problems are difficult in part because:

- changing a residue changes the chemical identity and thus the energy function,
- and the accessible rotamers and steric constraints can change dramatically.

The sequence-design stage in Part II interfaces the learned conformational model with a discrete generative model over sequences (see also **yi2025adflip** for a discrete flow-matching approach).

C.5 Paths and reaction coordinates

A *path* is a time-indexed curve $x(t) \in \mathbb{R}^{3N}$. In catalysis and enzyme mechanism studies, one often cares about paths that connect metastable regions (reactant to product). Two related viewpoints:

- **Reaction coordinate:** choose a low-dimensional observable $\xi(x)$ and study dynamics in ξ .
- **Transition path ensemble:** consider distributions over full paths conditioned on endpoints and (sometimes) on crossing events.

Transition path sampling (TPS) methods generate such ensembles by MCMC in path space. The case study in Part II aims to generate plausible transition paths *without* running expensive simulations, by learning a generative model of paths and conformations informed by an energy function.

C.6 Evaluating generated conformers and paths

Common evaluation questions include:

- (a) **Energy distribution:** do generated samples concentrate in low-energy regions under the target potential?
- (b) **Diversity:** do samples cover multiple basins rather than collapsing to one?

- (c) **Novelty vs training set:** are generated structures distinct from training examples (up to symmetry)?
- (d) **Functional metrics:** are active-site geometries and catalytic features preserved or improved?
- (e) **Path realism:** do generated paths avoid unphysical clashes and preserve connectivity constraints?

Toolkits such as `MDAnalysis` and its ensemble comparison module (e.g. `ENCORE` ([tiberti2015encore](#))) are frequently used to quantify diversity and clustering of conformational ensembles.

Remark C.2 (Geometry and learning objectives). In high-dimensional spaces, it is easy for a generative model to achieve good values on one metric (say, energy) while failing on another (say, physical plausibility). A recurring theme in scientific generative modeling is therefore to combine:

- *symmetry* (equivariance/invariance),
- *physics* (energies and constraints),
- and *data* (empirical distributions)

into a single training loop. Part II can be read as one concrete instance of this general recipe.

D Additional Notes on Iterated Energy-Based Flow Matching

This appendix is a didactic supplement to the research-style presentation in Section 18. The goal is to unpack, at a conceptual level, *why* energy-based flow matching is a reasonable strategy for sampling from Boltzmann densities, and how the “iterated” aspect fits into an optimization loop.

D.1 The basic inverse problem

We would like to sample from a target density

$$p^*(x) = \frac{1}{Z} \exp(-E(x)),$$

where $E : \mathbb{R}^d \rightarrow \mathbb{R}$ is an energy and the normalizer Z is unknown. In many scientific settings, evaluating $E(x)$ and its gradient $\nabla E(x)$ is feasible (or approximated by a learned surrogate), but direct sampling is expensive.

A normalizing flow provides a parameterized family of densities:

$$q_\theta = (F_\theta)_* p_0,$$

where p_0 is a simple base density and F_θ is an invertible map (for CNFs, F_θ is a time-1 flow map of an ODE). Sampling is easy because we can sample $x_0 \sim p_0$ and set $x = F_\theta(x_0)$.

The learning problem is then to choose θ so that $q_\theta \approx p^*$.

D.2 Reverse KL and why gradients of E help

A natural objective is the reverse KL divergence

$$\text{KL}(q_\theta \| p^\star) = \int q_\theta(x) \log \frac{q_\theta(x)}{p^\star(x)} dx.$$

Because $\log p^\star(x) = -E(x) - \log Z$, we can rewrite

$$\text{KL}(q_\theta \| p^\star) = \mathbb{E}_{x \sim q_\theta}[\log q_\theta(x) + E(x)] + \log Z.$$

The constant $\log Z$ is irrelevant for optimization in θ . Thus, *if* we can evaluate $\log q_\theta(x)$ for samples $x \sim q_\theta$, we can in principle minimize $\text{KL}(q_\theta \| p^\star)$ without knowing Z .

CNFs can evaluate $\log q_\theta(x)$ through the divergence integral (Appendix A). So why not just do this directly? In practice, high-dimensional optimization of reverse KL can be numerically delicate, and flow-matching objectives offer alternative training signals that behave better empirically.

A key structural fact is that gradients of $\log p^\star$ are available without Z :

$$\nabla \log p^\star(x) = -\nabla E(x).$$

This is the same “score” quantity that appears in score matching and diffusion models.

D.3 Flow matching as a way to learn velocities

Flow matching (Section 17 and [lipman2022flowmatching](#)) can be summarized as:

- Choose a family of intermediate marginals $\{\rho_t\}_{t \in [0,1]}$ connecting a base distribution to a target.
- Construct conditional velocities $u_t(\cdot \mid \text{context})$ whose average equals the true velocity field that realizes the marginals.
- Train a model velocity $v_\theta(t, x)$ by supervised regression to u_t .

If training succeeds, integrating the learned ODE $\dot{x}_t = v_\theta(t, x_t)$ transports the base to (approximately) the desired target.

D.4 What changes in the energy-based setting

In the standard supervised flow-matching setting, one assumes access to target samples (or paired endpoint samples) to generate the conditional velocities. In energy-based problems, *we do not have target samples*. Instead, we can:

- sample from the current model q_θ ,
- evaluate energies (and often energy gradients) at those samples,
- and use these evaluations to craft training signals that move q_θ toward lower energy.

Energy-based flow matching can be viewed as using the energy model to define a *pseudo-target drift* that resembles the score of $p^\star$. Since $\nabla \log p^\star = -\nabla E$, gradient information points in the direction of increasing probability mass.

D.5 Why “iterated”?

The word *iterated* reflects an outer loop:

$$q_{\theta_0} \longrightarrow q_{\theta_1} \longrightarrow \cdots \longrightarrow q_{\theta_K},$$

where each iteration uses the current model’s samples as a training distribution. Intuitively:

- (1) Early in training, q_{θ_k} may be far from p^* .
- (2) One uses energy evaluations to define a corrective velocity field (or a proxy for it) that nudges q_{θ_k} toward lower-energy regions.
- (3) After updating parameters, the new model $q_{\theta_{k+1}}$ samples a different region of state space, enabling a new corrective step.

In this sense iEFM is “off-policy”: it learns from its own rollouts.

D.6 Temperature conditioning as curriculum and regularization

For $p_T(x) \propto \exp(-U(x)/(k_B T))$, lower T concentrates mass and makes learning harder. A temperature ladder $\mathcal{T}_{\text{ladder}}$ serves two purposes:

- **Curriculum:** learn broad distributions first (high T) to discover relevant basins.
- **Regularization:** avoid early collapse by preventing the model from chasing extremely sharp minima before it has enough capacity.

In flow models, temperature conditioning can be implemented by conditioning the velocity network on T (or on $\beta = 1/(k_B T)$), as done in Part II.

D.7 Pseudo-code for an “outer loop” iEFM training pattern

The high-level pattern below is intentionally abstract (and does not commit to the specific engineering details in Part II), but it captures the structure of an iterated energy-based flow-matching loop.

Algorithm 5: Schematic iterated energy-based flow matching loop

Input: Base distribution p_0 ; energy E (and optionally ∇E); temperature ladder $\mathcal{T}_{\text{ladder}}$

Init: Initialize parameters θ for velocity field $v_\theta(t, x; T)$

for T in $\mathcal{T}_{\text{ladder}}$ (*high to low*) **do**

for *outer iteration* $k = 1, \dots, K$ **do**

Sample a minibatch $x \sim q_\theta(\cdot; T)$ by integrating the ODE flow from p_0 ;

Evaluate energy information: $E(x)$ (and possibly $\nabla E(x)$);

Construct a target conditional velocity signal $u_t(\cdot)$ using E and the current samples;

Update θ by minimizing a regression loss $\mathbb{E}[\|v_\theta(t, x; T) - u_t\|^2]$;

Return: Trained flow model $q_\theta(\cdot; T)$ (or a family across T).

Remark D.1 (Where the “math” lives). The crucial design choice is how to construct u_t from energy information so that matching it provably (or at least heuristically) decreases a divergence to the target. Different energy-based flow methods differ mostly in this construction. The original iEFM proposal ([woo2024iefm](#)) provides a concrete recipe in the Boltzmann setting.

D.8 A cautionary note

Energy-based training can fail in two common ways:

- **Energy misspecification:** if E (or its surrogate) is inaccurate in regions explored by the model, the learned flow may concentrate probability mass in unphysical artifacts.
- **Mode collapse:** strong energy gradients can pull samples into a single basin unless diversity is explicitly maintained (via temperature, noise injection, multiple initializations, or other diversity-promoting tricks).

These failure modes are not unique to iEFM; they are generic to energy-based generative modeling.

D.9 A variational viewpoint: Benamou–Brenier and kinetic energy

The continuity equation does more than describe how densities move under a velocity field: it also underlies a variational formulation of optimal transport. If ρ_0 and ρ_1 are two endpoint densities on $\mathbb{R}^d$, the (squared) 2-Wasserstein distance admits the dynamic formulation

$$W_2^2(\rho_0, \rho_1) = \inf_{\rho_t, v_t} \int_0^1 \int_{\mathbb{R}^d} \|v_t(x)\|^2 \rho_t(x) dx dt \quad \text{subject to} \quad \partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0,$$

where the infimum is taken over all density/velocity pairs connecting ρ_0 to ρ_1 . In words: among all ways to transport mass from ρ_0 to ρ_1 , optimal transport chooses the one with minimal kinetic energy.

While iEFM is not simply “optimal transport to a Boltzmann density,” this viewpoint provides useful intuition: learning a flow model amounts to choosing a path of marginals and a velocity field; different objectives implicitly bias the learned transport toward different paths (e.g. more OT-like, more diffusion-like, etc.).

D.10 Score matching and Stein identities

Because $\nabla \log p^*(x) = -\nabla E(x)$ is available without knowing Z , many objectives can be expressed in terms of ∇E . One example is *score matching*: in its simplest form, one tries to match the model score $\nabla \log q_\theta(x)$ to the target score $\nabla \log p^*(x)$:

$$\mathbb{E}_{x \sim q_\theta} [\|\nabla \log q_\theta(x) - \nabla \log p^*(x)\|^2] = \mathbb{E}_{x \sim q_\theta} [\|\nabla \log q_\theta(x) + \nabla E(x)\|^2].$$

This objective does not require Z , only gradients of E . It also highlights why energy-based learning often emphasizes gradients rather than values: normalization disappears after differentiating.

A related family of identities are *Stein identities*, which express expectations under p^* as divergences involving $\nabla \log p^*$. These identities can sometimes be used to derive learning signals that push a model toward the target distribution without explicit likelihood maximization.

D.11 From conformers to paths: what does the flow generate?

A CNF produces samples by integrating an ODE from $t = 0$ to $t = 1$. In the enzyme setting, the state x already encodes a full conformation. So what is a “path”?

There are at least two distinct objects:

- (1) **Generative path in latent time:** the curve $t \mapsto x_t$ traced by the ODE during sampling. This curve lives in conformation space and is guaranteed to be smooth in t (given smooth v_θ), but it is not automatically a physical trajectory.

- (2) **Physical (simulation) trajectory:** the time evolution of atomic coordinates under a physical dynamical model (Newtonian or Langevin). This curve depends on masses, friction, and stochastic forcing, and is typically used to define kinetics.

Part II uses the generative path as a *proposal* for transition-like motion (a plausible geometric interpolation between states), not as a literal molecular dynamics trajectory. Nevertheless, the energy-based training encourages these paths to pass through low-energy regions and to respect important geometric constraints.

Exercise D.2. In a toy 1D system with energy $U(x)$, compare:

- (a) the most probable path under overdamped Langevin dynamics conditioned on endpoints, and
- (b) the deterministic path obtained by integrating $\dot{x}_t = -\nabla U(x_t)$.

In what regimes (e.g. low vs high noise) do these paths resemble each other?

E Additional Implementation Notes

This appendix collects practical notes that are useful when implementing Mora-iFEM, without introducing new theory.

E.1 Choice of path discretization M

TPS operates on trajectories $(x_0, \dots, x_M)$. Larger M improves temporal resolution but increases dimensionality linearly. A practical starting point is $M \in \{16, 32, 64\}$ depending on system size.

E.2 Handling rigid-body modes

Because UMA energies are invariant to global rigid motions, training benefits from removing translation by centering and optionally fixing a reference frame (e.g. anchoring on ligand coordinates). Random rotations can be applied to prevent the CNF from overfitting to a fixed orientation.

E.3 Topology-dependent geometry penalties

L_{geom} in (15) requires chemical topology (bond graph). Topology can be obtained from the `.cif` parser or via standard chemistry toolkits. The penalty should be treated as a soft regularizer and monitored for stability.

E.4 Hydra configuration sketch

A new experiment configuration can include:

```
experiment: enzyme_tifem_uma
model:
  eq: {...}
  tps: {...}
  seq: {...}
temperature:
  ladder: [755, 500, 400, 350, 300]
  schedule: monotone_epoch
```

```

masking:
  tps:
    mode: residue | sidechain
    rho_min: 0.25
    rho_max: 0.50
    ramp_steps: 50000
  seq:
    rho_min: 0.25
    rho_max: 0.50
    ramp_steps: 50000

```

F Selected Solutions and Additional Worked Problems

This appendix collects worked solutions to several exercises and fills in computations that are easy to skip when first learning the material. The goal is not merely to record an answer, but to model the style of computation and proof that the main text uses.

F.1 Solution to the exercise in Section 7 on exact sequences (Exact sequences)

We analyze the sequence of linear maps

$$0 \longrightarrow \mathbb{R}^2 \xrightarrow{A} \mathbb{R}^3 \xrightarrow{B} \mathbb{R}^3 \xrightarrow{C} \mathbb{R}^2 \longrightarrow 0,$$

where (as given in the notes)

$$A = \begin{pmatrix} 2 & 3 \\ 0 & 1 \\ 2 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -3 & -1 \\ 1 & -3 & -1 \\ 1 & -3 & -1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

- (1) **Check that the sequence is a complex.** We need $BA = 0$ and $CB = 0$. A quick computation shows

$$BA = \begin{pmatrix} 1 & -3 & -1 \\ 1 & -3 & -1 \\ 1 & -3 & -1 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 0 & 1 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 2+0-2 & 3-3+0 \\ 2+0-2 & 3-3+0 \\ 2+0-2 & 3-3+0 \end{pmatrix} = 0,$$

so $\text{im}(A) \subseteq \ker(B)$. Similarly,

$$CB = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -3 & -1 \\ 1 & -3 & -1 \\ 1 & -3 & -1 \end{pmatrix} = \begin{pmatrix} 3 & -9 & -3 \\ 2 & -6 & -2 \end{pmatrix}.$$

This is *not* the zero matrix, so as written the sequence is *not even a complex* unless the intended B or C differs by a sign/convention. If we interpret the exercise literally with these matrices, then exactness fails already at $\mathbb{R}^3$ because $CB \neq 0$.

- (2) **If we correct C so that $CB = 0$.** A common fix is that the last map should annihilate the image of B , i.e. its rows should be orthogonal to $(1, 1, 1)$ and $(-3, -3, -3)$ and $(-1, -1, -1)$ simultaneously. Since $\text{im}(B)$ is spanned by $(1, 1, 1)$, it suffices that $C(1, 1, 1) = 0$. The given

C does not have this property (it sends $(1, 1, 1)$ to $(3, 2)$). With a corrected C , one can then test exactness by computing ranks:

$$\text{rank}(A) \leq 2, \quad \text{rank}(B) = 1, \quad \text{rank}(C) \leq 2.$$

Exactness at the middle $\mathbb{R}^3$ would force $\dim \ker(C) = \dim \text{im}(B) = 1$, hence $\text{rank}(C) = 2$. Exactness at the first $\mathbb{R}^3$ would force $\dim \ker(B) = \dim \text{im}(A) = \text{rank}(A)$, but $\dim \ker(B) = 2$ (since $\text{rank}(B) = 1$), so one would need $\text{rank}(A) = 2$. Thus the only way exactness could hold is with $\text{rank}(A) = 2$, $\text{rank}(B) = 1$, $\text{rank}(C) = 2$ and $CB = 0$.

Could this be (part of) the de Rham homology/cohomology of an oriented manifold?

De Rham cohomology groups $H^k(M)$ are finite-dimensional vector spaces over $\mathbb{R}$, but they do not typically appear as a short exact sequence of the form above unless M has very special topology and the maps are induced by geometric constructions (e.g. Mayer–Vietoris). Moreover, for a connected oriented closed manifold, $H^0(M) \cong \mathbb{R}$ and $H^n(M) \cong \mathbb{R}$, and Poincaré duality forces $b_k = b_{n-k}$. A random-looking sequence of vector spaces and maps is therefore very unlikely to arise as cohomology of an oriented manifold unless it satisfies these structural constraints.

F.2 Solution to the exercise in Section 7 on the exterior derivative (Exterior derivative and ambient dimension)

The exercise asks: take d of the form

$$\omega = a_1 dx \in \Omega^1(\mathbb{R}^4),$$

and ask whether the answer changes if we regard ω as a 1-form on $\mathbb{R}^5$.

Computation in $\mathbb{R}^4$. Write coordinates on $\mathbb{R}^4$ as (x_1, x_2, x_3, x_4) and interpret dx as (say) dx_1 . Then

$$\omega = a_1(x_1, x_2, x_3, x_4) dx_1.$$

By Definition 2.14,

$$d\omega = d(a_1) \wedge dx_1 = \left(\sum_{j=1}^4 \frac{\partial a_1}{\partial x_j} dx_j \right) \wedge dx_1 = \sum_{j=2}^4 \frac{\partial a_1}{\partial x_j} dx_j \wedge dx_1.$$

(The $j = 1$ term vanishes because $dx_1 \wedge dx_1 = 0$.)

What changes in $\mathbb{R}^5$? Now regard a_1 as a function of $(x_1, \dots, x_5)$. Then

$$d(a_1) = \sum_{j=1}^5 \frac{\partial a_1}{\partial x_j} dx_j, \quad \text{so} \quad d\omega = d(a_1) \wedge dx_1 = \sum_{j=2}^5 \frac{\partial a_1}{\partial x_j} dx_j \wedge dx_1.$$

Thus there is an *extra term*

$$\frac{\partial a_1}{\partial x_5} dx_5 \wedge dx_1$$

unless a_1 is independent of x_5 (for instance if we extend a_1 from $\mathbb{R}^4$ to $\mathbb{R}^5$ by making it constant in the x_5 direction). So the answer depends on how you extend the coefficient function to the larger space.

F.3 A worked pullback computation

We revisit Example 2.12 (pullback of a 2-form) and show a reliable method.

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be given by $f(u, v) = (u^2v, u, v)$ and let

$$\omega = z^2 dx \wedge dy + y^2 dx \wedge dz + x^2 dy \wedge dz.$$

Compute $f^*\omega$:

$$x = u^2v, \quad y = u, \quad z = v, \quad dx = 2uv du + u^2 dv, \quad dy = du, \quad dz = dv.$$

Then

$$\begin{aligned} f^*(dx \wedge dy) &= (2uv du + u^2 dv) \wedge du = u^2 dv \wedge du = -u^2 du \wedge dv, \\ f^*(dx \wedge dz) &= (2uv du + u^2 dv) \wedge dv = 2uv du \wedge dv, \quad f^*(dy \wedge dz) = du \wedge dv. \end{aligned}$$

Substituting coefficients,

$$f^*\omega = v^2(-u^2) du \wedge dv + u^2(2uv) du \wedge dv + (u^2v)^2 du \wedge dv,$$

so

$$f^*\omega = (-u^2v^2 + 2u^3v + u^4v^2) du \wedge dv.$$

This matches the computation in the main text.

F.4 A worked Hodge star computation in $\mathbb{R}^3$

Let $\omega = a_{12} dx_1 \wedge dx_2 + a_{13} dx_1 \wedge dx_3 + a_{23} dx_2 \wedge dx_3$. Using the convention that $dx_1 \wedge dx_2 \wedge dx_3$ is positively oriented, one has

$$\star(dx_1 \wedge dx_2) = dx_3, \quad \star(dx_1 \wedge dx_3) = -dx_2, \quad \star(dx_2 \wedge dx_3) = dx_1,$$

hence

$$\star\omega = a_{12} dx_3 - a_{13} dx_2 + a_{23} dx_1,$$

as in Example 2.20.

F.5 Further exercises for practice

Exercise F.1. Let $\alpha = f dx + g dy + h dz$ be a 1-form on $\mathbb{R}^3$. Compute $d\alpha$ explicitly and verify that, under the identifications of Section 2.5, $d\alpha$ corresponds to $\text{curl}(\nabla f)$.

Exercise F.2. Let M be a compact oriented n -manifold and let ω be a closed $(n-1)$ -form. Show that the functional $H_{\text{dR}}^1(M) \rightarrow \mathbb{R}$ given by $\eta \mapsto \int_M \eta \wedge \omega$ is well-defined on cohomology classes.

F.6 Solution: degree of the map $z \mapsto z^k$ on S^1

Let $f : S^1 \rightarrow S^1$ be $f(e^{it}) = e^{ikt}$ for an integer k . We claim $\deg(f) = k$.

One way to see this is by integration (Proposition 9.23). Let $\omega = d\theta$ be the standard 1-form on S^1 in the angular coordinate θ . Then

$$f^*(d\theta) = d(k\theta) = k d\theta.$$

Therefore

$$\int_{S^1} f^*(d\theta) = \int_{S^1} k d\theta = k \int_{S^1} d\theta = k \cdot 2\pi.$$

By Proposition 9.23,

$$\int_{S^1} f^*(d\theta) = \deg(f) \int_{S^1} d\theta = \deg(f) \cdot 2\pi,$$

so $\deg(f) = k$.

F.7 Solution: a nowhere-vanishing vector field on T^2

Write the torus as $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ with coordinates (θ^1, θ^2) . The constant vector field

$$X = \frac{\partial}{\partial \theta^1}$$

on $\mathbb{R}^2$ is $\mathbb{Z}^2$ -invariant and descends to a smooth vector field on T^2 . It never vanishes, so T^2 admits a nowhere-vanishing vector field. This is consistent with Poincaré–Hopf: $\chi(T^2) = 0$, and a vector field without zeros has total index 0.

More generally, every torus T^n admits n linearly independent nowhere-vanishing vector fields coming from the coordinate directions.

F.8 Solution: submersions are open maps

Let $f : M \rightarrow N$ be a submersion and let $U \subseteq M$ be open. We show $f(U)$ is open in N .

Pick $q \in f(U)$ and choose $p \in U$ with $f(p) = q$. By the constant rank theorem (submersion case), there are charts $(\phi : U_p \rightarrow \mathbb{R}^m)$ around p and $(\psi : V_q \rightarrow \mathbb{R}^n)$ around q such that, in these coordinates,

$$\psi \circ f \circ \phi^{-1}(x_1, \dots, x_m) = (x_1, \dots, x_n).$$

In particular, the map is locally a projection $\mathbb{R}^m \rightarrow \mathbb{R}^n$, which sends open sets to open sets. Since $U \cap U_p$ is open in U_p , its image under f contains a neighborhood of q in N . Thus every point of $f(U)$ is an interior point, so $f(U)$ is open.

F.9 A worked example: $H_{\text{dR}}^1(\mathbb{R}^2 \setminus \{0\})$

Let $U = \mathbb{R}^2 \setminus \{0\}$ with coordinates (x, y) . Consider the 1-form

$$\omega = \frac{-y \, dx + x \, dy}{x^2 + y^2}.$$

A computation shows $d\omega = 0$ on U . (For instance, write $\omega = d\theta$ in polar coordinates $x = r \cos \theta$, $y = r \sin \theta$; then $\omega = d\theta$ and $d\omega = 0$.) If $\omega = df$ were exact, then $\int_{S^1} \omega = 0$ for any loop, but on the unit circle one has

$$\int_{S^1} \omega = \int_0^{2\pi} d\theta = 2\pi \neq 0,$$

so ω is not exact. Therefore $[\omega] \neq 0$ in $H_{\text{dR}}^1(U)$.

Using Mayer–Vietoris with the cover of U by two overlapping angular sectors (each contractible) shows that $H_{\text{dR}}^1(U) \cong \mathbb{R}$. Thus $[\omega]$ is a generator, and it measures the winding number of loops around the origin.

G A quick reference of identities and conventions

This page collects several formulas used repeatedly throughout the notes. (Conventions vary between authors; when comparing sources, always check signs and orientation conventions.)

G.1 Algebra of forms

Let $\alpha \in \Omega^r(M)$ and $\beta \in \Omega^s(M)$.

- Graded-commutativity: $\alpha \wedge \beta = (-1)^{rs} \beta \wedge \alpha$.
- Leibniz rule for d : $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^r \alpha \wedge d\beta$.
- $d^2 = 0$.
- Interior product: $\iota_X(\alpha \wedge \beta) = \iota_X \alpha \wedge \beta + (-1)^r \alpha \wedge \iota_X \beta$.
- Cartan's magic formula: $\mathcal{L}_X \alpha = d(\iota_X \alpha) + \iota_X(d\alpha)$.

G.2 Pullbacks

For a smooth map $f : M \rightarrow N$ and forms ω, η on N :

- $f^*(\omega + \eta) = f^*\omega + f^*\eta$.
- $f^*(\omega \wedge \eta) = f^*\omega \wedge f^*\eta$.
- $f^*(d\omega) = d(f^*\omega)$.
- If $g : N \rightarrow P$, then $(g \circ f)^* = f^* \circ g^*$.

G.3 Hodge star (Euclidean conventions)

On $\mathbb{R}^n$ with the standard orientation and inner product, the Hodge star $\star : \Lambda^k(\mathbb{R}^n)^* \rightarrow \Lambda^{n-k}(\mathbb{R}^n)^*$ satisfies

$$\alpha \wedge \star \beta = \langle \alpha, \beta \rangle dx_1 \wedge \cdots \wedge dx_n.$$

In particular, $\star \star \alpha = (-1)^{k(n-k)} \alpha$ for $\alpha \in \Lambda^k$.

G.4 Vector calculus in $\mathbb{R}^3$ via forms

With the identifications described in Section 2.5 (metric dual and Hodge star):

- gradient: ∇f corresponds to df ;
- curl: $\text{curl}(X)$ corresponds to $\star d(X^\flat)$;
- divergence: $\text{div}(X)$ corresponds to $\star d \star (X^\flat)$;
- Laplacian: $\Delta f = \text{div}(\nabla f)$ corresponds to $\star d \star df$.

G.5 Stokes' theorem

If M is an oriented compact n -manifold with boundary and $\omega \in \Omega^{n-1}(M)$, then

$$\int_M d\omega = \int_{\partial M} \omega,$$

where ∂M is oriented by the outward-normal-first convention.

G.6 Homological algebra essentials

Let $(C_\bullet, d)$ be a chain complex in an abelian category (often $\text{Vect}_{\mathbb{R}}$):

- $Z_n(C) = \ker(d_n)$ (cycles), $B_n(C) = \text{im}(d_{n+1})$ (boundaries), $H_n(C) = Z_n/B_n$.
- A chain map $f : C \rightarrow D$ is a collection of maps $f_n : C_n \rightarrow D_n$ with $df = fd$.
- A chain homotopy $s : f \simeq g$ is a degree +1 map $s_n : C_n \rightarrow D_{n+1}$ such that $f - g = ds + sd$.
- If $f \simeq g$, then $f_* = g_*$ on homology.
- A short exact sequence of complexes $0 \rightarrow C' \rightarrow C \rightarrow C'' \rightarrow 0$ induces a long exact sequence in homology via the connecting homomorphism constructed in the snake lemma.

G.7 Cohomology conventions

We write $\Omega^k(M)$ for smooth k -forms, $Z^k(M) = \ker(d : \Omega^k \rightarrow \Omega^{k+1})$, $B^k(M) = \text{im}(d : \Omega^{k-1} \rightarrow \Omega^k)$, and $H_{\text{dR}}^k(M) = Z^k(M)/B^k(M)$.

For a connected oriented closed n -manifold:

- $H_{\text{dR}}^0(M) \cong \mathbb{R}$ (constants),
- $H_{\text{dR}}^n(M) \cong \mathbb{R}$ generated by an orientation form (Poincaré duality),
- $\chi(M) = \sum_{k=0}^n (-1)^k \dim H_{\text{dR}}^k(M)$,
- $\dim H_{\text{dR}}^k(M) = \dim H_{\text{dR}}^{n-k}(M)$ (Poincaré duality).

G.8 Orientation conventions and signs

- An orientation on an n -manifold M is an equivalence class of nowhere-vanishing n -forms, where $\omega \sim \omega'$ if $\omega = a\omega'$ for some smooth $a > 0$.
- If M has boundary, the induced orientation on ∂M is *outward normal first*: a positively oriented basis $(v_1, \dots, v_{n-1})$ of $T_p(\partial M)$ is one for which $(\nu, v_1, \dots, v_{n-1})$ is positively oriented in $T_p M$, where ν is an outward-pointing normal vector.
- For a diffeomorphism $F : M \rightarrow M$, F is orientation preserving iff $\det(dF_p) > 0$ for all p .

G.9 Change of variables for integrals of forms

Let $f : M^n \rightarrow N^n$ be a diffeomorphism of oriented manifolds and let $\omega \in \Omega^n(N)$ have compact support. Then

$$\int_M f^* \omega = \int_N \omega.$$

In local coordinates this is the usual Jacobian change-of-variables formula.

More generally, if f is a proper smooth map between oriented n -manifolds, then

$$\int_M f^* \omega = \deg(f) \int_N \omega$$

when ω is an n -form and $\deg(f)$ is defined (for example when M is compact and N is connected).

G.10 Intersection numbers in dimension n

If M^n is oriented and $A^p, B^q \subseteq M$ are oriented embedded submanifolds with $p + q = n$ and $A \pitchfork B$, then $A \cap B$ is a finite set of points. Each $x \in A \cap B$ has a sign ± 1 determined by whether the direct sum orientation on $T_x A \oplus T_x B$ matches the orientation of $T_x M$. The *intersection number* is

$$A \cdot B := \sum_{x \in A \cap B} \text{sign}(x) \in \mathbb{Z}.$$

Intersection numbers are stable under homotopies through transverse intersections.

H Roadmap for further study

These notes provide a first coherent pass through multilinear algebra, differential forms, manifolds, and the algebraic tools (exact sequences and diagram chases) that organize topological information. A natural next step is to deepen each thread:

- **Smooth manifolds and differential topology:** transversality as a general-position principle, tubular neighborhoods, intersection theory, and Morse theory (often continuing to cobordism and characteristic classes).
- **Forms and geometry:** Riemannian metrics, connections, Hodge theory, and the analytic proof of Poincaré duality on compact oriented manifolds.
- **Algebraic topology:** singular homology/cohomology, CW complexes, spectral sequences, and the de Rham theorem as one instance of “cohomology theories agree”.
- **Homological algebra:** derived functors (Ext, Tor), projective and injective resolutions, and how exactness properties encode topology and geometry.

The texts in the bibliography are excellent entry points, and **bott-tu** is especially recommended for seeing differential forms used systematically in algebraic topology.

I Annotated bibliography

- **mcinerney2013** is a gentle entry point for differential geometry and smooth manifolds, with many concrete computations and examples.
- **docarmo1994** is a classic focused introduction to differential forms, Stokes’ theorem, and applications.
- **tu2011** develops manifolds carefully and emphasizes geometry/topology interplay; the exercises are excellent.
- **lee2013** is a comprehensive modern reference for smooth manifolds and differential topology, including vector bundles and transversality.
- **guilleminpollack** is a standard differential topology text, famous for its clear treatment of transversality, degree, and intersection theory.
- **bott-tu** connects differential forms to algebraic topology and provides a full proof of the de Rham theorem, plus many computations.

As you continue studying, it is worth reading at least one source that emphasizes *geometry* (metrics and connections) and one that emphasizes *topology* (homology/cohomology and exact sequences). The power of differential topology is that many arguments can be run in either language once you are fluent in both.